



Deutsche Bank
Research

APAC Automation Market: Where we are in the cycle? A study based on historical patterns

Iris Zheng, Head of APAC Automation & Industrials | (+852) 2203 5884 | iris.zheng@db.com

Xu Liu, APAC Automation & Industrials | (+852) 2203 5885 | xu.liu@db.com

28 September 2026

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.



I. Where we are in the cycle? A study based on historical patterns

Executive summary	2
Key charts	4

II. Historical patterns of Japan machine tool orders

Further order upside, YoY likely already peaking	9
Global	10
Japan (domestic)	11
Foreign (non-Japan)	12
China	13
Other Asia	14
North America	15
United States	16
Europe	17
Automation market growth by region	18

III. Japan: Correlation of share price vs Japan machine tool orders

Share-price correlation with order growth	20
Order YoY deceleration could be mild	22
Japanese company past four upcycles analyses	23
Share-price correction	23
FANUC (6954.T): HOLD	24
THK (6481.T): SELL	27
Yaskawa Electric (6506.T): BUY	30
SMC (6273.T): BUY	33
Harmonic Drive Systems (6324.T): BUY	36

IV. Taiwan: Airtac (1590.TW; BUY) sales upcycles analyses

Sales upcycles patterns and end-market drivers	40
Share-price correlation and past cycles	42
Share-price correction	45

V. Taiwan: Hiwin (2049.TW; BUY) sales upcycles analyses

Sales upcycles patterns and end-market drivers	47
Share-price correlation and past cycles	49
Share-price correction	52

VI. Link to previous reports and Appendix

Where we are in the cycle? A study based on historical patterns



Amid increasing concerns over the automation market cycle from investors, we study the historical cycle datapoints and share price movements to understand the cycle patterns and implications to share prices of automation companies.

Further upside potential for automation orders and sales, but YoY growth is likely peaking with a mild deceleration ahead

Japan machine tool orders data implies that, as of June 2026, we are likely **70-80% through the current upcycle, leaving a further 20-30% of upside potential for the orders**, based on our analysis of historical patterns. That said, as the market is more than halfway through this phase, we believe **the YoY growth rate is likely peaking**. Conversely, our **blue-sky scenario**—which accounts for a potential super-upcycle driven by structurally higher demand from AI and reshoring—suggests we are only over 60% through the current upcycle.

By region, our upcycle maturity indicator shows that the upcycle is most mature in **China** (over 80% through, having been the first region to record an order recovery since 2H23, with accelerating growth since 2Q24) and **North America** (over 70% through, driven by tariff-related demand since April 2025). In comparison, **Japan** (60-70% through), **Europe** (50-60% through), and **Other Asia** (~50% through) are at relatively earlier stages. By end-market, **Electricals** grew the fastest (over 150%), followed by **Aerospace & Shipping** (over 100%), whilst **Industrial machinery** (over 70%) and **Automotive** (over 80%) also made positive contributions.

We have also analysed the cyclical patterns of Airtac and Hiwin. **Airtac's** current upcycle stage is difficult to assess, as its cumulative sales improvement has already exceeded historical levels. We estimate that Airtac is likely ~70% through the upcycle, with YoY sales growth potentially peaking in 2Q26 or 3Q26E. However, the 20th anniversary of the iPhone and subsequent new product launches in 2027E could offer further upside. In comparison, **Hiwin** (30-35% through) remains at an early stage of its upcycle, and in our view, its YoY sales growth is unlikely to peak until late 2026E or 1Q27E.

Higher orders/sales justify higher share prices, but potentially slowing YoY growth is testing near-term share price momentum

For Japanese factory automation (FA) companies, **share prices show a high correlation with both absolute order values and the YoY growth of Japanese machine tool orders**. Furthermore, **share price returns are closely correlated with YoY order growth**. This implies that higher order values justify higher share prices; however, short-term share price returns tend to track order YoY growth momentum. We observe similar correlations between share prices and sales value/growth for Airtac and Hiwin. We believe the correlation partly explains the widespread share price correction seen since 2Q26, which has occurred amid concerns that YoY order/sales growth is peaking, whilst absolute orders and sales continue to expand.

We note that the share price returns of **Airtac** (37%), **FANUC** (34%), **Yaskawa** (34%), **THK** (29%), and **Hiwin** (29%) exhibit a correlation of ~30% or above with YoY order/sales growth. This makes their share price momentum sensitive to changes in YoY growth. In comparison, **SMC** (17%) and **Harmonic Drive** (10%) show a weaker correlation.



Recent share price correction appears overdone; order growth could remain resilient post-peak

The market seems to assume that Japanese machine tool order growth peaked in August 2026 at 65% YoY (51% adjusted for FX; DBE, as detailed August datapoints not yet available). Consequently, the share prices of the five Japanese FA companies under our coverage **have corrected by close to 30%** on average in September 2026 from their respective peaks. **The magnitude of this correction is more severe** than the 15-20% decline observed 1M post-peak during the past two cycles; historically, a correction of ~30% was not reached until 3-6M after the peak. We believe this suggests **the current sell-off could be overdone**, particularly as **near-term order growth deceleration could prove milder** than in past cycles due to sustained AI-related demand.

Similarly, for **Airtac**, we expect YoY sales growth to peak in 2Q26/3Q26E, followed by a mild deceleration, whilst its share price has corrected more severely than previous cycles. For **Hiwin**, we believe sales growth could peak in 2H26E or early 2027E, with only a mild deceleration thereafter—a outlook that appears to have been more than priced in.

Order of preference: SMC (Buy) > Harmonic Drive (Buy) > Yaskawa (Buy) > FANUC (Hold) > THK (Sell) in Japan; Hiwin (Buy) > Airtac (Buy) in Taiwan

As a rule of thumb, **we prefer companies** that: **(1) exhibit lower correlation with YoY order and sales growth trends**, making them more resilient to market slowdown concerns; and **(2) can achieve superior alpha** through favourable end-market exposure (e.g. semiconductors), thematic growth (e.g. humanoid robots), and company-specific measures.

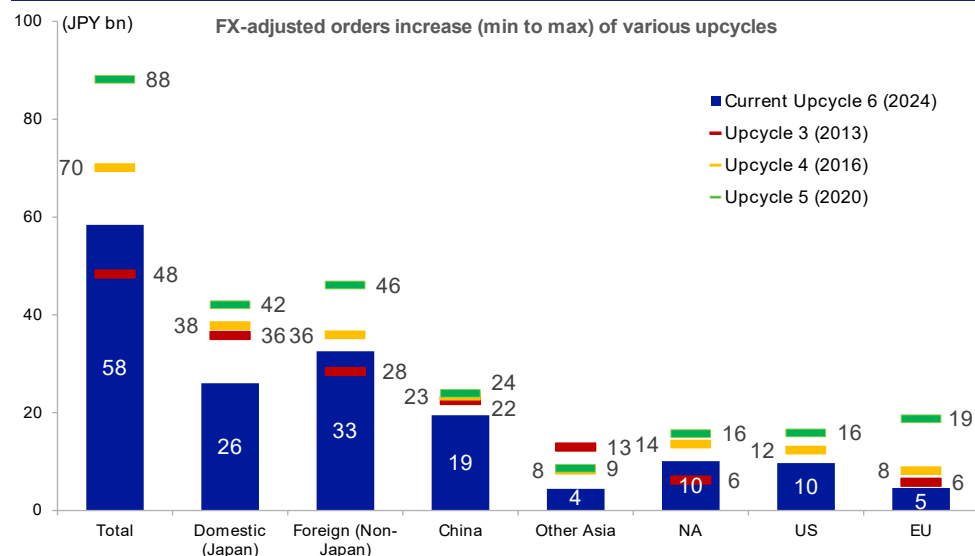
In Japan, our order of preference is: **SMC** (Buy, share price ¥69,620) > **Harmonic Drive** (Buy, share price ¥6,300) > **Yaskawa** (Buy, share price ¥4,480) > **FANUC** (Hold, share price ¥5,985) > **THK** (Sell, share price ¥6,429). Among these, SMC, THK, Yaskawa, and Harmonic Drive have more significant semiconductor exposure, whilst Harmonic Drive is well-positioned to capitalise on growth in the humanoid robot market. Heading into **the upcoming CY3Q26E reporting season**, we expect strong results from THK and SMC, whilst FANUC and Yaskawa are likely to be impacted by delivery bottlenecks.

In Taiwan, our preference is: **Hiwin** (Buy, share price TWD325.5) > **Airtac** (Buy, share price TWD1,210). Hiwin has greater exposure to the machine tool and semiconductor end-markets, and we expect higher average selling prices to further benefit margins in 2H26E, with expanded capacity supporting sales in 2027E. Airtac should enjoy extended resilient growth, though its YoY growth momentum is likely to fade from 4Q26E.

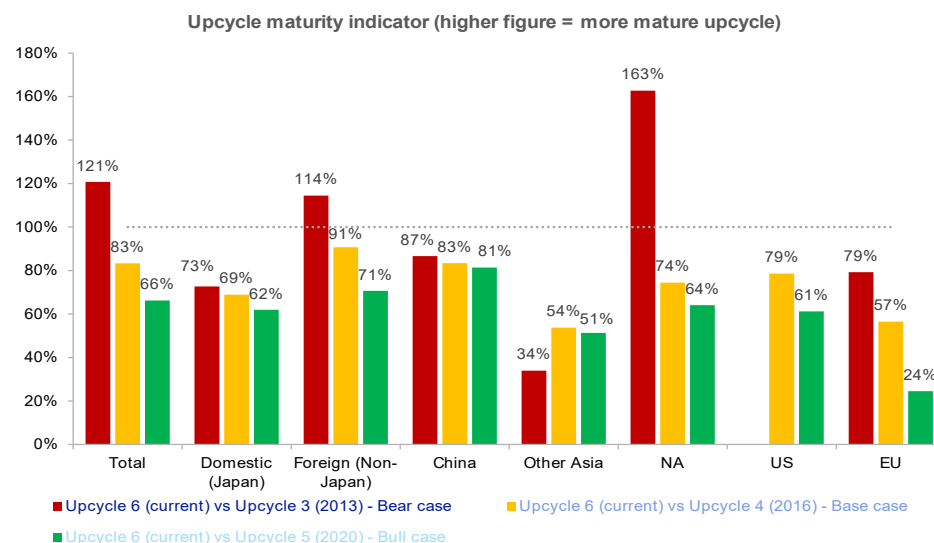
Key charts (1)



Upcycle pattern: Order increase in current upcycle vs past upcycles*



Upcycle maturity indicator: total orders already 70-80% through the upcycle



*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

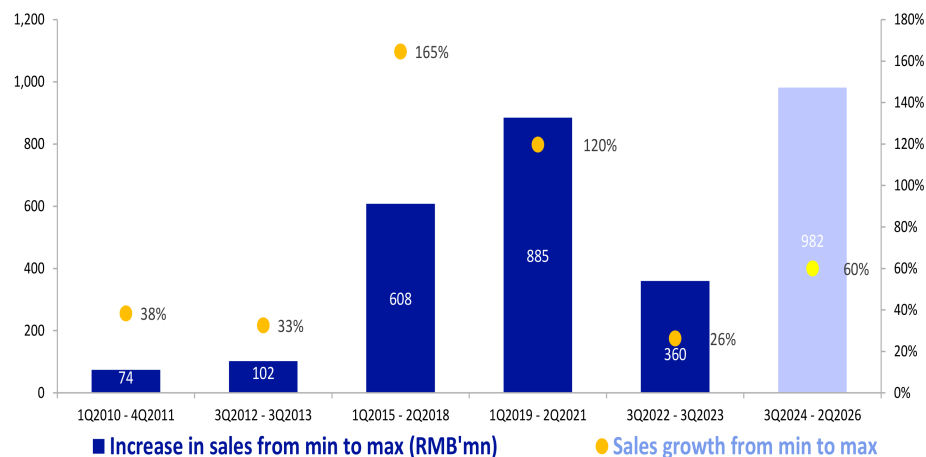
Share price performance: Correlates with order value and YoY growth of Japan machine tool orders, especially for FANUC and THK

Japan machine tool order metric (since 2015)	Stock lead	FANUC	THK	Harmonic Drive	Yaskawa	SMC
Reported values in JPY						
Japan machine tool orders value vs stock price	Same month	0.61	0.61	0.06	0.31	0.22
Japan machine tool orders value vs stock price	1 month	0.67	0.65	0.10	0.33	0.25
Japan machine tool orders value vs stock price	3 months	0.66	0.64	0.11	0.32	0.25
Japan machine tool orders YoY vs stock price	Same month	0.58	0.45	0.38	0.32	0.27
Japan machine tool orders YoY vs stock price	1 month	0.61	0.48	0.44	0.33	0.28
Japan machine tool orders YoY vs stock price	3 months	0.59	0.48	0.50	0.31	0.27
Japan machine tool orders YoY vs 1M stock return	Same month	-0.02	-0.03	-0.07	-0.01	-0.02
Japan machine tool orders YoY vs 1M stock return	1 month	0.04	0.03	-0.04	0.04	0.02
Japan machine tool orders YoY vs 1M stock return	3 months	0.14	0.12	0.05	0.17	0.06
Japan machine tool orders 3MMA YoY vs 3M stock return	Same month	0.02	0.00	-0.06	0.06	0.00
Japan machine tool orders 3MMA YoY vs 3M stock return	3 months	0.20	0.21	0.03	0.25	0.12
Japan machine tool orders 3MMA YoY vs 3M stock return	6 months	0.34	0.29	0.10	0.34	0.17
FX adjusted values						
FX-adjusted Japan machine tool orders value vs stock price	Same month	0.45	0.38	0.11	0.17	0.00
FX-adjusted Japan machine tool orders value vs stock price	1 month	0.53	0.44	0.16	0.18	0.02
FX-adjusted Japan machine tool orders value vs stock price	3 months	0.56	0.46	0.19	0.15	0.01
FX-adjusted Japan machine tool orders YoY vs stock price	Same month	0.51	0.38	0.35	0.23	0.17
FX-adjusted Japan machine tool orders YoY vs stock price	1 month	0.54	0.40	0.40	0.24	0.18
FX-adjusted Japan machine tool orders YoY vs stock price	3 months	0.55	0.42	0.46	0.20	0.17
FX-adjusted Japan machine tool orders YoY vs 1M stock return	Same month	-0.03	-0.03	-0.09	-0.02	-0.03
FX-adjusted Japan machine tool orders YoY vs 1M stock return	1 month	0.02	0.02	-0.03	0.05	0.01
FX-adjusted Japan machine tool orders YoY vs 1M stock return	3 months	0.10	0.07	0.01	0.13	0.06
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	Same month	0.00	0.00	-0.08	0.06	-0.03
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	3 months	0.17	0.16	-0.01	0.22	0.09
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	6 months	0.30	0.26	0.09	0.31	0.11

Key charts (2)



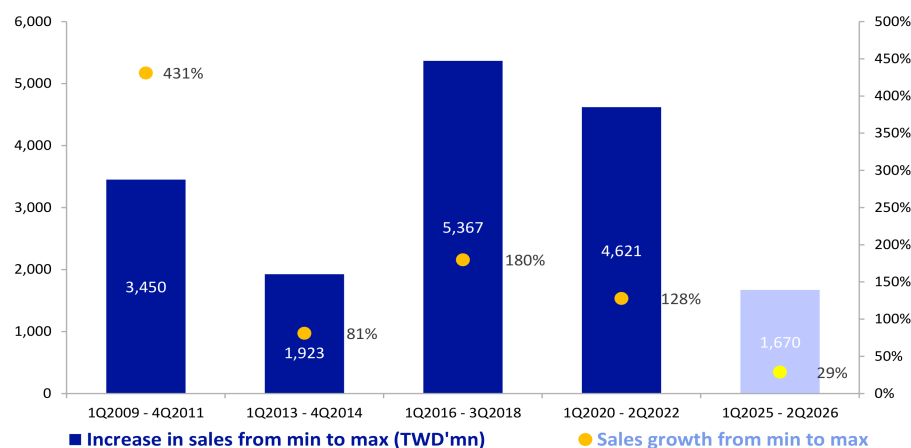
Upcycle pattern: Airtac has delivered sales increase exceeding past upcycles; however, upside remains given its new product launches



Upcycle maturity indicator: total orders already 70-80% through the upcycle

Relationship (Airtac) (since 2011)	Stock lead	Correlation
Quarterly sales (in TWD)		
Quarterly sales (in TWD) vs share price	Same Quarter	0.93
Quarterly sales (in TWD) vs share price	1Q Lead	0.92
Quarterly Sales (in TWD) vs 3M share price return	Same Quarter	-0.06
Quarterly Sales (in TWD) vs 3M share price return	1Q Lead	-0.08
Quarterly sales (in TWD) YoY		
Quarterly sales (in TWD) YoY vs share price	Same Quarter	0.05
Quarterly sales (in TWD) YoY vs share price	1Q Lead	-0.03
Quarterly sales (in TWD) YoY vs 3M share price return	Same Quarter	0.28
Quarterly sales (in TWD) YoY vs 3M share price return	1Q Lead	0.32
Quarterly sales (adjusted in RMB) YoY		
Quarterly sales (adjusted in RMB) YoY vs share price	Same Quarter	0.01
Quarterly sales (adjusted in RMB) YoY vs share price	1Q Lead	-0.07
Quarterly sales (adjusted in RMB) YoY vs 3M share price return	Same Quarter	0.30
Quarterly sales (adjusted in RMB) YoY vs 3M share price return	1Q Lead	0.37

Upcycle 4 (1Q20 to 2Q22)



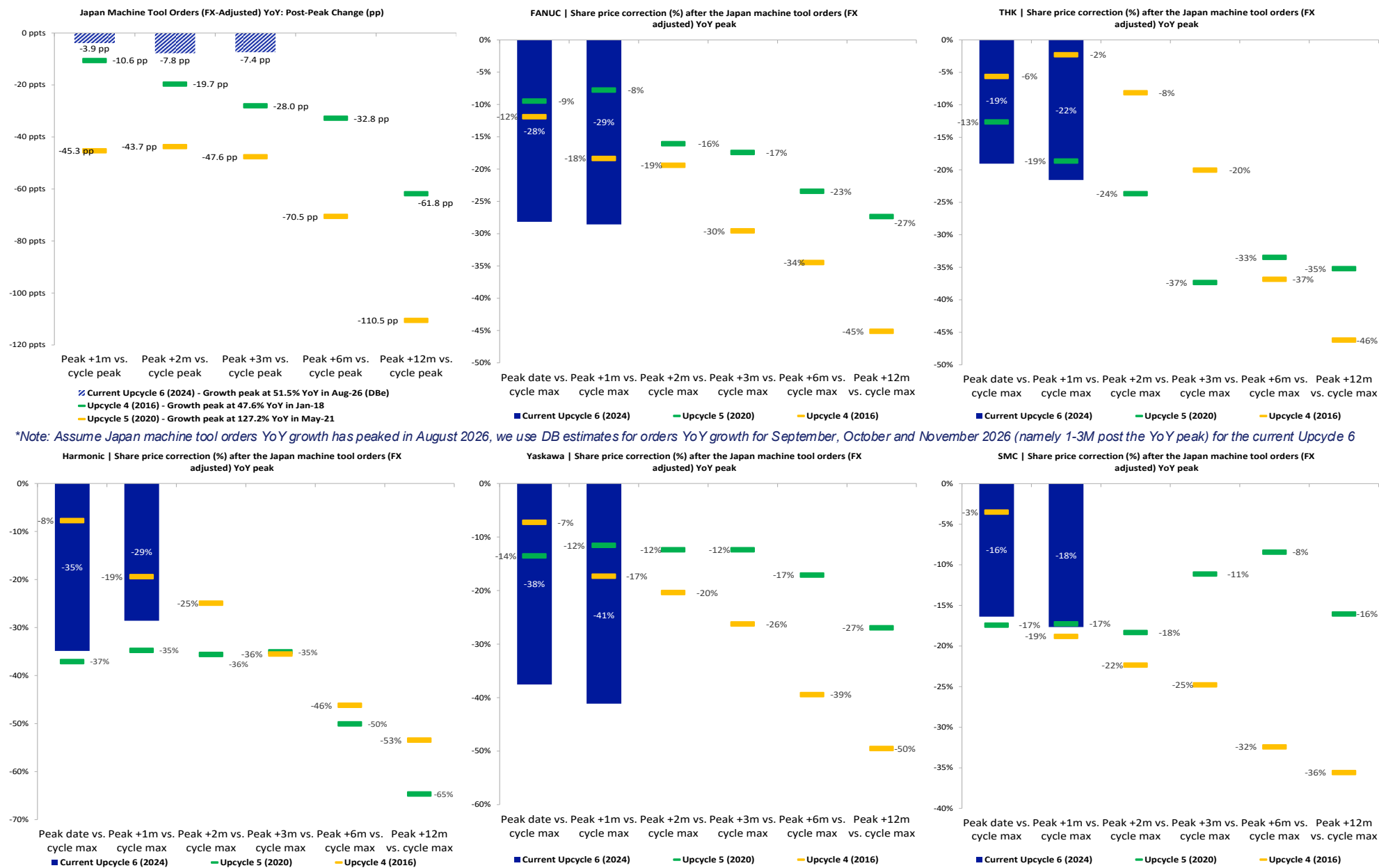
Current upcycle 5 (1Q25 to 2Q26)

Relationship (Hiwin) (since 2012)	Stock lead	Correlation
Quarterly sales		
Quarterly sales vs share price	Same Quarter	0.54
Quarterly sales vs share price	1Q Lead	0.60
Quarterly Sales vs 3M share price return	Same Quarter	-0.05
Quarterly Sales vs 3M share price return	1Q Lead	0.05
Quarterly sales YoY		
Quarterly sales YoY vs share price	Same Quarter	0.44
Quarterly sales YoY vs share price	1Q Lead	0.34
Quarterly sales YoY vs 3M share price return	Same Quarter	0.16
Quarterly sales YoY vs 3M share price return	1Q Lead	0.29

Key charts (3)



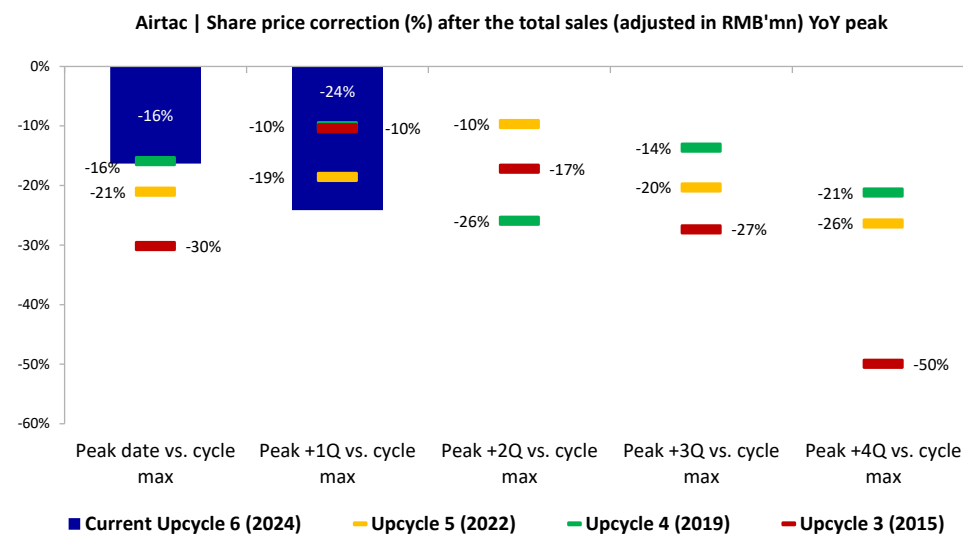
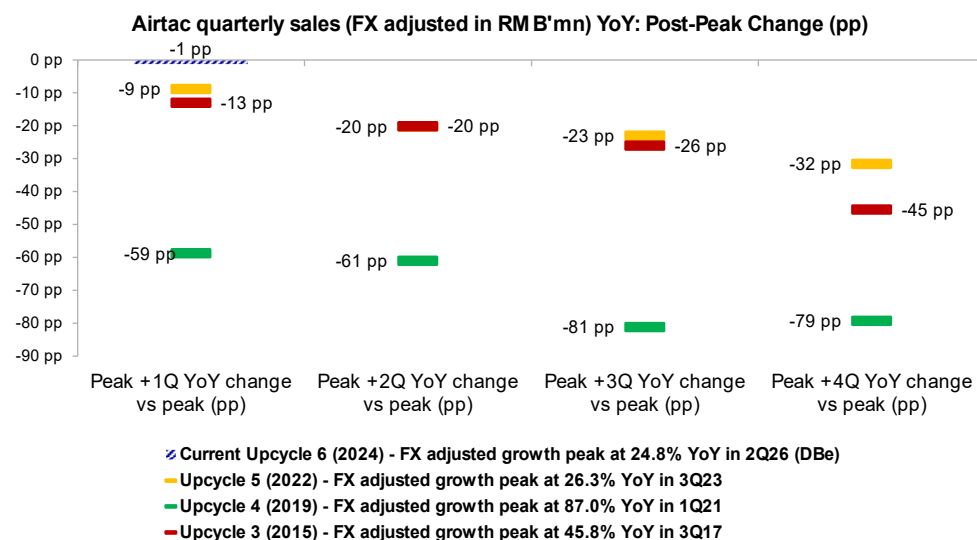
Japan FA names share prices correction appears overdone against potentially mild order growth deceleration: Share price has corrected more severely than the declines observed one month post-peak during the past two cycles



Key charts (4)

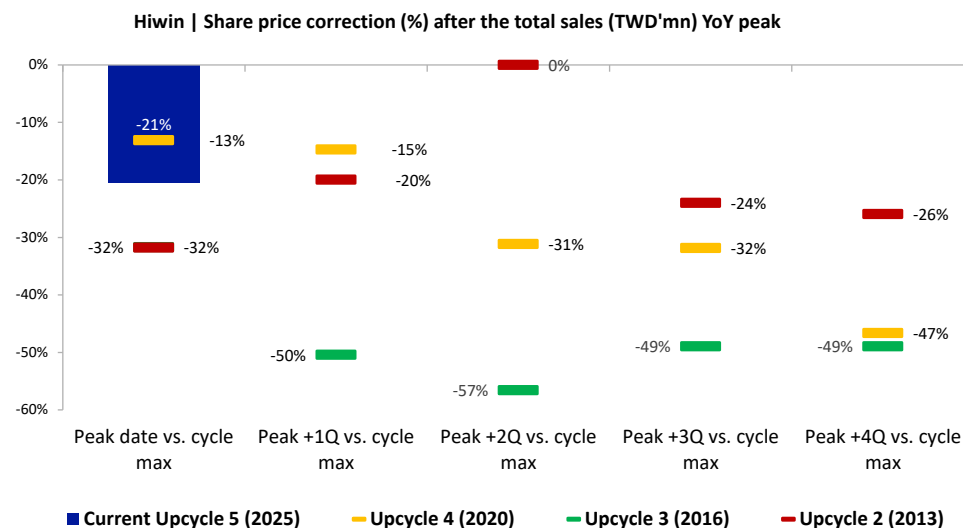
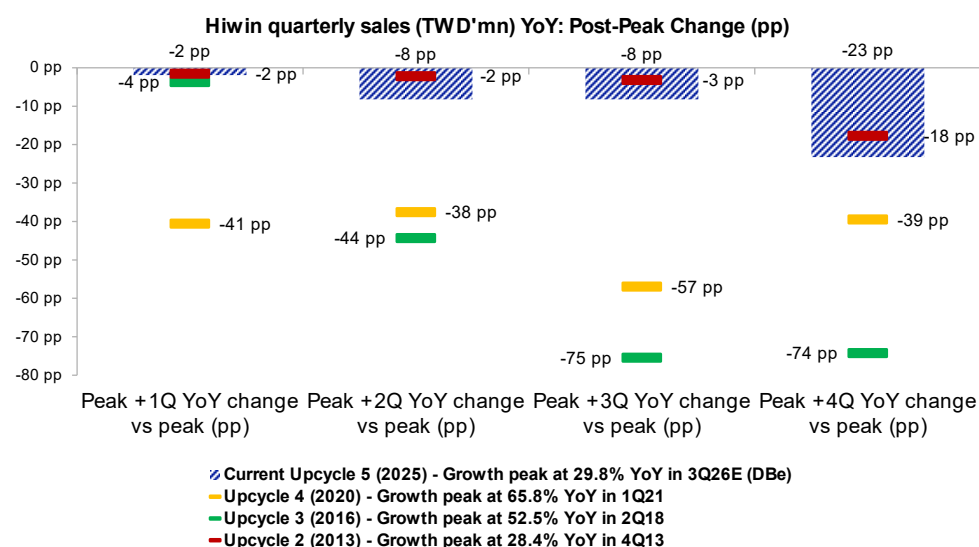


Airtac share prices correction appears overdone against potentially mild sales growth deceleration: Share price has corrected more severely than the declines observed one quarter post-peak during the past cycles



*Note: Assume Airtac FX-adjusted sales YoY growth has peaked in 2Q26, we use DB estimates for sales YoY growth 3Q26E (namely 1Q post the YoY peak) for the current Upcycle 6

Hiwin share prices correction appears overdone against potentially mild sales growth deceleration: Share price has corrected broadly inline with the declines observed one quarter post-peak during the past cycles



*Note: Assume Hiwin sales YoY growth to peak in 3Q26E, we use DB estimates for sales YoY growth 4Q26E, 1Q26E (namely 1Q post the YoY peak) for the current Upcycle 6



Historical patterns of Japan machine tool orders

Japan machine tool orders: Further upside for orders, but YoY growth likely already peaking

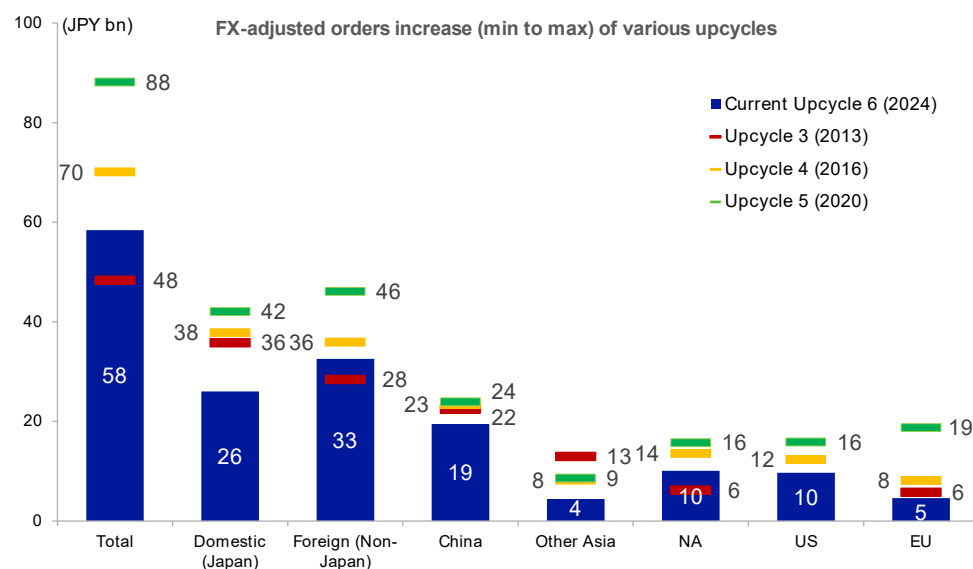


We have observed **six upcycles** in Japan machine tool orders since 2000: the 1st wave commenced in 2002, the 2nd in 2009, the 3rd in 2013, the 4th in 2016, the 5th in 2020, and **the current 6th cycle in 2024**. Here, we analyse these historical patterns and their implications for the current upcycle. We adopt **Upcycle 4 (2016) as our base case**. We use Upcycle 5 (2020) as our bull case, as we classify it as a super-cycle where COVID-19 lockdowns initially depressed demand, whilst subsequent global quantitative easing measures led to widespread and synchronised order strength globally.

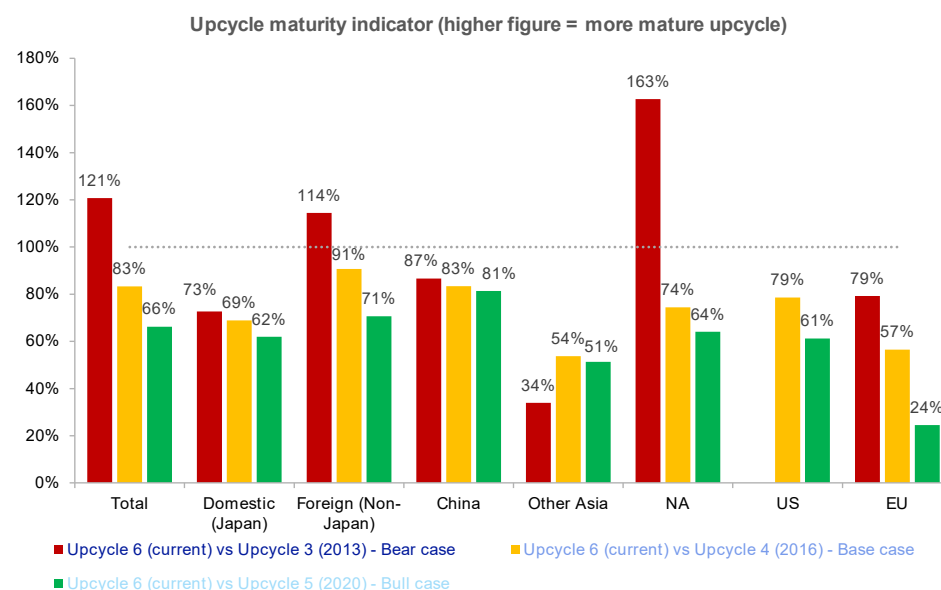
Our upcycle maturity indicator suggests that **Japan machine tool orders are already 70–80% through the current upcycle**, comparing the order increase of this cycle with historical upcycles. Consequently, whilst **we see further upside for machine tool orders, the rate of growth is likely already peaking**.

The current expansion has been driven mainly by robust orders from **China** (now over 80% through its upcycle) and **North America** (over 70% through). In comparison, **Japan** (at 60–70% through) should deliver further headroom, especially given the ongoing robust demand for semiconductor manufacturing equipment. Similarly, we see potential upside for the **Other Asia region** (at 50% through). Meanwhile, **Europe** (at 50–60% through) has so far shown only a mild recovery.

Upcycle pattern: Order increase in current upcycle vs past upcycles



Upcycle maturity indicator: total orders already 70-80% through the upcycle

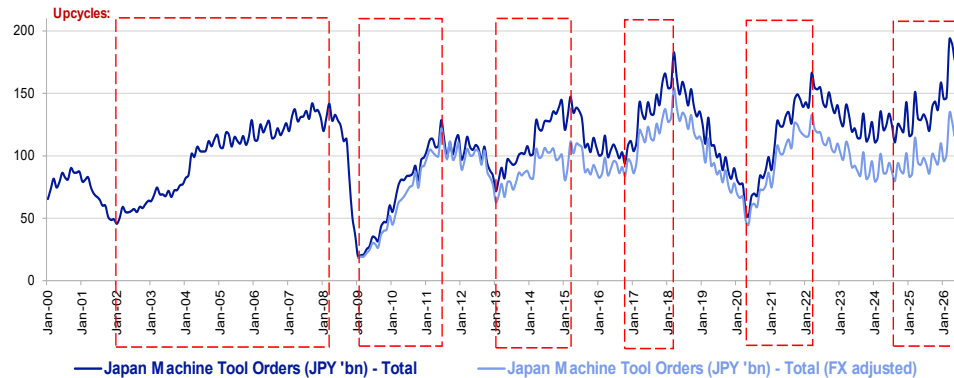


*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

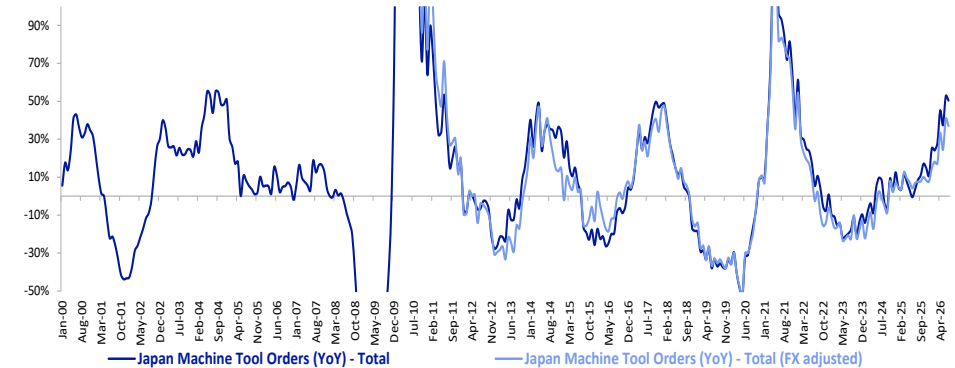
Japan machine tool orders: Global orders – Likely 70-80% through the upcycle with 20-30% further upside



Japan machine tool orders: Global orders – order value



Japan machine tool orders: Global orders – YoY growth

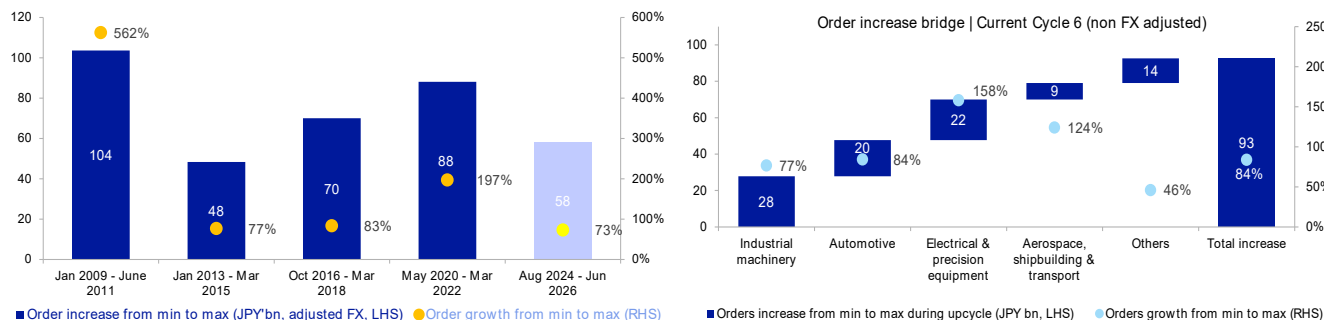


Japan machine tool orders: Global orders – pattern from past upcycles*

Total	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Jan 2002 - Mar 2008	Jan 2009 - June 2011	Jan 2013 - Mar 2015	Oct 2016 - Mar 2018	May 2020 - Mar 2022	Aug 2024 - Jun 2026
Upcycle duration (months)	75	30	27	18	23	23
Reported orders (JPY'bn)						
Start month	Jan-2002	Jan-2009	Jan-2013	Oct-2016	May-2020	Aug-2024
Min - Orders (JPY'bn)	46	19	72	94	51	111
End month	Mar-2008	Jun-2011	Mar-2015	Mar-2018	Mar-2022	Jun-2026
Max - Orders (JPY'bn)	142	129	147	183	166	203
Orders increase from min to max (JPY'bn)	96	110	76	89	115	93
Orders growth from min to max	211%	575%	106%	95%	224%	84%
Adjusted FX orders (JPY'bn)						
Start month		Jan-2009	Jan-2013	Oct-2016	May-2020	Aug-2024
Min - Orders (JPY'bn, adjusted FX)		18	63	84	45	80
End month		Jun-2011	Mar-2015	Mar-2018	Mar-2022	Jun-2026
Max - Orders (JPY'bn, adjusted FX)		122	111	154	133	138
Orders increase from min to max (JPY'bn)		104	48	70	88	58
Orders growth from min to max		562%	77%	83%	197%	73%
Orders increase from min to max - Upcycle 6 as % of ...		56%	121%	83%	66%	

The current upcycle commenced in mid-2024, but growth remained mild until a clear acceleration emerged in 2Q26. We believe total Japanese machine tool orders are likely 70–80% through this upcycle, with 20–30% further upside, given that the order increase from trough to peak during this cycle stands at ¥58bn—still below the ¥70–88bn levels observed in the previous two upcycles. Admittedly, the order expansion of ¥88bn during the 2020 upcycle was exceptionally strong, as COVID-19 lockdowns had initially depressed baseline orders.

By end-market, industrial machinery, electrical, and automotive have been the main contributors to order increase during the current upcycle. The electrical end-market has shown the most rapid expansion. Aerospace & shipping also exhibited strong growth, albeit from a lower base.

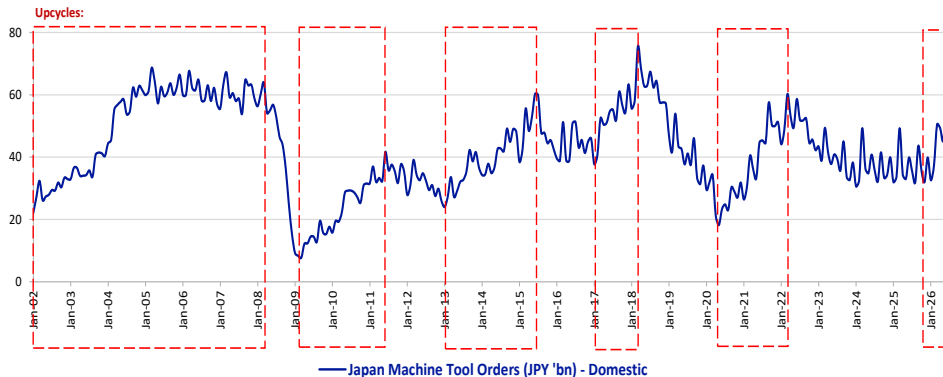


*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

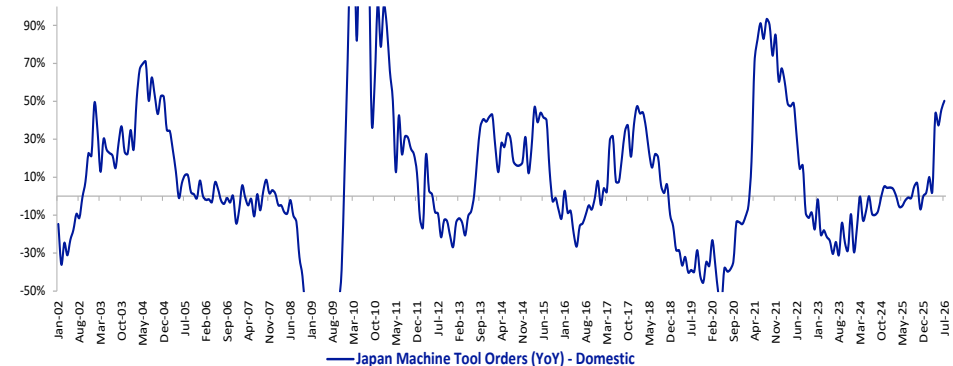
Japan machine tool orders: Japan (domestic) orders – More than 30% further upside left fueled by AI demand



Japan machine tool orders: Japan orders – order value

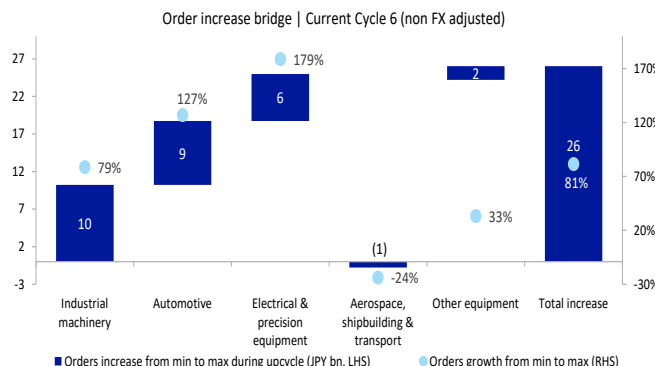
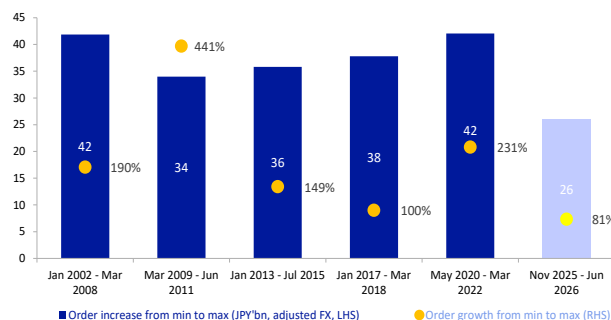


Japan machine tool orders: Japan orders – YoY growth



Japan machine tool orders: Japan orders – pattern from past upcycles

Domestic (Japan)	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle	Jan 2002 - Mar 2008	Mar 2009 - Jun 2011	Jan 2013 - Jul 2015	Jan 2017 - Mar 2018	May 2020 - Mar 2022	Nov 2025 - Jun 2026
Upcycle duration (months)	75	28	31	15	23	8
Reported orders (JPY'bn)						
Start month	Jan-2002	Mar-2009	Jan-2013	Jan-2017	May-2020	Nov-2025
Min - Orders (JPY'bn)	22	8	24	38	18	32
End month	Mar-2008	Jun-2011	Jul-2015	Mar-2018	Mar-2022	Jun-2026
Max - Orders (JPY'bn)	64	42	60	76	60	58
Orders increase from min to max (JPY'bn)	42	34	36	38	42	26
Orders growth from min to max	190%	441%	149%	100%	231%	81%
Orders increase from min to max - Upcycle 6 as % of ...		77%	73%	69%	62%	



Japan was late in entering the upcycle in this round, only starting to deliver higher orders from late 2025 and meaningful growth from 2Q26 onwards.

Consequently, we believe the Japanese market is only 60–70% through this upcycle, leaving more than 30% further upside.

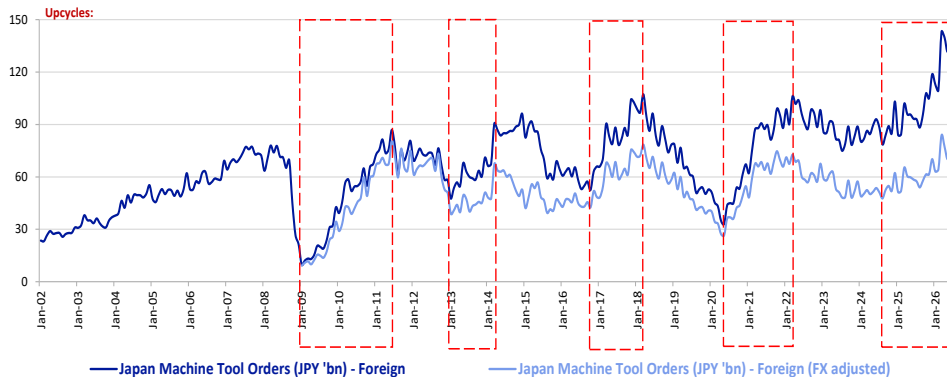
By end-market, the electrical sector has shown the strongest growth during the current upcycle, whilst automotive has also achieved a robust recovery.

We remain constructive on Japan, as we believe AI-driven demand and overseas orders will continue to support growth in the region.

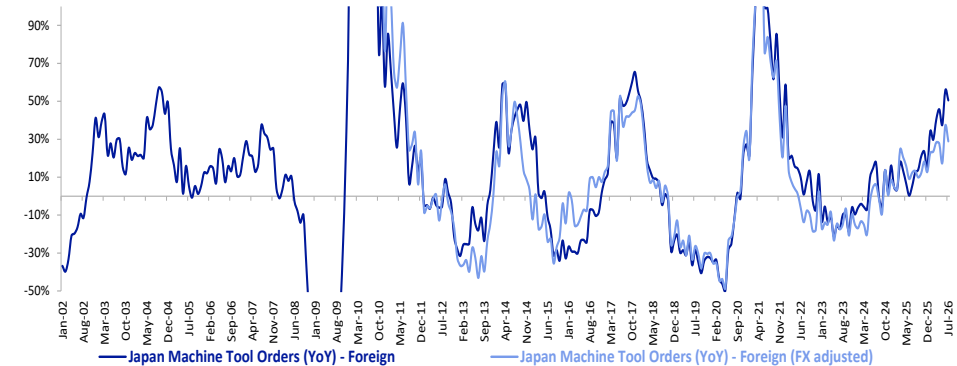
Japan machine tool orders: Foreign (non-Japan) orders – Close to peaking due to high orders from NA and China



Japan machine tool orders: Foreign orders – order value

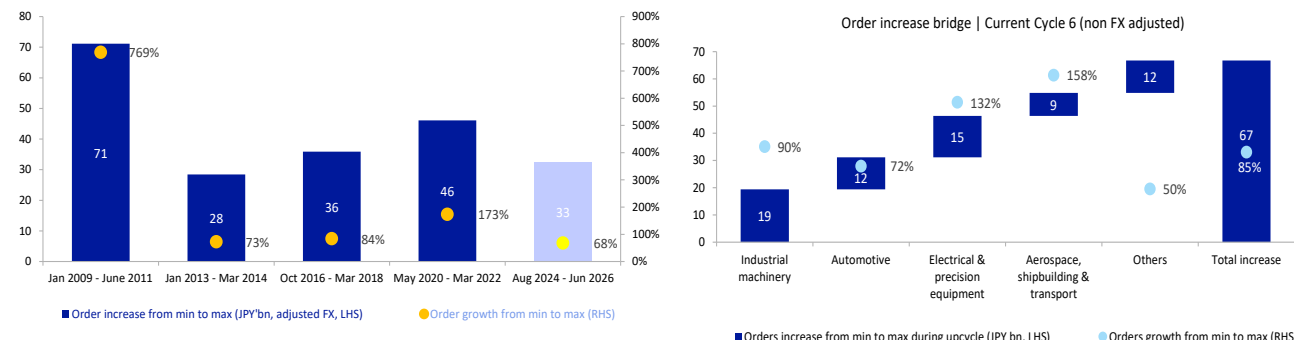


Japan machine tool orders: Foreign orders – YoY growth



Japan machine tool orders: Foreign orders – pattern from past upcycles*

Foreign (Non-Japan)	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle	Jan 2009 - June 2011	Jan 2013 - Mar 2014	Oct 2016 - Mar 2018	May 2020 - Mar 2022	Aug 2024 - Jun 2026	
Upcycle duration (months)	30	15	18	23	23	
Reported orders (JPY'bn)						
Start month	Jan-2009	Jan-2013	Oct-2016	May-2020	Aug-2024	
Min - Orders (JPY'bn)	10	48	53	33	79	
End month	Jun-2011	Mar-2014	Mar-2018	Mar-2022	Jun-2026	
Max - Orders (JPY'bn)	87	90	107	106	145	
Orders increase from min to max (JPY'bn)	77	43	55	73	67	
Orders growth from min to max	782%	90%	104%	221%	85%	
Adjusted FX orders (JPY'bn)						
Start month	Jan-2009	Jan-2013	Oct-2016	May-2020	Aug-2024	
Min - Orders (JPY'bn, adjusted FX)	9	39	43	27	48	
End month	Jun-2011	Mar-2014	Mar-2018	Mar-2022	Jun-2026	
Max - Orders (JPY'bn, adjusted FX)	80	67	79	73	80	
Orders increase from min to max (JPY'bn)	71	28	36	46	33	
Orders growth from min to max	769%	73%	84%	173%	68%	
Orders increase from min to max - Upcycle 6 as % of ...	46%	114%	91%	71%		



Foreign (non-Japan) orders entered an upcycle in mid-2024, albeit with only mild growth initially as recovering orders from China were offset by continued contractions in the US and Europe. Subsequently, growth accelerated in April 2025 (as US demand surged due to Liberation Day tariffs), gaining further momentum in 2H25 (when European demand turned positive) and late 4Q25 (driven by strong expansion in Other Asia).

Historical patterns suggest that foreign orders are already ~90% through this upcycle. That said, we believe that robust electrical-related demand could potentially extend the duration of the current upcycle.

By end-market, both the electrical and the aerospace & shipping sectors have delivered over 100% growth during the current upcycle.

*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

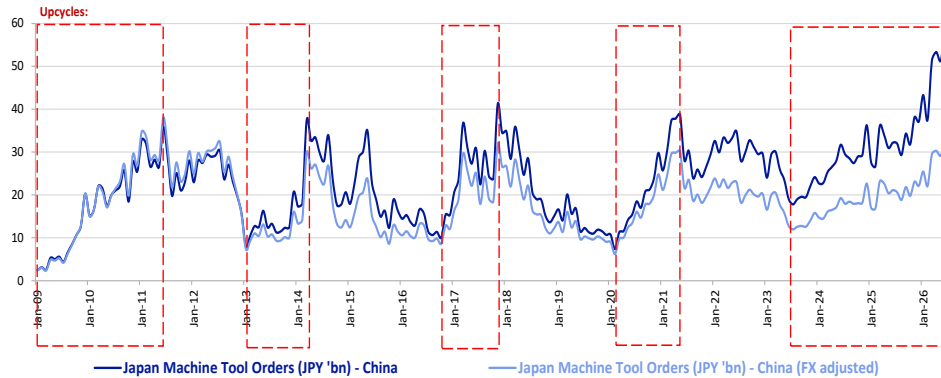
Deutsche Bank. Research Iris Zheng | (+852) 2203 5884 | iris.zheng@db.com

Source: JMTBA, Bloomberg Finance LP, Deutsche Bank 12

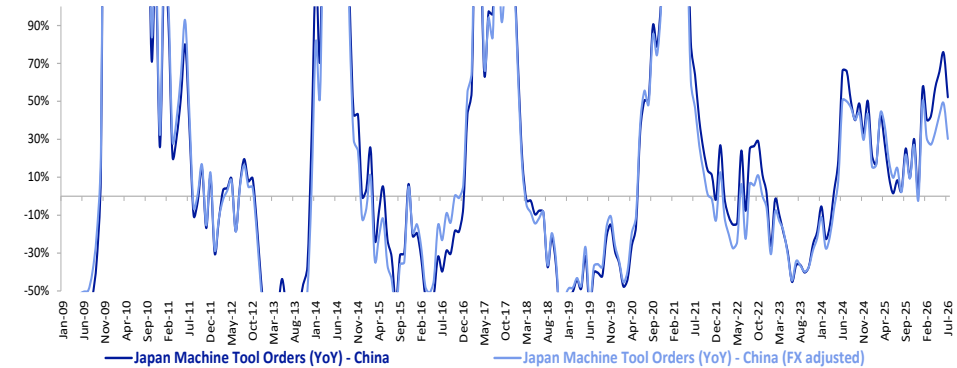
Japan machine tool orders: China orders – Already 80-90% through the upcycle



Japan machine tool orders: China orders – order value



Japan machine tool orders: China orders – YoY growth



Japan machine tool orders: China orders – pattern from past upcycles*

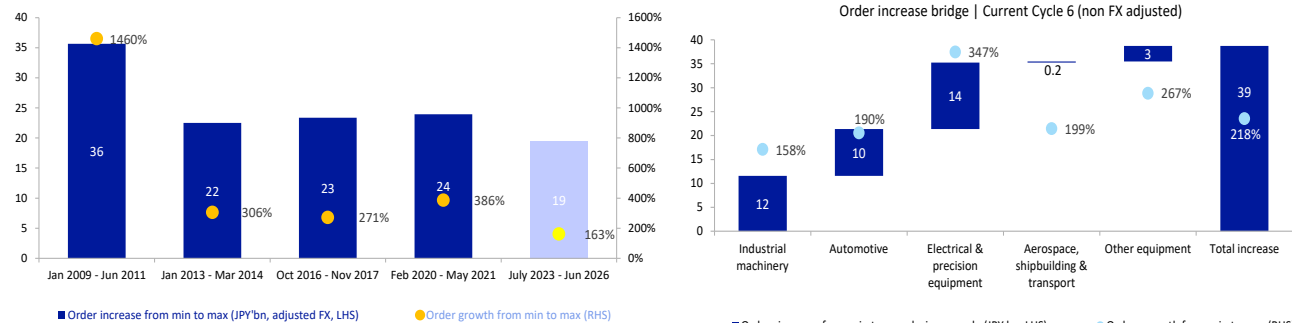
China	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle	Jan 2009 - Jun 2011	Jan 2013 - Mar 2014	Oct 2016 - Nov 2017	Feb 2020 - May 2021	July 2023 - Jun 2026	
Upcycle duration (months)	30	15	14	16	36	
Reported orders (JPY'bn)						
Start month	Jan-2009	Jan-2013	Oct-2016	Feb-2020	Jul-2023	
Min - Orders (JPY'bn)	2	8	10	7	18	
End month	Jun-2011	Mar-2014	Nov-2017	May-2021	Jun-2026	
Max - Orders (JPY'bn)	36	37	41	39	56	
Orders increase from min to max (JPY'bn)	33	29	31	31	39	
Orders growth from min to max	1368%	371%	310%	425%	218%	
Adjusted FX orders (JPY'bn)						
Start month	Jan-2009	Jan-2013	Oct-2016	Feb-2020	Jul-2023	
Min - Orders (JPY'bn, adjusted FX)	2	7	9	6	12	
End month	Jun-2011	Mar-2014	Nov-2017	May-2021	Jun-2026	
Max - Orders (JPY'bn, adjusted FX)	38	30	32	30	31	
Orders increase from min to max (JPY'bn)	36	22	23	24	19	
Orders growth from min to max	1460%	306%	271%	386%	163%	
Orders increase from min to max - Upcycle 6 as % of ...	55%	87%	83%	81%		

China orders hit a trough in mid-2023, with growth turning positive from 2H24 and establishing China as one of the first regions to enter an upcycle during this round.

Given the substantial magnitude of this recovery, we believe the Chinese upcycle is already 80–90% complete.

By end-market, the electrical sector has shown the strongest growth, whilst industrial machinery and automotive have also delivered more than 100% expansion from their respective troughs.

Refer to our China automation market monthly channel checks for more details – [link to report](#).



*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

Deutsche Bank. Research

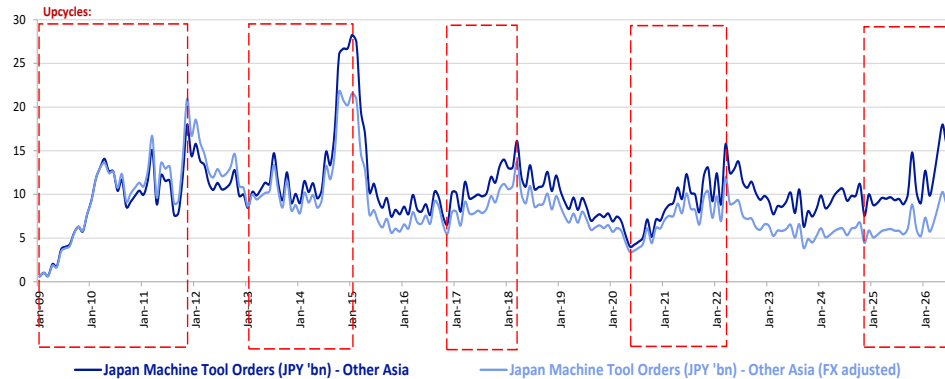
Iris Zheng | (+852) 2203 5884 | iris.zheng@db.com

Source: JMTBA, Bloomberg Finance LP, Deutsche Bank 13

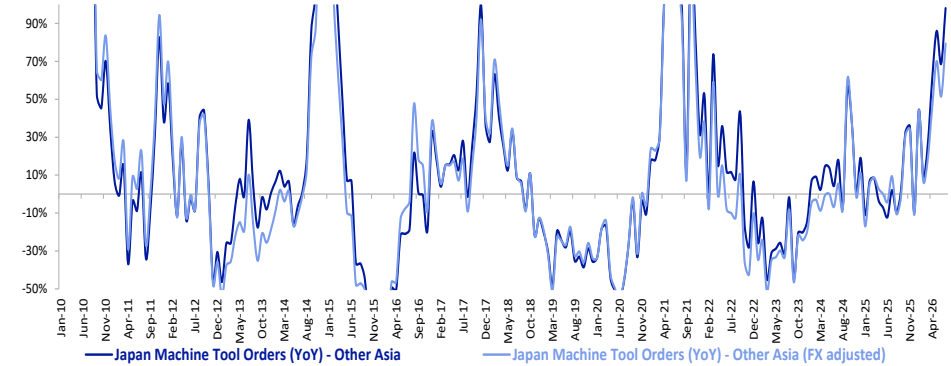
Japan machine tool orders: Other Asia orders – Strong momentum to continue lifted by AI demand



Japan machine tool orders: Other Aisa orders – order value

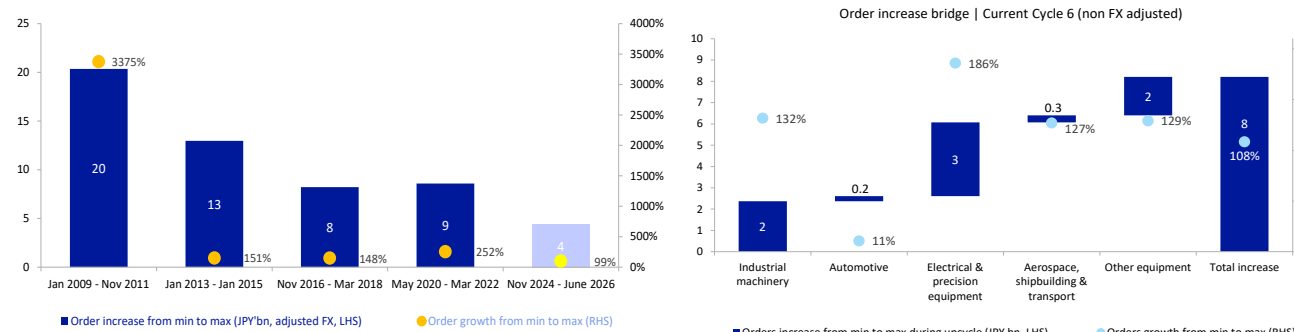


Japan machine tool orders: Other Aisa orders – YoY growth



Japan machine tool orders: Other Aisa orders – pattern from past upcycles*

Other Asia	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle		Jan 2009 - Nov 2011	Jan 2013 - Jan 2015	Nov 2016 - Mar 2018	May 2020 - Mar 2022	Nov 2024 - June 2026
Upcycle duration (months)		35	25	17	23	20
Reported orders (JPY'bn)						
Start month		Jan-2009	Jan-2013	Nov-2016	May-2020	Nov-2024
Min - Orders (JPY'bn)		1	8	7	4	8
End month		Nov-2011	Jan-2015	Mar-2018	Mar-2022	Jun-2026
Max - Orders (JPY'bn)		18	28	16	16	16
Orders increase from min to max (JPY'bn)		17	20	9	12	8
Orders growth from min to max		2882%	233%	143%	289%	108%
Adjusted FX orders (JPY'bn)						
Start month		Jan-2009	Jan-2013	Nov-2016	May-2020	Nov-2024
Min - Orders (JPY'bn, adjusted FX)		1	9	6	3	4
End month		Nov-2011	Jan-2015	Mar-2018	Mar-2022	Jun-2026
Max - Orders (JPY'bn, adjusted FX)		21	22	14	12	9
Orders increase from min to max (JPY'bn)		20	13	8	9	4
Orders growth from min to max		3375%	151%	148%	252%	99%
Orders increase from min to max - Upcycle 6 as % of ...		22%	34%	54%	51%	



*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

Deutsche Bank. Research Iris Zheng | (+852) 2203 5884 | iris.zheng@db.com

Whilst orders in the Other Asia region bottomed out in late 2024, meaningful growth acceleration did not emerge until 4Q25, when semiconductor demand surged in key markets such as Taiwan and South Korea.

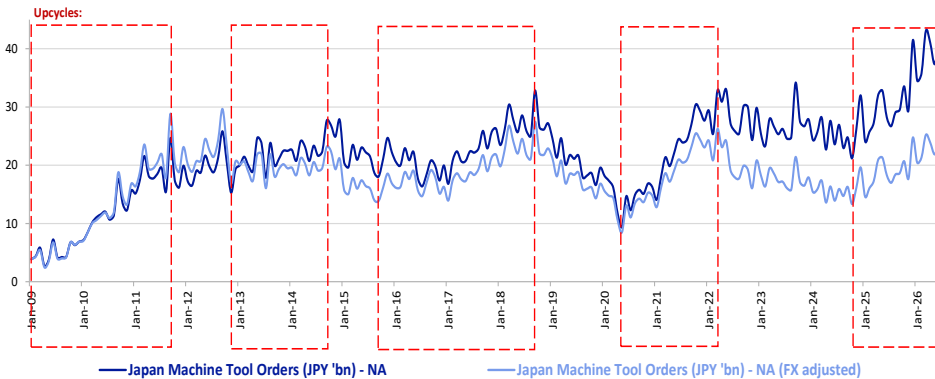
Based on historical patterns, we believe Other Asia is only 40–50% through its current upcycle, and this strong momentum is highly likely to continue, supported by robust AI demand.

By end-market, the electrical sector has grown the fastest, complemented by robust expansion in industrial machinery during the current upcycle.

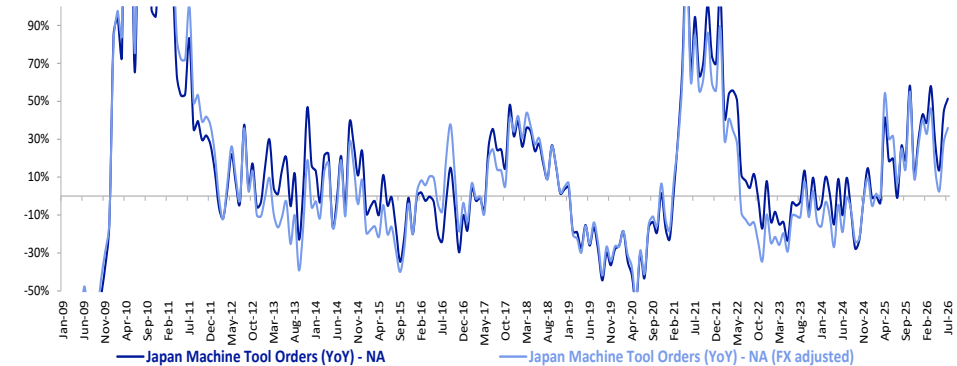
Japan machine tool orders: NA orders – Reshoring driving demand



Japan machine tool orders: NA orders – order value

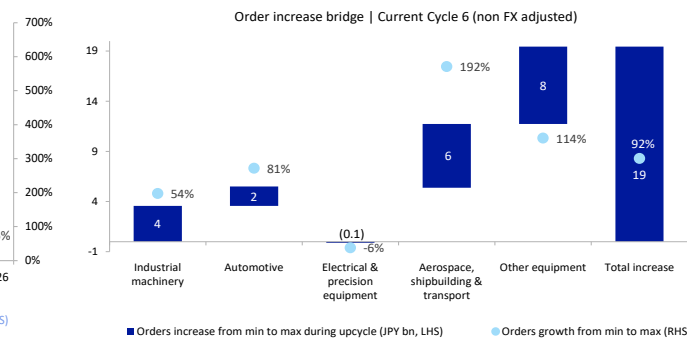
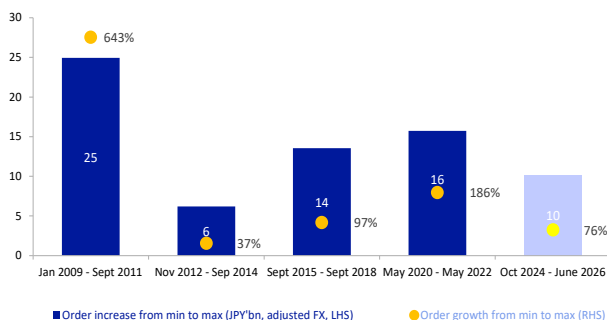


Japan machine tool orders: NA orders – YoY growth



Japan machine tool orders: NA orders – pattern from past upcycles*

NA	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle	Jan 2009 - Sept 2011	Nov 2012 - Sep 2014	Sept 2015 - Sept 2018	May 2020 - May 2022	Oct 2024 - June 2026	
Upcycle duration (months)	33	23	37	25	21	
Reported orders (JPY'bn)						
Start month	Jan-2009	Nov-2012	Sept-2015	May-2020	Oct-2024	
Min - Orders (JPY'bn)	4	15	18	9	21	
End month	Sep-2011	Sep-2014	Sep-2018	May-2022	Jun-2026	
Max - Orders (JPY'bn)	25	28	33	33	41	
Orders increase from min to max (JPY'bn)	21	12	15	24	19	
Orders growth from min to max	535%	81%	81%	251%	92%	
Adjusted FX orders (JPY'bn)						
Start month	Jan-2009	Nov-2012	Sept-2015	May-2020	Oct-2024	
Min - Orders (JPY'bn, adjusted FX)	4	17	14	8	13	
End month	Sep-2011	Sep-2014	Sep-2018	May-2022	Jun-2026	
Max - Orders (JPY'bn, adjusted FX)	29	23	27	24	23	
Orders increase from min to max (JPY'bn)	25	6	14	16	10	
Orders growth from min to max	643%	37%	97%	186%	76%	
Orders increase from min to max - Upcycle 6 as % of ...	40%	163%	74%	64%		



Reshoring, nearshoring, and the introduction of Liberation Day tariffs have significantly boosted demand from North America from April 2025 onwards within the current upcycle.

Historical patterns suggest the region is already 60–70% through the current upcycle. However, we note that ongoing reshoring and favourable domestic tax policies could lead to a structurally stronger and more prolonged upcycle for North America.

By end-market, the aerospace & shipping sector has shown the strongest growth during the current upcycle. Whilst electrical demand was robust in late 2025 and 1Q26, it subsided somewhat in 2Q26.

*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

Deutsche Bank. Research

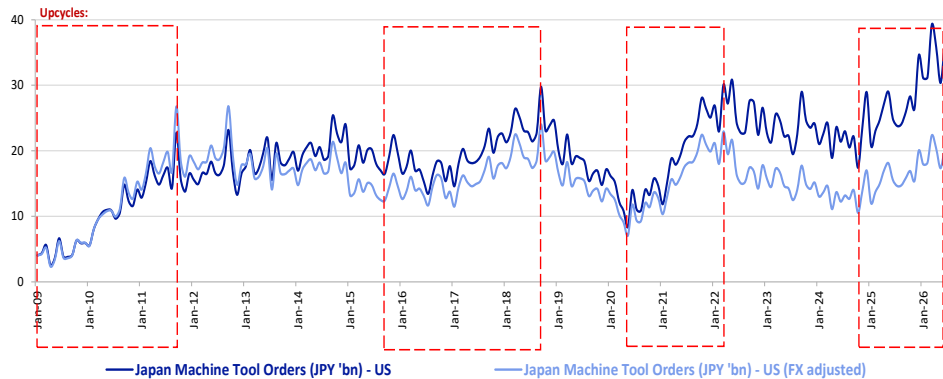
Iris Zheng | (+852) 2203 5884 | iris.zheng@db.com

Source: JMTBA, Bloomberg Finance LP, Deutsche Bank 15

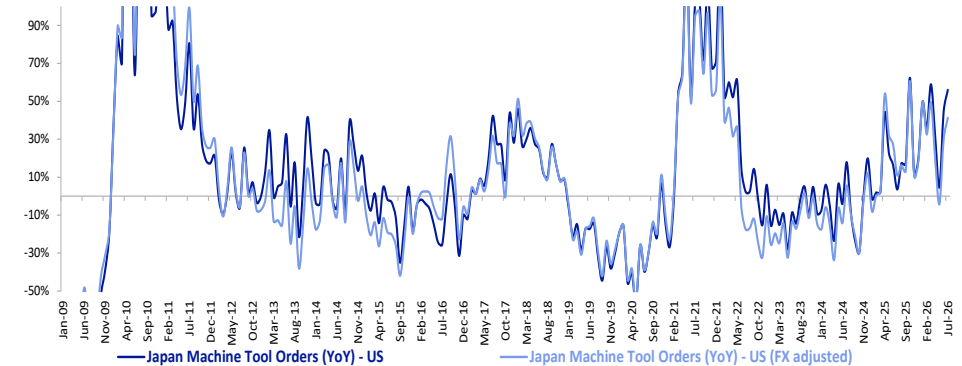
Japan machine tool orders: US orders – Potentially further upside on reshoring demand



Japan machine tool orders: US orders – order value

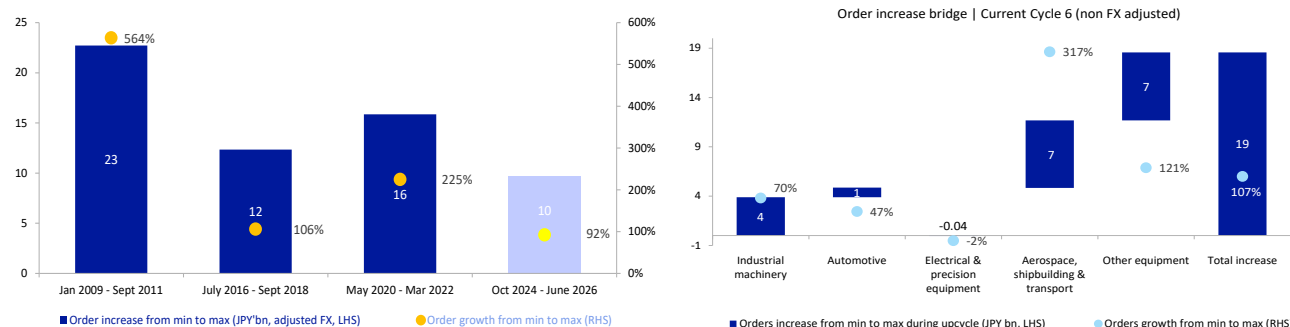


Japan machine tool orders: US orders – YoY growth



Japan machine tool orders: US orders – pattern from past upcycles*

US	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle	Jan 2009 - Sept 2011		July 2016 - Sept 2018		May 2020 - Mar 2022	Oct 2024 - June 2026
Upcycle duration (months)	33		27		23	21
Reported orders (JPY'bn)						
Start month	Jan-2009		Jul-2016		May-2020	Oct-2024
Min - Orders (JPY'bn)	4		13		8	17
End month	Sep-2011		Sep-2018		Mar-2022	Jun-2026
Max - Orders (JPY'bn)	23		30		30	36
Orders increase from min to max (JPY'bn)	19		16		22	19
Orders growth from min to max	464%		122%		259%	107%
Adjusted FX orders (JPY'bn)						
Start month	Jan-2009		Jul-2016		May-2020	Oct-2024
Min - Orders (JPY'bn, adjusted FX)	4		12		7	11
End month	Sep-2011		Sep-2018		Mar-2022	Jun-2026
Max - Orders (JPY'bn, adjusted FX)	27		24		23	20
Orders increase from min to max (JPY'bn)	23		12		16	10
Orders growth from min to max	564%		106%		225%	92%
Orders increase from min to max - Upcycle 6 as % of ...	43%		79%		61%	



Historically, the US witnessed limited growth during the 2012–14 period. However, orders registered clear and robust upcycles over 2016–18 and 2020–22, supported by the country's reshoring policies, in our view.

Historical patterns suggest the US is 60–70% through the current upcycle, although we believe there remains upside given the potential for structurally higher order volumes driven by political and trade considerations.

By end-market, the aerospace & shipping sector is the strongest contributor. Whilst electrical demand was strong in late 2025 and 1Q26, it subsided in 2Q26.

*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

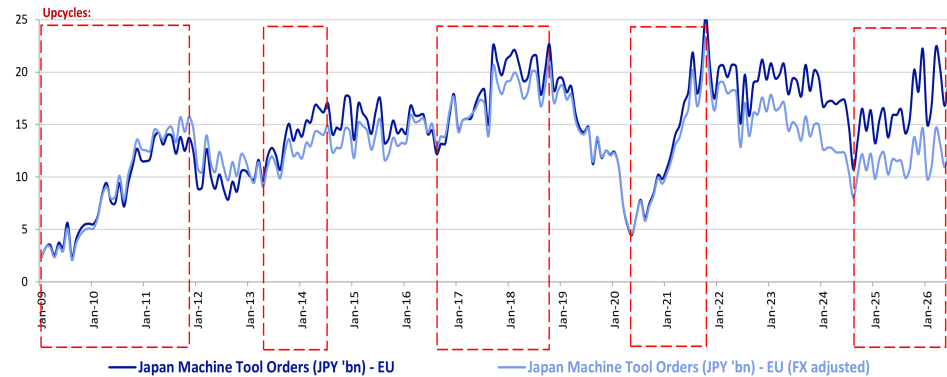
Deutsche Bank. Research Iris Zheng | (+852) 2203 5884 | iris.zheng@db.com

Source: JMTBA, Bloomberg Finance LP, Deutsche Bank 16

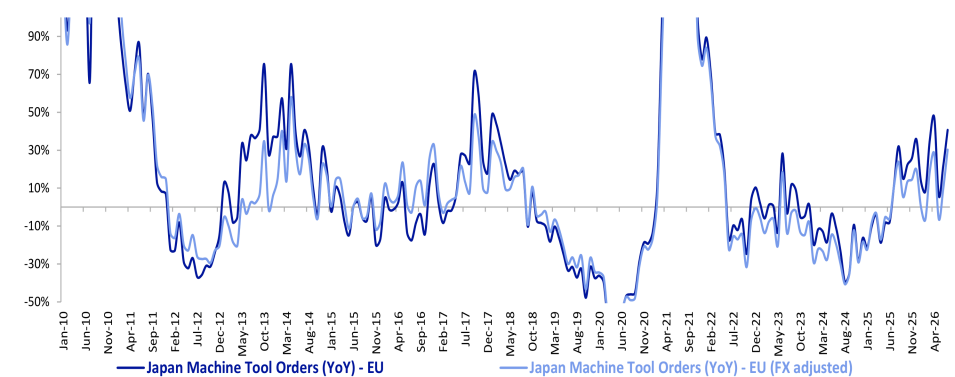
Japan machine tool orders: EU orders – Mild recovery so far, but structural challenges draw upside uncertainties



Japan machine tool orders: EU orders – order value



Japan machine tool orders: EU orders – YoY growth



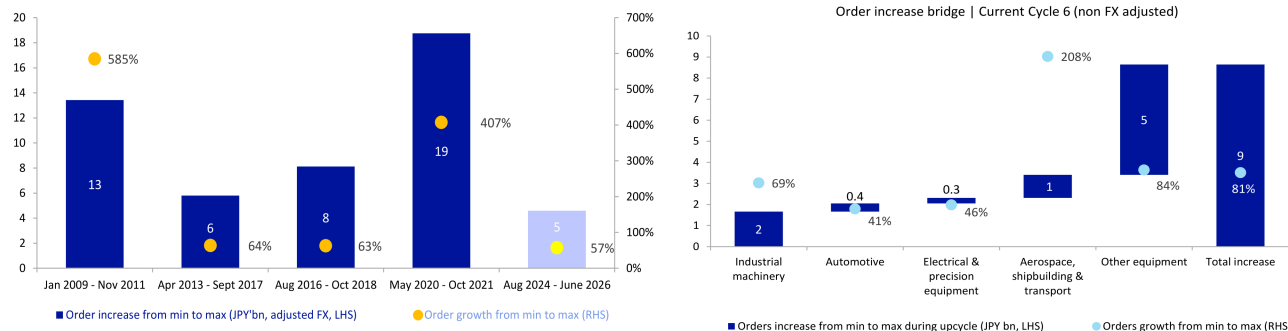
Japan machine tool orders: EU orders – pattern from past upcycles*

EU	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle		Jan 2009 - Nov 2011	Apr 2013 - Sept 2017	Aug 2016 - Oct 2018	May 2020 - Oct 2021	Aug 2024 - June 2026
Upcycle duration (months)		35	16	27	18	23
Reported orders (JPY'bn)						
Start month		Jan-2009	Apr-2013	Aug-2016	May-2020	Aug-2024
Min - Orders (JPY'bn)		2	10	12	4	11
End month		Nov-2011	Jul-2014	Oct-2018	Oct-2021	Jun-2026
Max - Orders (JPY'bn)		14	17	23	25	19
Orders increase from min to max (JPY'bn)		11	7	10	21	9
Orders growth from min to max		499%	77%	86%	469%	81%
Adjusted FX orders (JPY'bn)						
Start month		Jan-2009	Apr-2013	Aug-2016	May-2020	Aug-2024
Min - Orders (JPY'bn, adjusted FX)		2	9	13	5	8
End month		Nov-2011	Jul-2014	Oct-2018	Oct-2021	Jun-2026
Max - Orders (JPY'bn, adjusted FX)		16	15	21	23	13
Orders increase from min to max (JPY'bn)		13	6	8	19	5
Orders growth from min to max		585%	64%	63%	407%	57%
Orders increase from min to max - Upcycle 6 as % of ...		34%	79%	57%	24%	

Orders from Europe troughed in mid-2024, but YoY growth did not turn positive until 2H25.

The recovery during this upcycle has been modest compared to historical cycles, implying there is further room for expansion. That said, the structural challenges facing Europe's manufacturing sector introduce uncertainties regarding the potential upside, in our view.

By end-market, the aerospace & shipping sector has delivered the highest growth during this upcycle, albeit off a very low base.

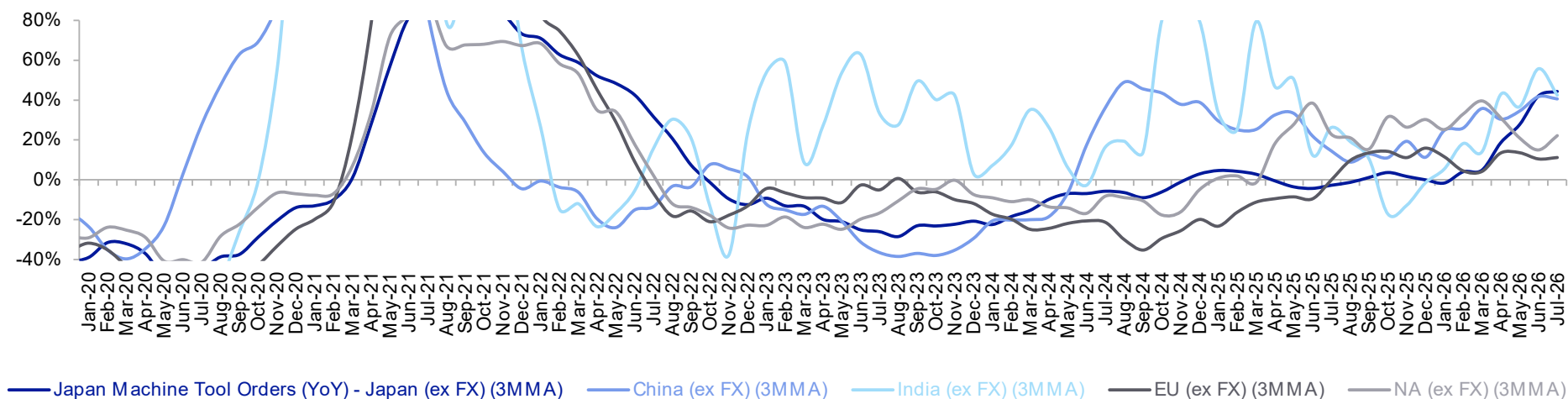


*Note: (1) we use constant currency to adjust for FX impacts; (2) detailed regional and end-market data for August 2026 not yet available so we use the 2nd highest monthly data of June 2026 for the current Upcycle 6

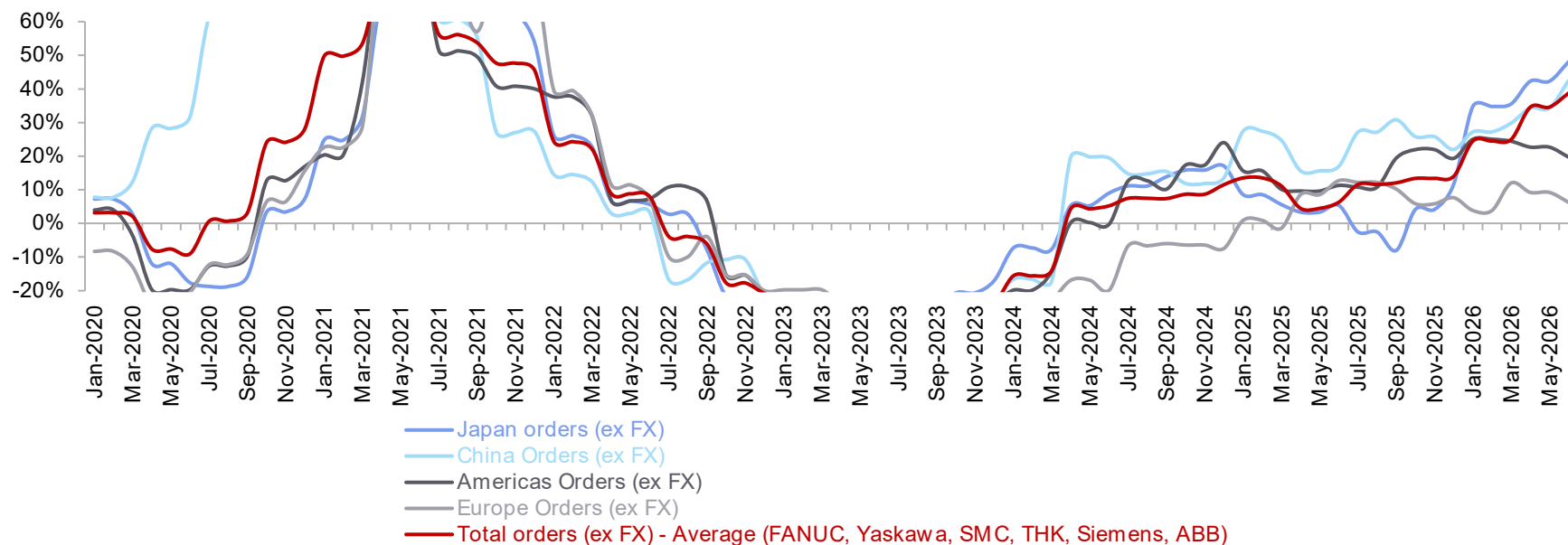
Automation market growth by region



Japan machine tool orders (data as of July 2026) : Growth accelerated across Japan, Asia and Europe in 2Q26, but decelerated in North America on a higher base



Automation orders from major global automation players by region (quarterly data as of CY2Q26)





Correlation of share price vs Japan machine tool orders

Share price performance: Slowing YoY growth of Japan machine tool orders has led to share price correction



Our correlation analysis shows that the share prices of Japanese factory automation (FA) players are highly correlated with both the absolute order value and the YoY growth momentum of Japanese machine tool orders. **Since 2015, this correlation with YoY growth has visibly strengthened.**

The share prices of **FANUC** and **THK** exhibit the highest correlation with Japanese machine tool orders, rendering these two names the most sensitive to fluctuations in the monthly data. **Harmonic Drive**, **Yaskawa**, and **SMC** also show a strong correlation with the YoY growth momentum of Japanese machine tool orders.

Driven by market concerns that the YoY growth of Japanese machine tool orders is peaking, alongside other macro and company-specific factors, the share prices of the five Japanese FA names under our coverage have already corrected significantly from their recent peaks.

We favour names that offer structural growth beyond the core machine tool order cycle; these include **SMC** and **Yaskawa** for their high exposure to the semiconductor sector, which is likely to enjoy a prolonged upcycle in this round, and **Harmonic Drive** for its long-term humanoid robotics opportunities.

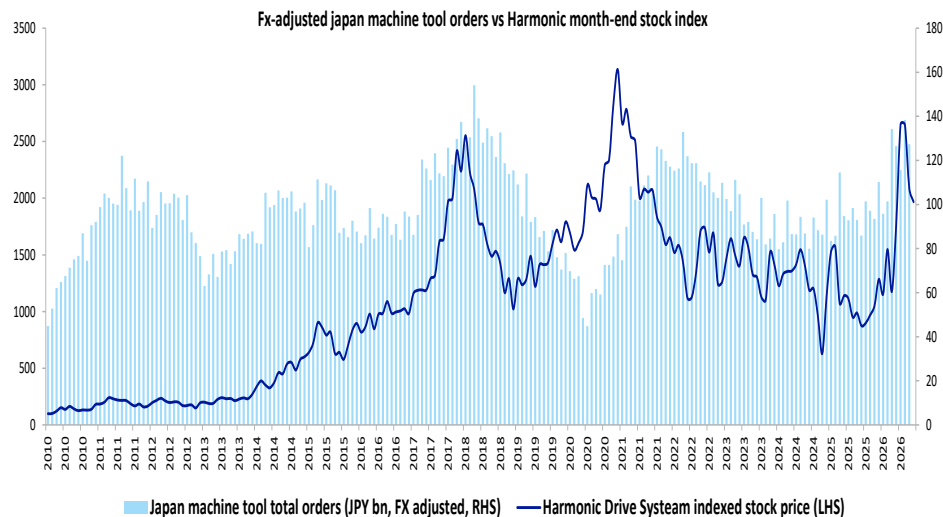
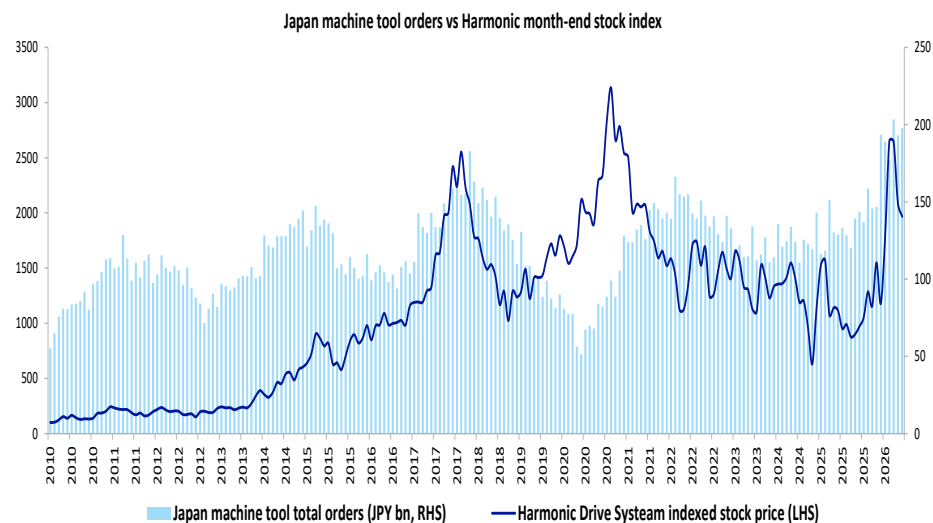
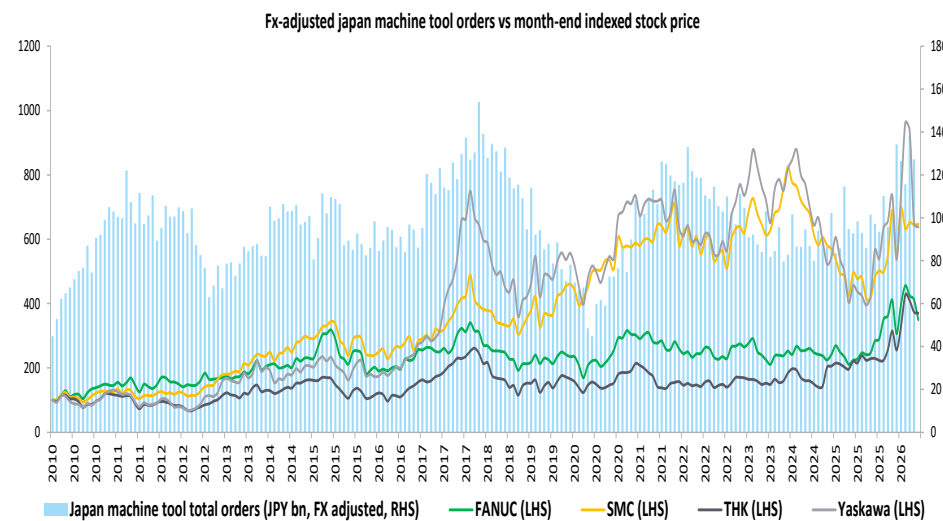
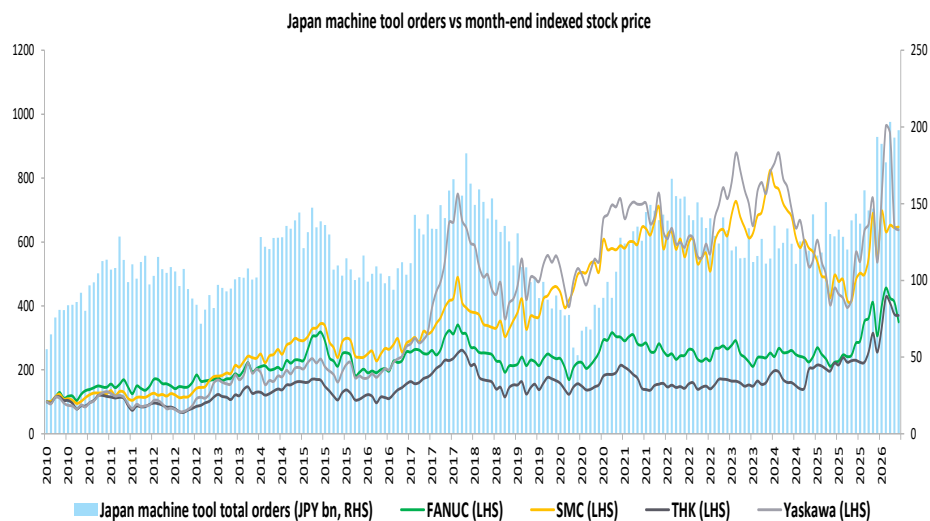
Share price performance: Highly correlates with order value and YoY growth of Japan machine tool orders

Japan machine tool order metric (since 2015)	Stock lead	FANUC	THK	Harmonic Drive	Yaskawa	SMC
Reported values in JPY						
Japan machine tool orders value vs stock price	Same month	0.61	0.61	0.06	0.31	0.22
Japan machine tool orders value vs stock price	1 month	0.67	0.65	0.10	0.33	0.25
Japan machine tool orders value vs stock price	3 months	0.66	0.64	0.11	0.32	0.25
Japan machine tool orders YoY vs stock price	Same month	0.58	0.45	0.38	0.32	0.27
Japan machine tool orders YoY vs stock price	1 month	0.61	0.48	0.44	0.33	0.28
Japan machine tool orders YoY vs stock price	3 months	0.59	0.48	0.50	0.31	0.27
Japan machine tool orders YoY vs 1M stock return	Same month	-0.02	-0.03	-0.07	-0.01	-0.02
Japan machine tool orders YoY vs 1M stock return	1 month	0.04	0.03	-0.04	0.04	0.02
Japan machine tool orders YoY vs 1M stock return	3 months	0.14	0.12	0.05	0.17	0.06
Japan machine tool orders 3MMA YoY vs 3M stock return	Same month	0.02	0.00	-0.06	0.06	0.00
Japan machine tool orders 3MMA YoY vs 3M stock return	3 months	0.20	0.21	0.03	0.25	0.12
Japan machine tool orders 3MMA YoY vs 3M stock return	6 months	0.34	0.29	0.10	0.34	0.17
FX adjusted values						
FX-adjusted Japan machine tool orders value vs stock price	Same month	0.45	0.38	0.11	0.17	0.00
FX-adjusted Japan machine tool orders value vs stock price	1 month	0.53	0.44	0.16	0.18	0.02
FX-adjusted Japan machine tool orders value vs stock price	3 months	0.56	0.46	0.19	0.15	0.01
FX-adjusted Japan machine tool orders YoY vs stock price	Same month	0.51	0.38	0.35	0.23	0.17
FX-adjusted Japan machine tool orders YoY vs stock price	1 month	0.54	0.40	0.40	0.24	0.18
FX-adjusted Japan machine tool orders YoY vs stock price	3 months	0.55	0.42	0.46	0.20	0.17
FX-adjusted Japan machine tool orders YoY vs 1M stock return	Same month	-0.03	-0.03	-0.09	-0.02	-0.03
FX-adjusted Japan machine tool orders YoY vs 1M stock return	1 month	0.02	0.02	-0.03	0.05	0.01
FX-adjusted Japan machine tool orders YoY vs 1M stock return	3 months	0.10	0.07	0.01	0.13	0.06
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	Same month	0.00	0.00	-0.08	0.06	-0.03
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	3 months	0.17	0.16	-0.01	0.22	0.09
FX-adjusted Japan machine tool orders 3MMA YoY vs 3M stock return	6 months	0.30	0.26	0.09	0.31	0.11

Share price performance: Broadly trending together with Japan machine tool orders



Share price of Japanese factory automation names broadly trend together with Japan machine tool orders, especially for FANUC and THK



Japan machine tool order YoY growth is likely peaking, but deceleration could prove mild



Given that the share price returns of Japanese FA companies are correlated with YoY growth in Japanese machine tool orders, we focus on YoY order growth to assess its implications for share prices. Historically, we find that **the share prices of Japanese FA companies tend to peak concurrently with, or a few months prior to, the peak in YoY order growth.**

For Japanese machine tool orders, FX-adjusted YoY growth has historically declined rapidly post-peak; however, we believe **this deceleration could prove milder during the current cycle**, supported by the ongoing AI demand cycle.

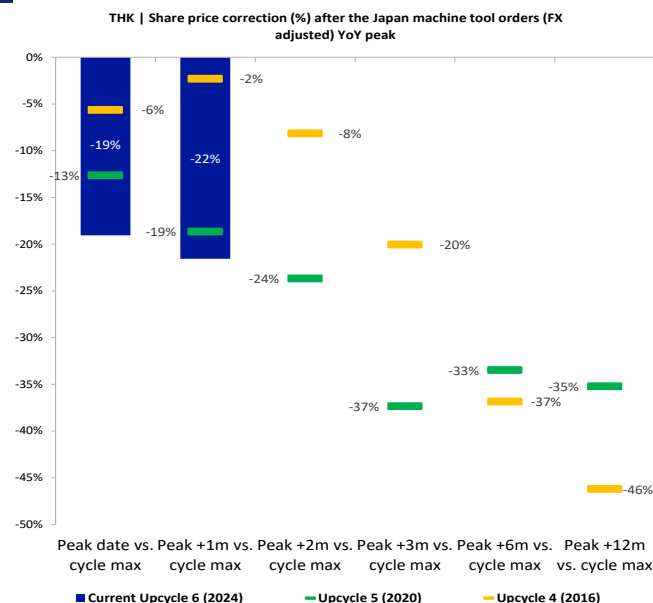
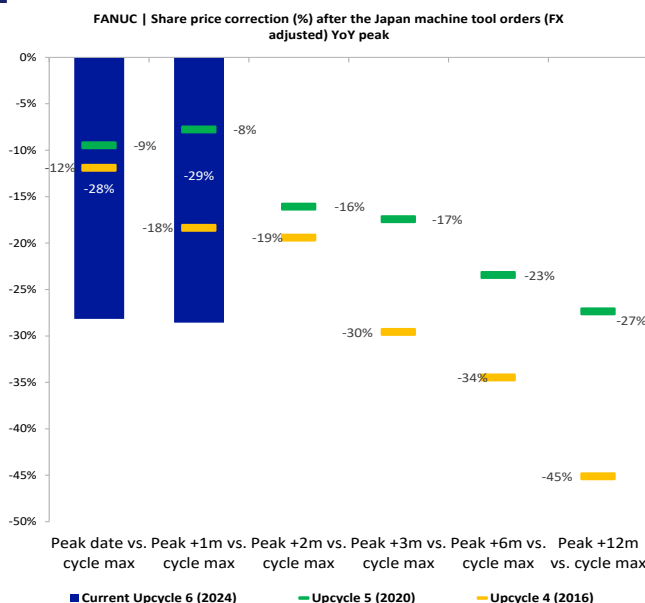
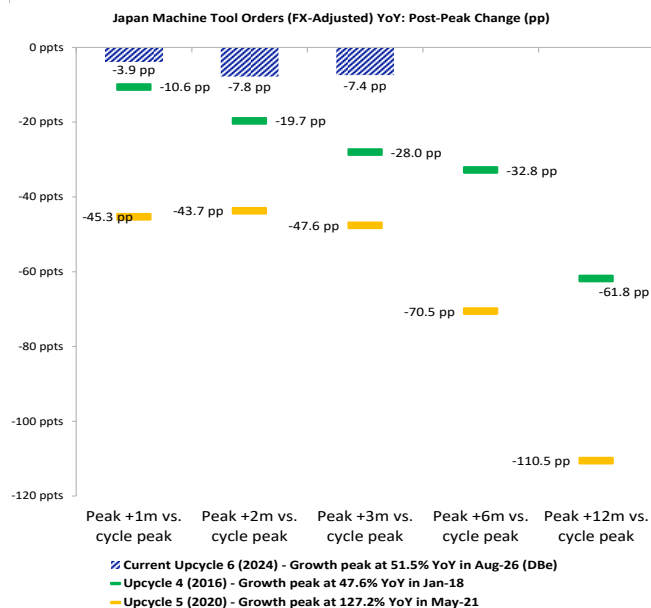
In light of our expectation of a mild deceleration in YoY order growth, we believe **the share price correction for Japanese FA companies appears overdone**, as share prices have corrected more severely than the declines observed one month post-peak during the previous two cycles.

	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBE)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBE)
Peak - orders YoY	47.6%	127.2%	51.5%
Peak +1 month	Feb-2018	Jun-2021	Sep-2026 (DBE)
Peak +1 month - orders YoY	37.0%	81.9%	47.6%
Peak +1 month · YoY change vs peak (pp)	-10.6 pp	-45.3 pp	-3.9 pp
Peak +2 months	Mar-2018	Jul-2021	Oct-2026 (DBE)
Peak +2 months - orders YoY	27.9%	83.5%	43.7%
Peak +2 months · YoY change vs peak (pp)	-19.7 pp	-43.7 pp	-7.8 pp
Peak +3 months	Apr-2018	Aug-2021	Nov-2026 (DBE)
Peak +3 months - orders YoY	19.6%	79.6%	44.1%
Peak +3 months · YoY change vs peak (pp)	-28.0 pp	-47.6 pp	-7.4 pp
Peak +6 months	Jul-2018	Nov-2021	
Peak +6 months - orders YoY	14.8%	56.7%	
Peak +6 months · YoY change vs peak (pp)	-32.8 pp	-70.5 pp	
Peak +12 months	Jan-2019	May-2022	
Peak +12 months - orders YoY	(14.2%)	16.7%	
Peak +12 months · YoY change vs peak (pp)	-61.8 pp	-110.5 pp	

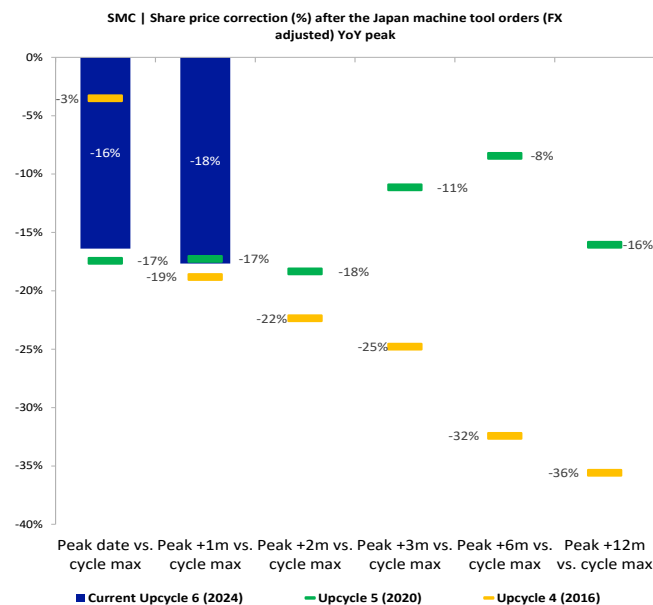
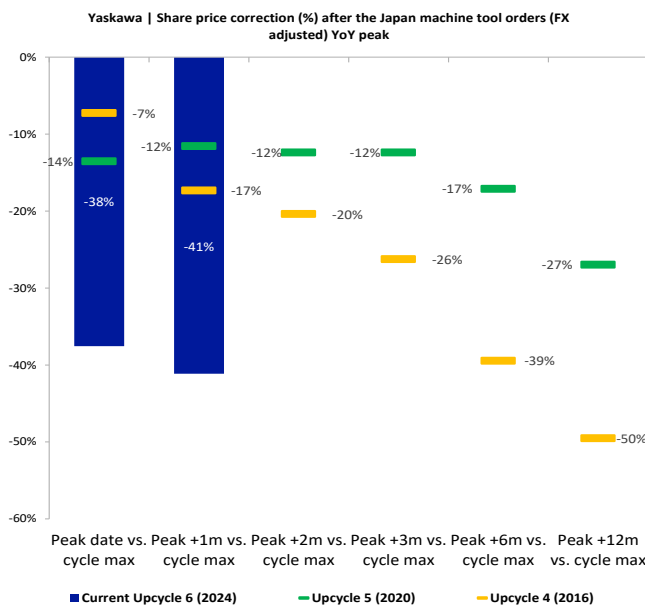
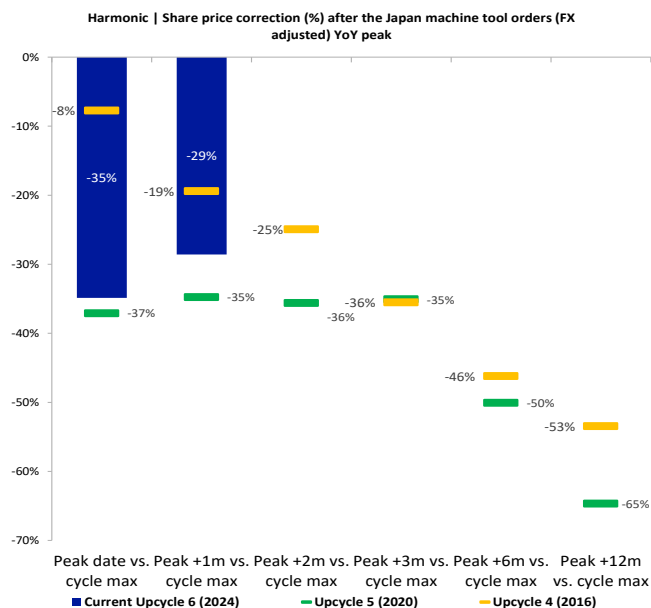
Share price performance: Correction appears overdone



Japan FA names share prices correction appears overdone against potentially mild order growth deceleration: Share price has corrected more severely than the declines observed one month post-peak during the past two cycles

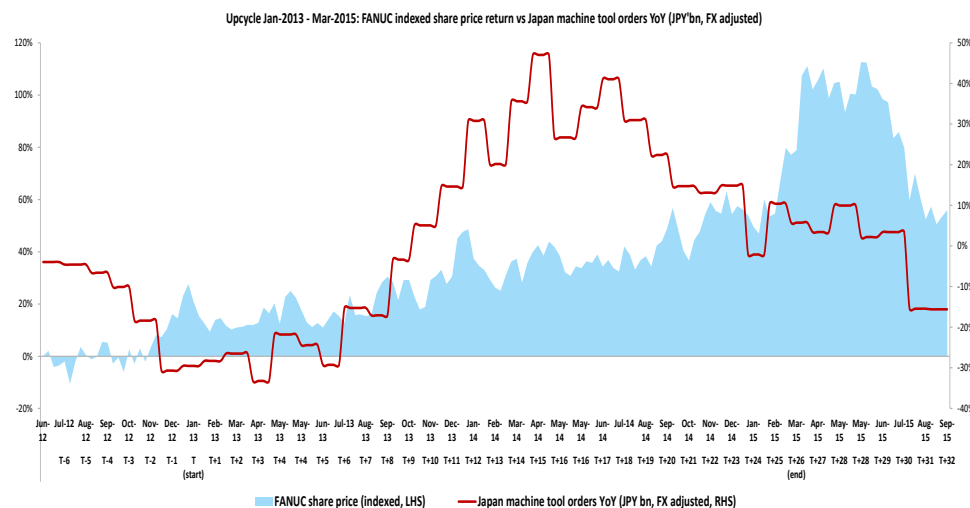
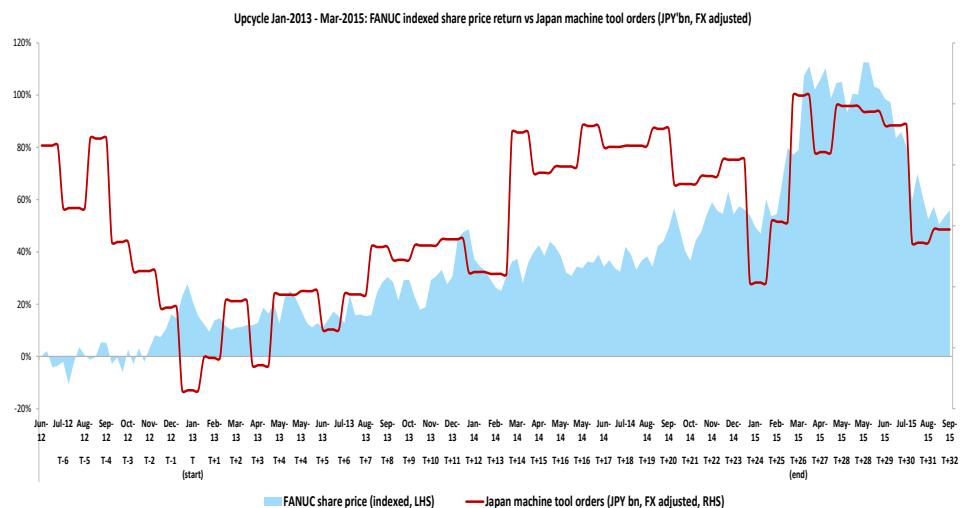


*Note: Assume Japan machine tool orders YoY growth has peaked in August 2026, we use DB estimates for orders YoY growth for September, October and November 2026 (namely 1-3M post the YoY peak) for the current Upcycle 6

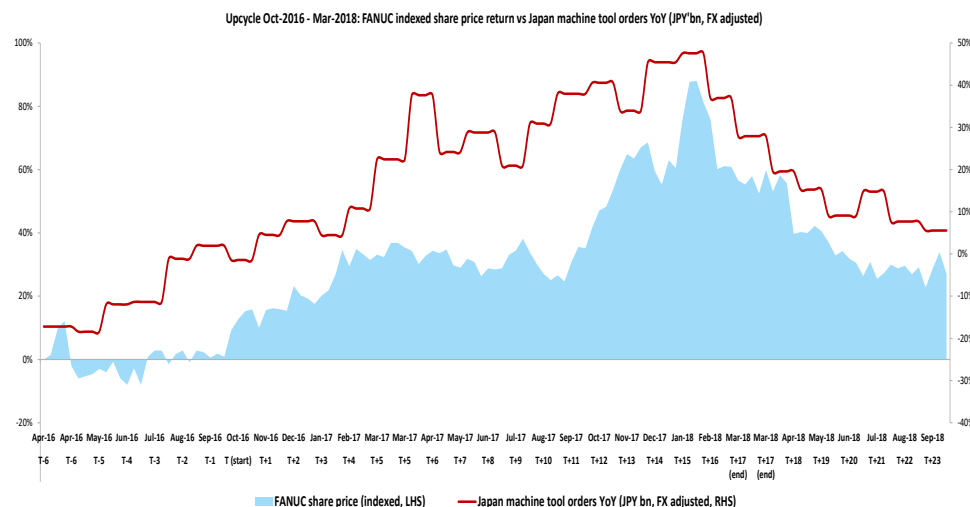
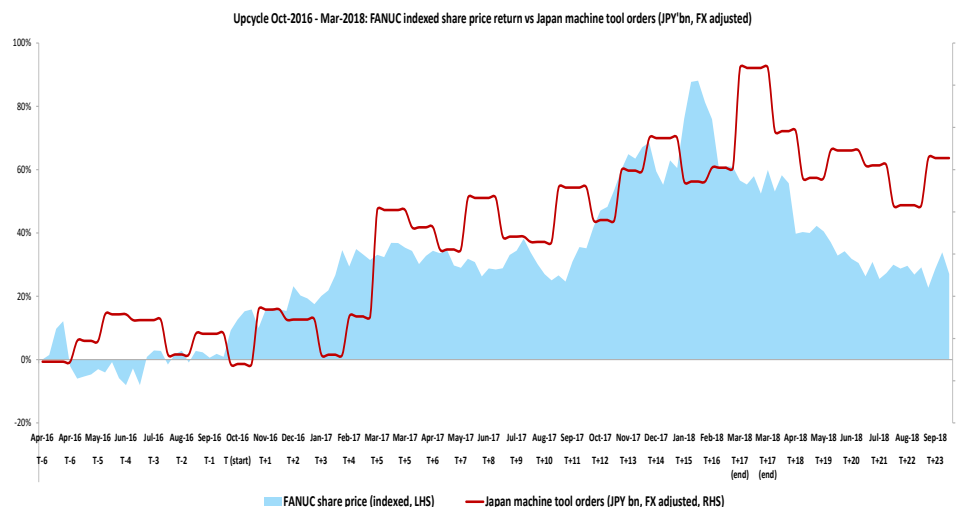




Upcycle 3 (Jan-2013 to Mar-2015): FANUC share price peaked at the time when orders peaked, despite slowing YoY order growth

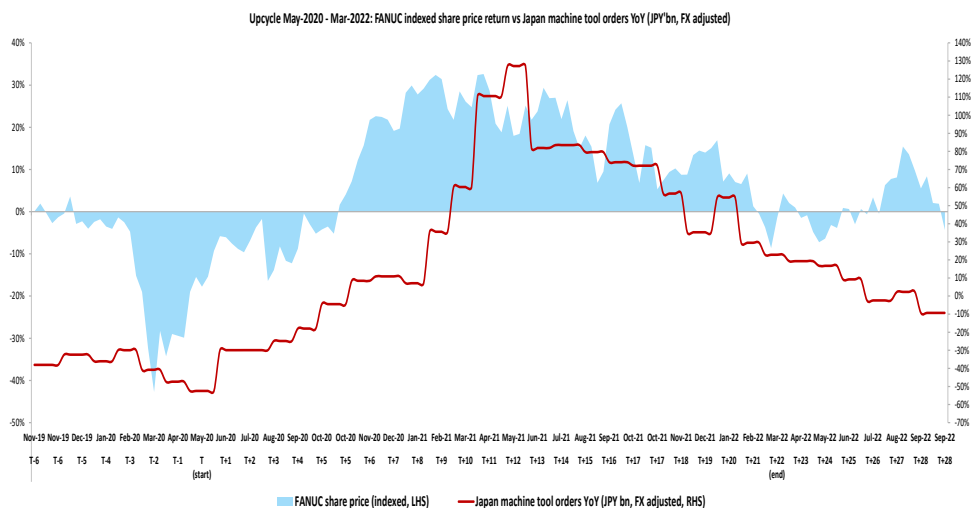
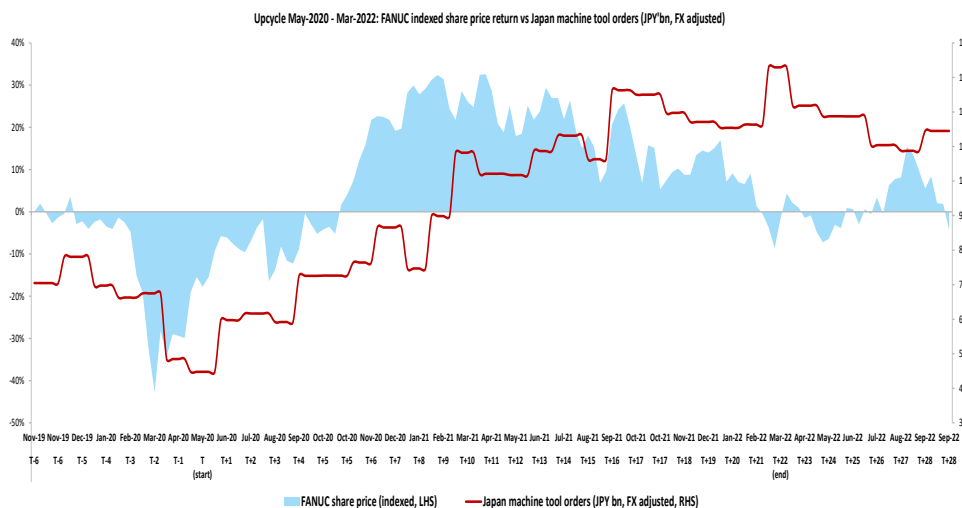


Upcycle 4 (Oct-2016 to Mar-2018): FANUC share price peaked when YoY growth peaked, and 2M prior to the orders peak

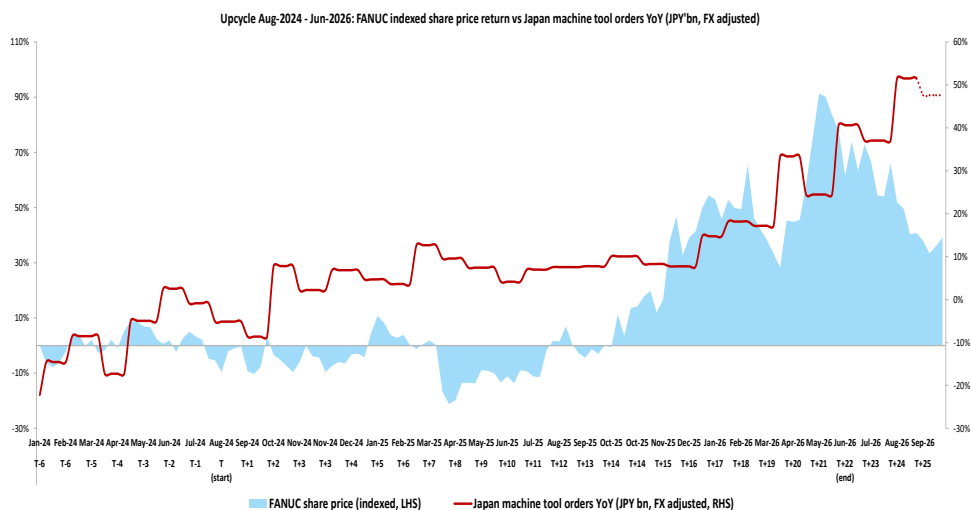
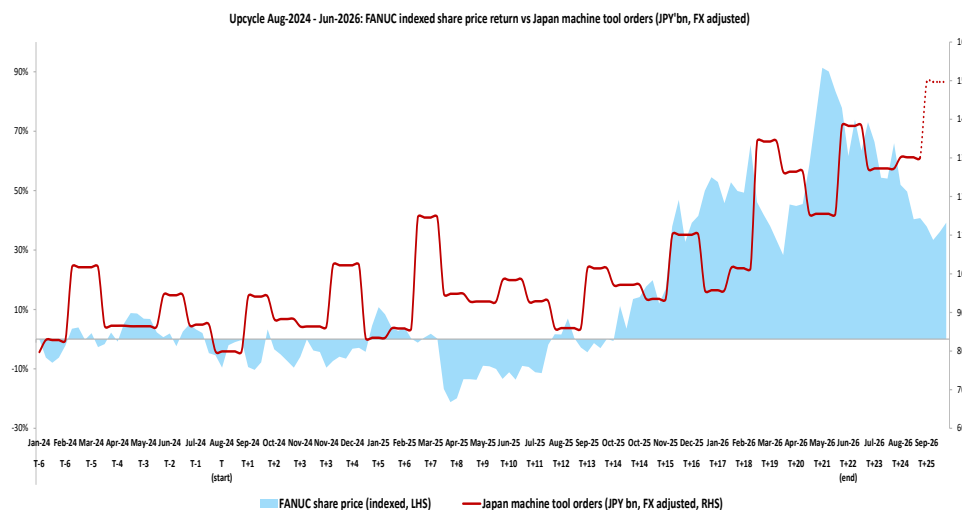




Upcycle 5 (May-2020 to Mar-2022): FANUC share price peaked when YoY peaked and 10M ahead of orders peaking



Upcycle 6 (Aug-2024 to Jun-2026): FANUC share price already retreating ahead of the slowdown of YoY growth



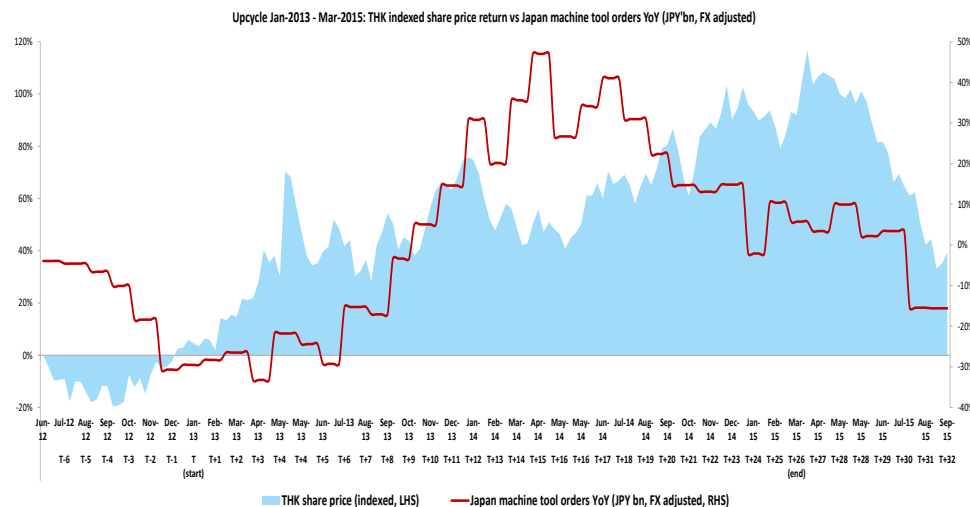
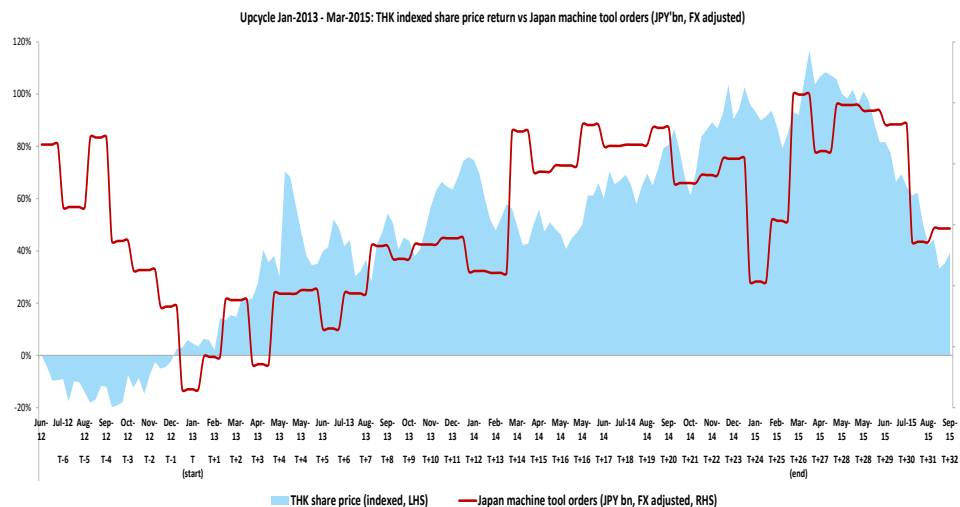


Under the assumption that Japan machine tool orders YoY growth has peaked in August 2026, FANUC's share price has corrected by close to -30% from its current cycle peak in September, more severe than the share price corrections observed in the past two upcycles

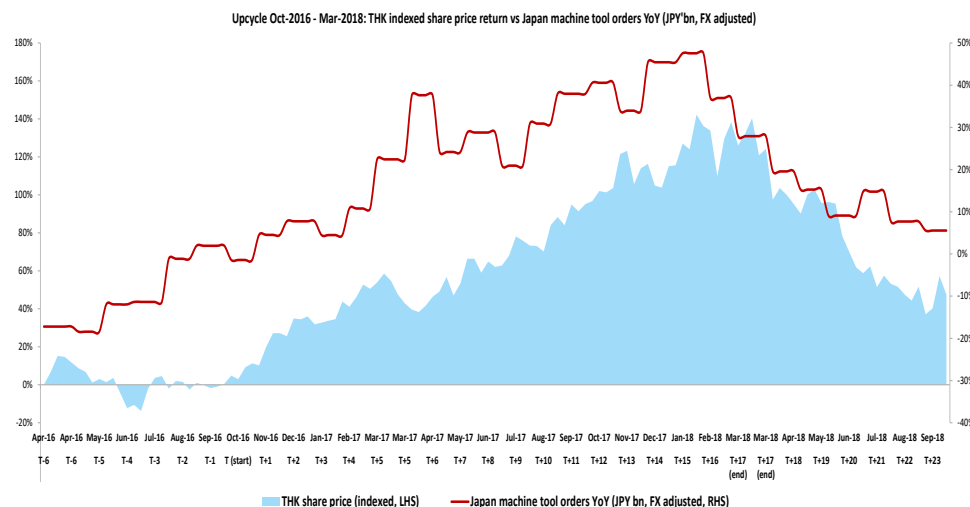
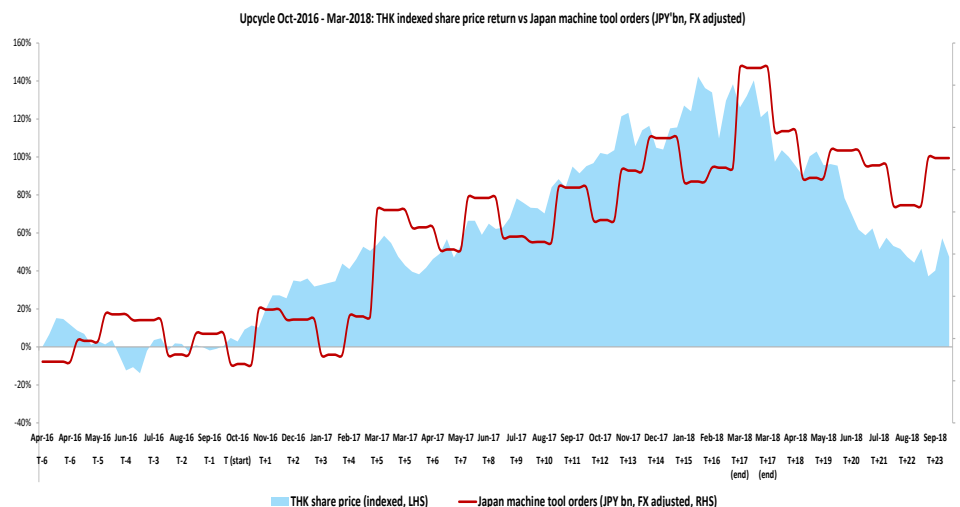
FANUC	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBE)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBE)
Peak - orders YoY	47.6%	127.2%	51.5%
FANUC share price performance			
Min - Share price (JPY) across the cycle	3,100	2,427	3,114
Date of minimum	24-Jun-2016 (T-4 month)	19-Mar-2020 (T-2 month)	09-Apr-2025 (T+8 month)
Max - Share price (JPY) across the cycle	6,690	5,810	8,428
Date of maximum	16-Jan-2018 (T+15 month)	18-Feb-2021 (T+9 month)	25-May-2026 (T+21 month)
Increase in share price from min to max (JPY)	3,590	3,383	5,314
Increase in share price from min to max (%)	116%	139%	171%
Japan machine tool orders (FX adjusted) YoY peak	31-Jan-2018 (T+15)	31-May-2021 (T+12)	31-Aug-2026 (T+24)
Share price (JPY) at the peak date	5,895	5,260	6,055
Peak-date share price vs. cycle Min share price (JPY)	2,795	2,833	2,941
Peak-date share price vs. cycle Min share price (%)	90%	117%	94%
Peak-date share price vs. cycle Max share price (JPY)	-795	-550	-2,373
Peak-date share price vs. cycle Max share price (%)	-12%	-9%	-28%
Peak +1 month	28-Feb-2018	30-Jun-2021	25-Sept-2026 (latest date)
Peak +1 month - share price (JPY)	5,462	5,359	5,985
Peak +1 month share price vs. cycle Min share price (JPY)	2,362	2,932	2,871
Peak +1 month share price vs. cycle Min share price (%)	76%	121%	92%
Peak +1 month share price vs. cycle Max share price (JPY)	-1,228	-451	-2,443
Peak +1 month share price vs. cycle Max share price (%)	-18%	-8%	-29%
Peak +2 months	30-Mar-2018	30-Jul-2021	
Peak +2 months - share price (JPY)	5,392	4,876	
Peak +2 months share price vs. cycle Min share price (JPY)	2,292	2,449	
Peak +2 months share price vs. cycle Min share price (%)	74%	101%	
Peak +2 months share price vs. cycle Max share price (JPY)	-1,298	-934	
Peak +2 months share price vs. cycle Max share price (%)	-19%	-16%	
Peak +3 months	27-Apr-2018	31-Aug-2021	
Peak +3 months - share price (JPY)	4,712	4,798	
Peak +3 months share price vs. cycle Min share price (JPY)	1,612	2,371	
Peak +3 months share price vs. cycle Min share price (%)	52%	98%	
Peak +3 months share price vs. cycle Max share price (JPY)	-1,978	-1,012	
Peak +3 months share price vs. cycle Max share price (%)	-30%	-17%	
Peak +6 months	31-Jul-2018	30-Nov-2021	
Peak +6 months - share price (JPY)	4,384	4,449	
Peak +6 months share price vs. cycle Min share price (JPY)	1,284	2,022	
Peak +6 months share price vs. cycle Min share price (%)	41%	83%	
Peak +6 months share price vs. cycle Max share price (JPY)	-2,306	-1,361	
Peak +6 months share price vs. cycle Max share price (%)	-34%	-23%	
Peak +12 months	31-Jan-2019	31-May-2022	
Peak +12 months - share price (JPY)	3,672	4,221	
Peak +12 months share price vs. cycle Min share price (JPY)	572	1,794	
Peak +12 months share price vs. cycle Min share price (%)	18%	74%	
Peak +12 months share price vs. cycle Max share price (JPY)	-3,018	-1,589	
Peak +12 months share price vs. cycle Max share price (%)	-45%	-27%	



Upcycle 3 (Jan-2013 to Mar-2015): THK share price peaked at the time when orders peaked, despite slowing YoY growth

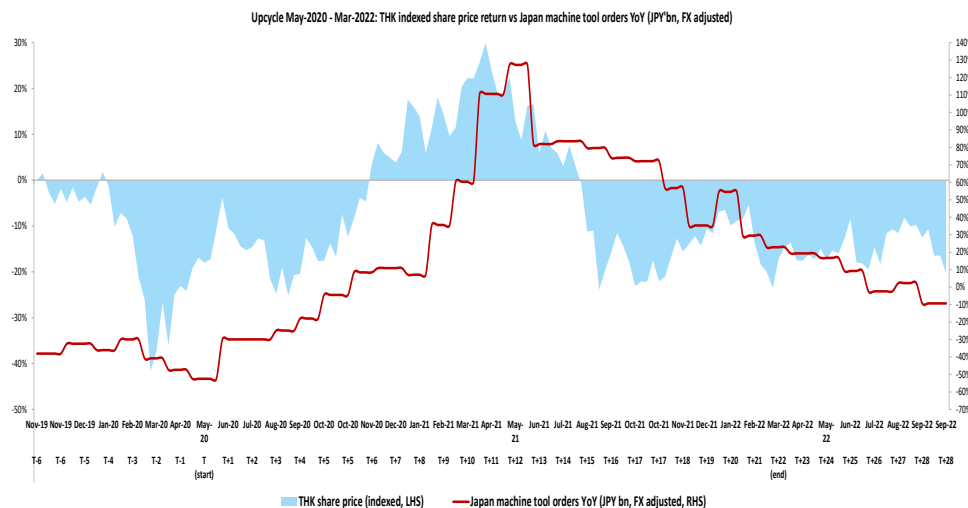
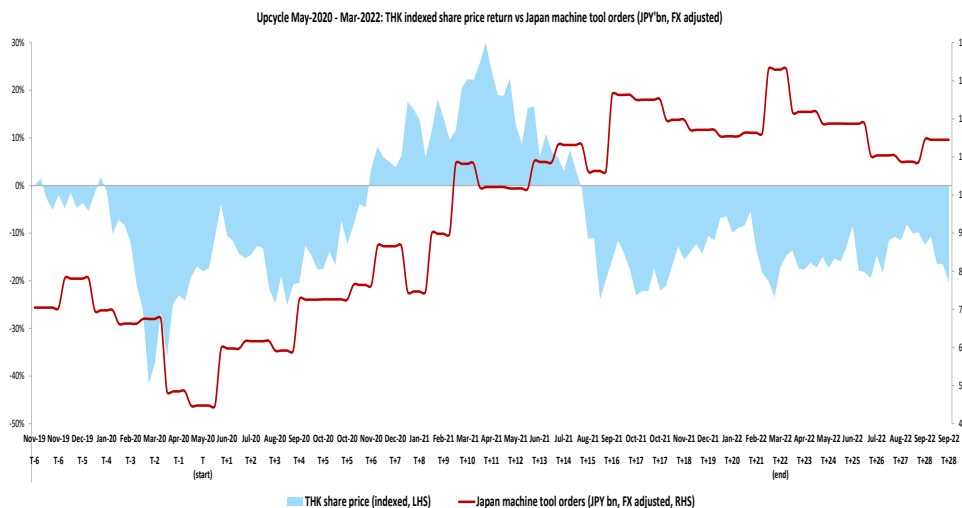


Upcycle 4 (Oct-2016 to Mar-2018): THK share price peaked when orders and YoY growth peaked

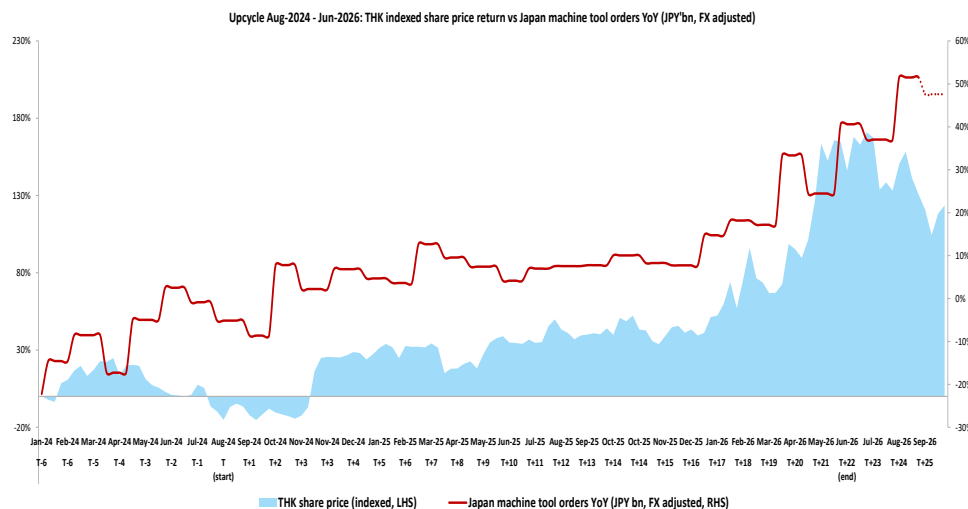
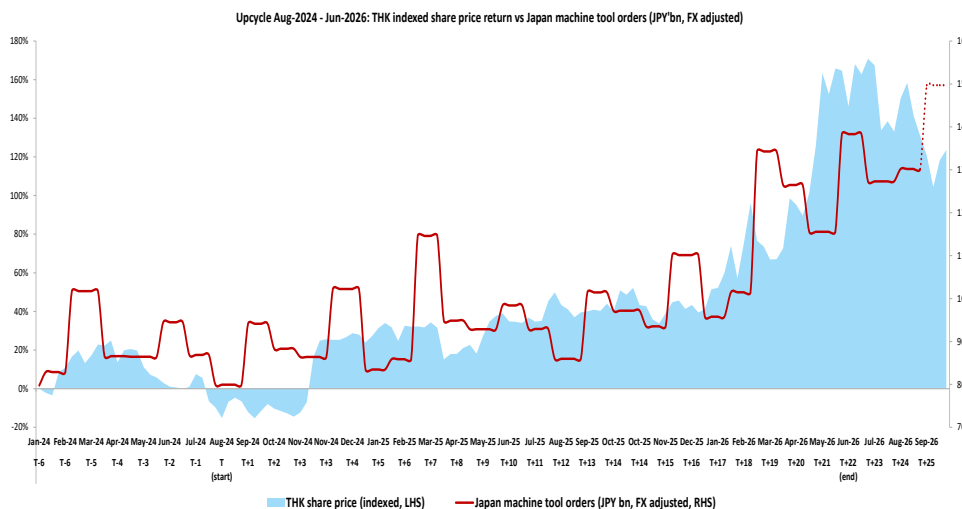




Upcycle 5 (May-2020 to Mar-2022): THK share price peaked 1M prior to the YoY growth peak and 11M prior to the order peak



Upcycle 6 (Aug-2024 to Jun-2026): THK peaked 1M prior to the YoY growth peaking (scale issue)





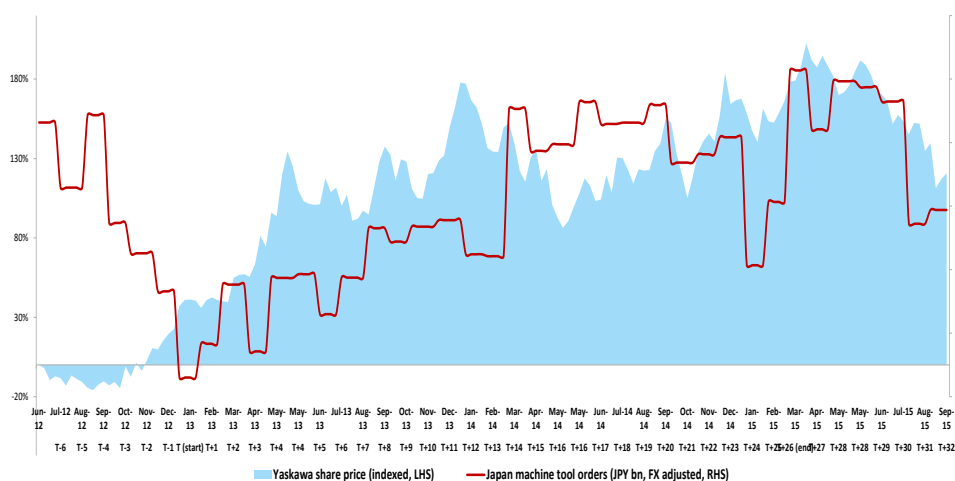
Under the assumption that Japan machine tool orders YoY growth has peaked in August 2026, THK's share price has corrected by over -20% from its current cycle peak in September, close to the share price correction observed in the past Upcycle 5

THK	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBE)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBE)
Peak - orders YoY	47.6%	127.2%	51.5%
THK share price performance			
Min - Share price (JPY) across the cycle	1,687	1,781	2,155
Date of minimum	06-Jul-2016 (T-3 month)	16-Mar-2020 (T-2 month)	05-Aug-2024 (T month)
Max - Share price (JPY) across the cycle	4,790	4,075	8,196
Date of maximum	13-Mar-2018 (T+17 month)	09-Apr-2021 (T+11 month)	22-Jun-2026 (T+22 month)
Increase in share price from min to max (JPY)	3,103	2,294	6,042
Increase in share price from min to max (%)	184%	129%	280%
Japan machine tool orders (FX adjusted) YoY peak			
Share price (JPY) at the peak date	31-Jan-2018 (T+15)	31-May-2021 (T+12)	31-Aug-2026 (T+24)
Peak-date share price vs. cycle Min share price (JPY)	4,520	3,560	6,634
Peak-date share price vs. cycle Min share price (%)	2,833	1,779	4,480
Peak-date share price vs. cycle Max share price (JPY)	168%	100%	208%
Peak-date share price vs. cycle Max share price (%)	-270	-515	-1,562
Peak-date share price vs. cycle Max share price (%)	-6%	-13%	-19%
Peak +1 month			
Peak +1 month - share price (JPY)	28-Feb-2018	30-Jun-2021	25-Sept-2026 (latest date)
Peak +1 month share price vs. cycle Min share price (JPY)	4,680	3,315	6,429
Peak +1 month share price vs. cycle Min share price (%)	2,993	1,534	4,275
Peak +1 month share price vs. cycle Max share price (JPY)	177%	86%	198%
Peak +1 month share price vs. cycle Max share price (%)	-110	-760	-1,767
Peak +1 month share price vs. cycle Max share price (%)	-2%	-19%	-22%
Peak +2 months			
Peak +2 months - share price (JPY)	30-Mar-2018	30-Jul-2021	
Peak +2 months share price vs. cycle Min share price (JPY)	4,400	3,110	
Peak +2 months share price vs. cycle Min share price (%)	2,713	1,329	
Peak +2 months share price vs. cycle Max share price (JPY)	161%	75%	
Peak +2 months share price vs. cycle Max share price (%)	-390	-965	
Peak +2 months share price vs. cycle Max share price (%)	-8%	-24%	
Peak +3 months			
Peak +3 months - share price (JPY)	27-Apr-2018	31-Aug-2021	
Peak +3 months share price vs. cycle Min share price (JPY)	3,830	2,553	
Peak +3 months share price vs. cycle Min share price (%)	2,143	772	
Peak +3 months share price vs. cycle Max share price (JPY)	127%	43%	
Peak +3 months share price vs. cycle Max share price (%)	-960	-1,522	
Peak +3 months share price vs. cycle Max share price (%)	-20%	-37%	
Peak +6 months			
Peak +6 months - share price (JPY)	31-Jul-2018	30-Nov-2021	
Peak +6 months share price vs. cycle Min share price (JPY)	3,025	2,711	
Peak +6 months share price vs. cycle Min share price (%)	1,338	930	
Peak +6 months share price vs. cycle Max share price (JPY)	79%	52%	
Peak +6 months share price vs. cycle Max share price (%)	-1,765	-1,364	
Peak +6 months share price vs. cycle Max share price (%)	-37%	-33%	
Peak +12 months			
Peak +12 months - share price (JPY)	31-Jan-2019	31-May-2022	
Peak +12 months share price vs. cycle Min share price (JPY)	2,577	2,640	
Peak +12 months share price vs. cycle Min share price (%)	890	859	
Peak +12 months share price vs. cycle Max share price (JPY)	53%	48%	
Peak +12 months share price vs. cycle Max share price (%)	-2,213	-1,435	
Peak +12 months share price vs. cycle Max share price (%)	-46%	-35%	

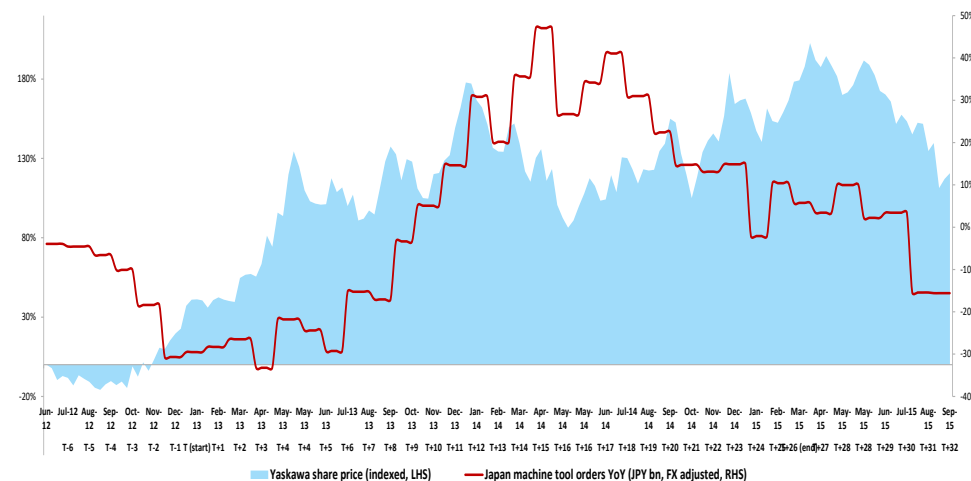


Upcycle 3 (Jan-2013 to Mar-2015): Yaskawa share price peaked at the time when orders peaked, despite slowing YoY growth

Upcycle Jan-2013 - Mar-2015: Yaskawa indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)

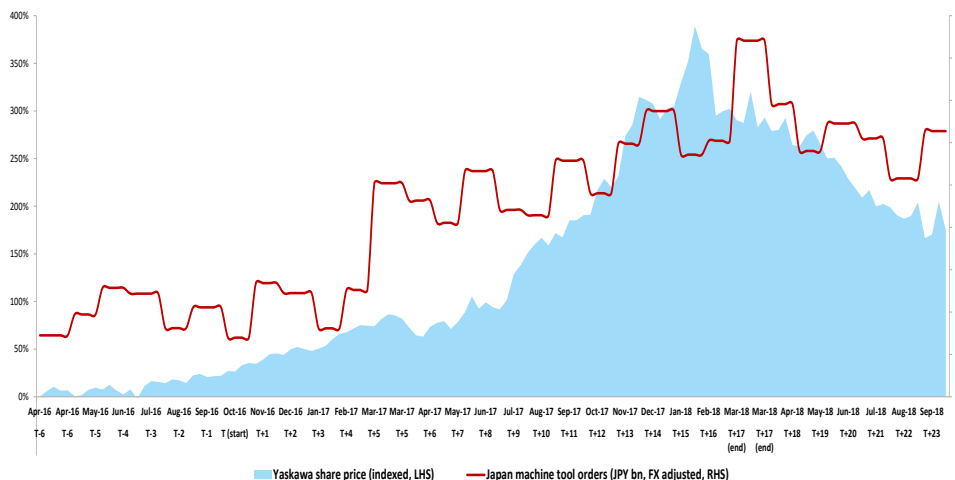


Upcycle Jan-2013 - Mar-2015: Yaskawa indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)

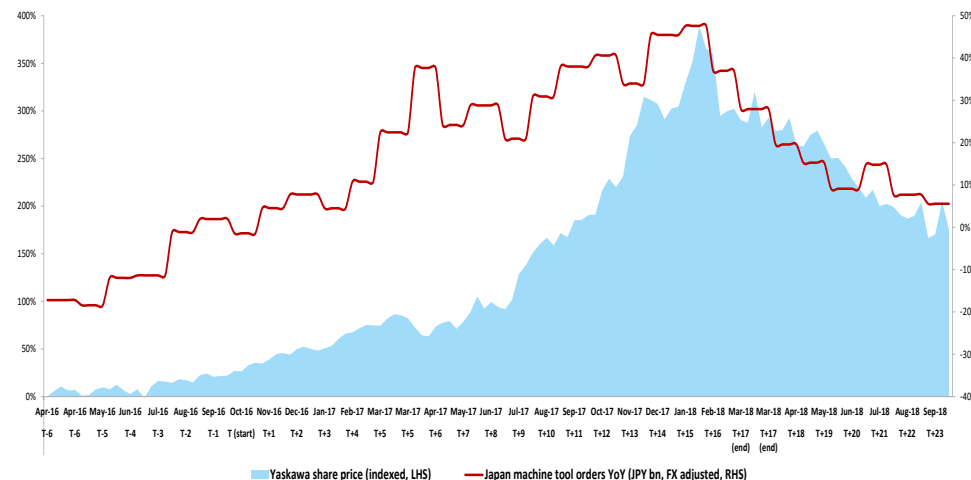


Upcycle 4 (Oct-2016 to Mar-2018): Yaskawa share price peaked when YoY growth peaked and 3M prior to the order peak

Upcycle Oct-2016 - Mar-2018: Yaskawa indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)

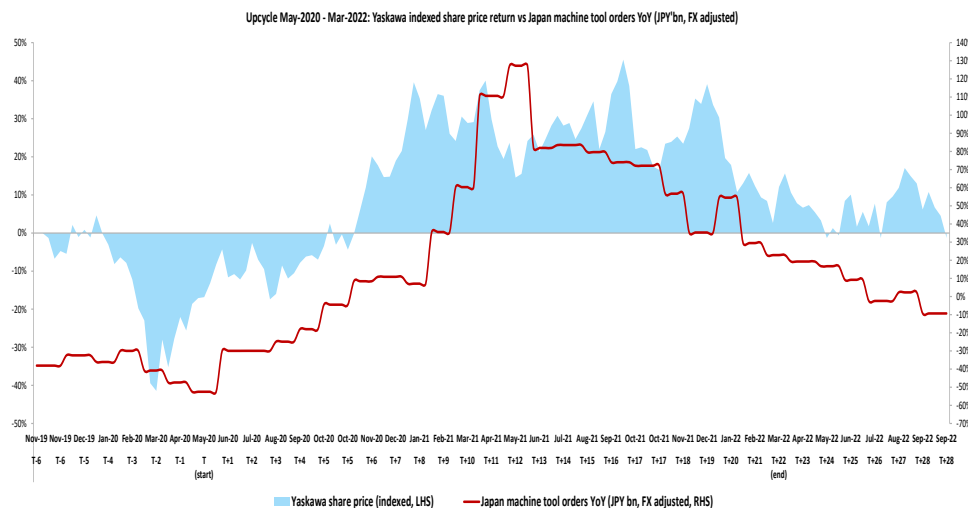
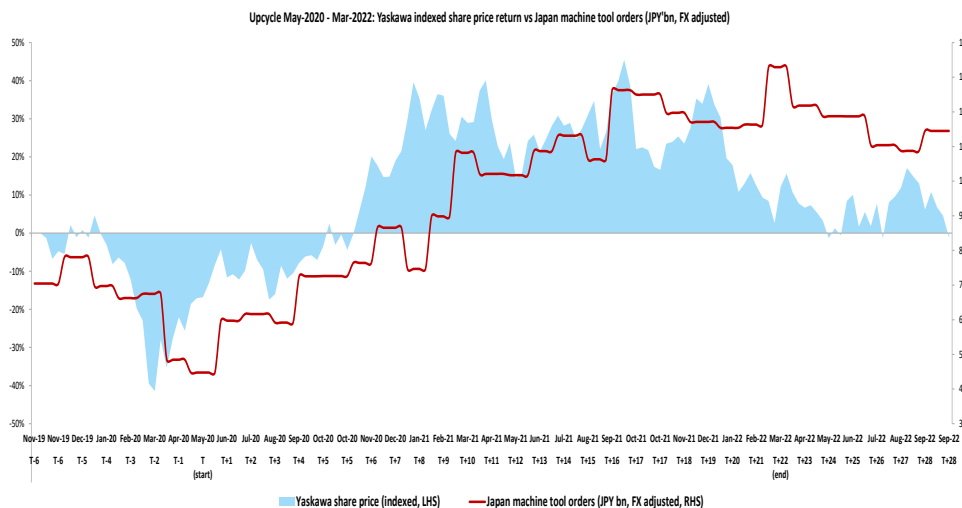


Upcycle Oct-2016 - Mar-2018: Yaskawa indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)

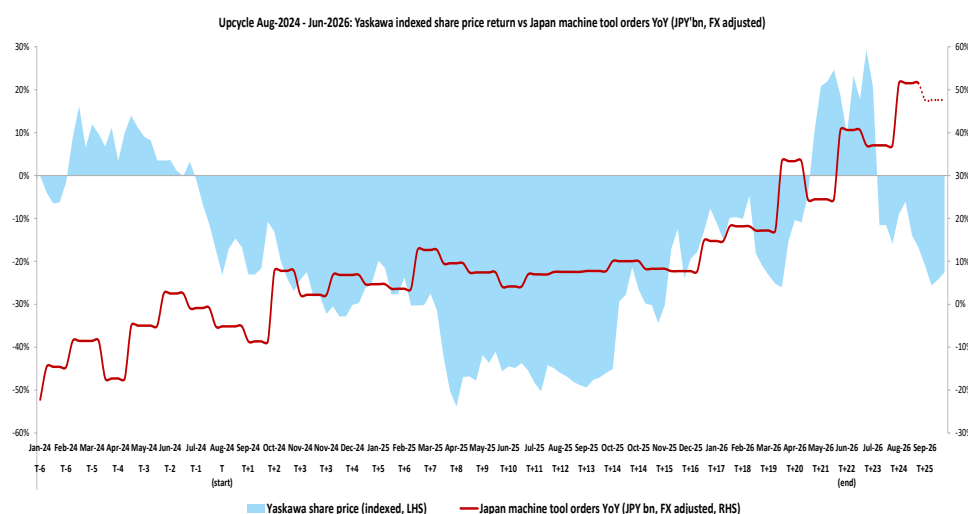
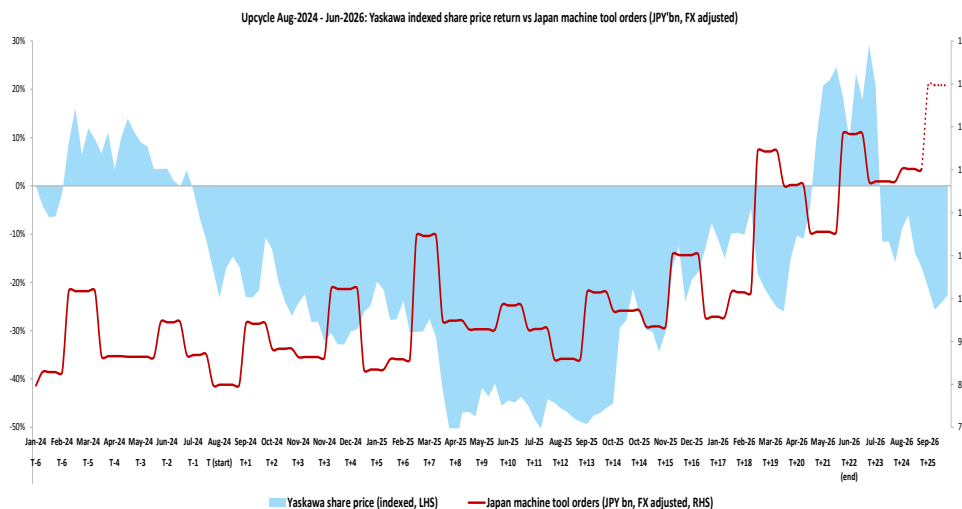




Upcycle 5 (May-2020 to Mar-2022): Yaskawa share price continued to rally despite slowing YoY growth for 5M



Upcycle 6 (Aug-2024 to Jun-2026): Yaskawa share price peaked when YoY growth peaked; the ERP system issue mentioned in July added further pressure



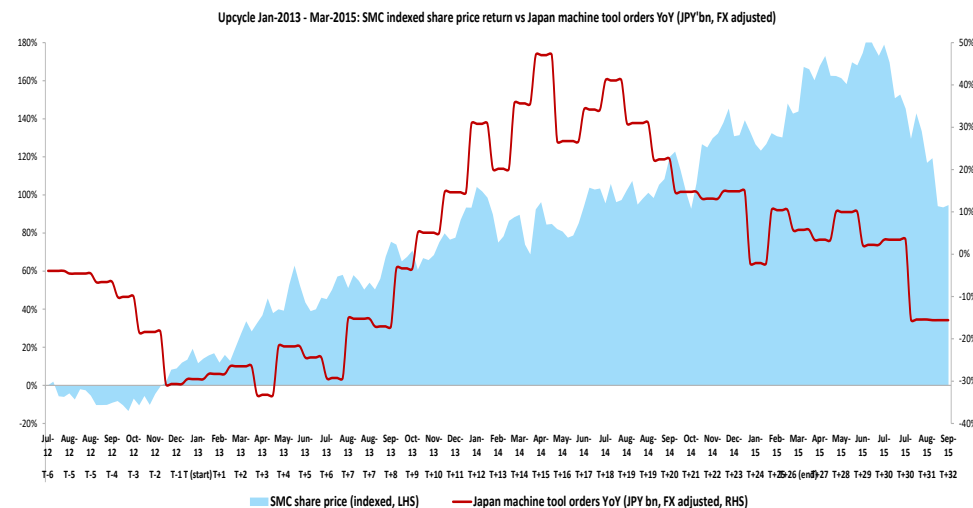
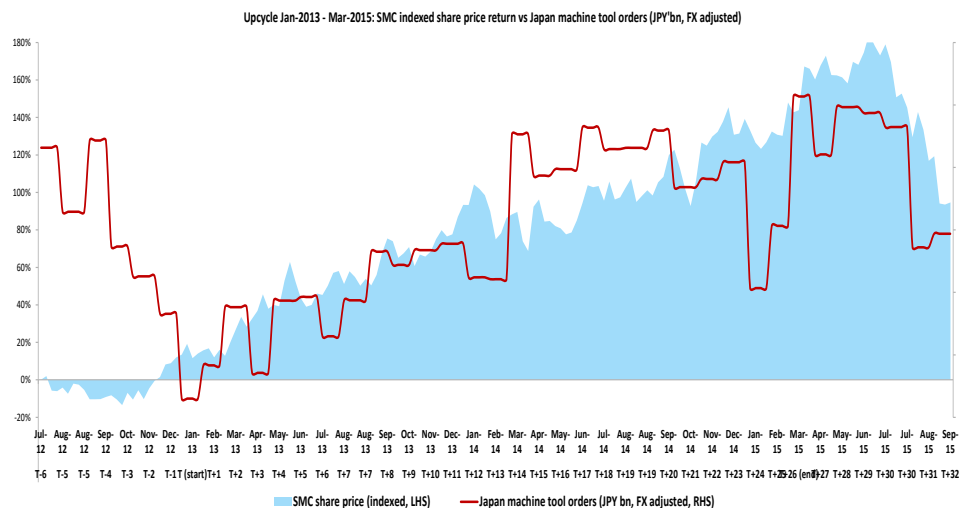


Under the assumption that Japan machine tool orders YoY growth has peaked in August 2026, Yaskawa's share price has corrected by more than -40% from its current cycle peak in September, clearly more severe than the share price corrections observed in the past two upcycles

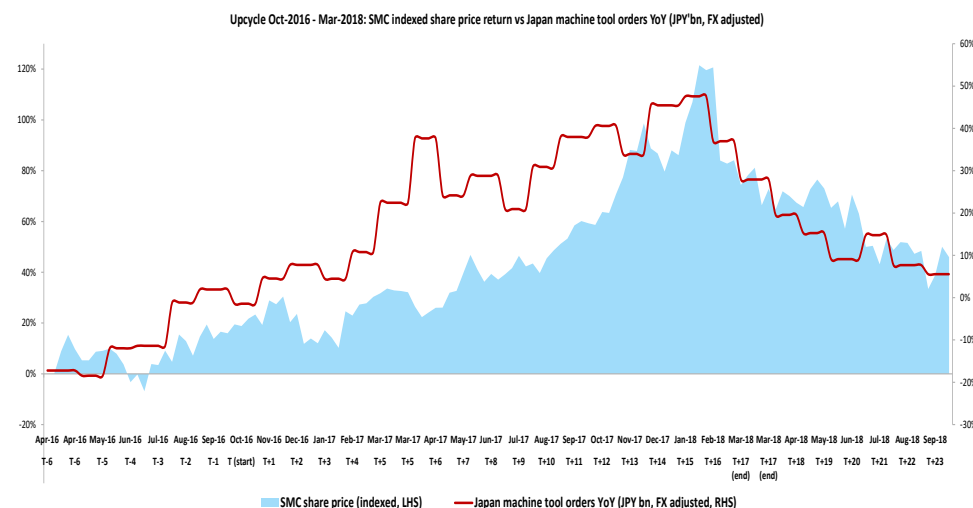
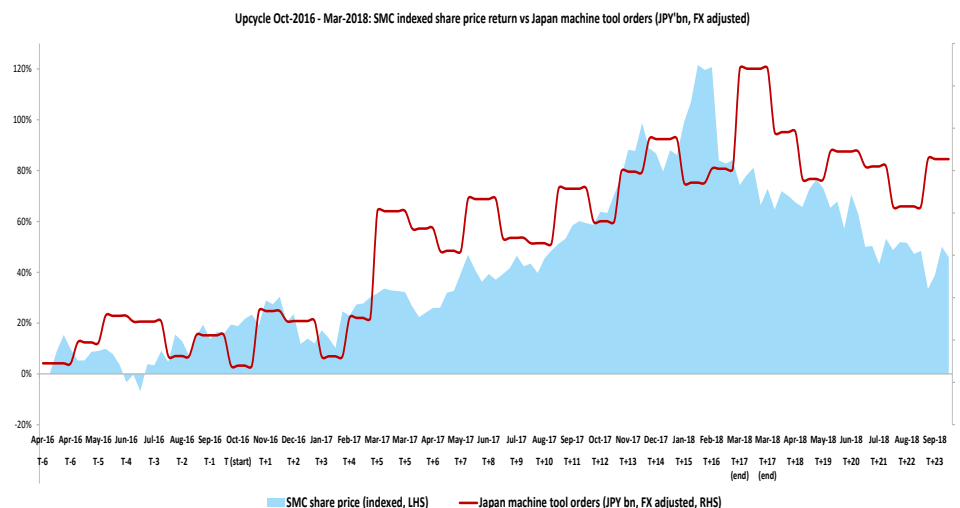
Yaskawa Electric	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBE)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBE)
Peak - orders YoY	47.6%	127.2%	51.5%
Yaskawa Electric share price performance			
Min - Share price (JPY) across the cycle	1,204	2,429	2,606
Date of minimum	08-Jul-2016 (T-3 month)	18-Mar-2020 (T-2 month)	22-Apr-2025 (T+8 month)
Max - Share price (JPY) across the cycle	6,060	6,140	7,655
Date of maximum	23-Jan-2018 (T+15 month)	17-Sep-2021 (T+16 month)	22-Jun-2026 (T+22 month)
Increase in share price from min to max (JPY)	4,856	3,711	5,050
Increase in share price from min to max (%)	403%	153%	194%
Japan machine tool orders (FX adjusted) YoY peak	31-Jan-2018 (T+15)	31-May-2021 (T+12)	31-Aug-2026 (T+24)
Share price (JPY) at the peak date	5,620	5,310	4,780
Peak-date share price vs. cycle Min share price (JPY)	4,416	2,881	2,175
Peak-date share price vs. cycle Min share price (%)	367%	119%	83%
Peak-date share price vs. cycle Max share price (JPY)	-440	-830	-2,875
Peak-date share price vs. cycle Max share price (%)	-7%	-14%	-38%
Peak +1 month	28-Feb-2018	30-Jun-2021	25-Sept-2026 (latest date)
Peak +1 month - share price (JPY)	5,010	5,430	4,480
Peak +1 month share price vs. cycle Min share price (JPY)	3,806	3,001	1,875
Peak +1 month share price vs. cycle Min share price (%)	316%	124%	72%
Peak +1 month share price vs. cycle Max share price (JPY)	-1,050	-710	-3,175
Peak +1 month share price vs. cycle Max share price (%)	-17%	-12%	-41%
Peak +2 months	30-Mar-2018	30-Jul-2021	
Peak +2 months - share price (JPY)	4,825	5,380	
Peak +2 months share price vs. cycle Min share price (JPY)	3,621	2,951	
Peak +2 months share price vs. cycle Min share price (%)	301%	121%	
Peak +2 months share price vs. cycle Max share price (JPY)	-1,235	-760	
Peak +2 months share price vs. cycle Max share price (%)	-20%	-12%	
Peak +3 months	27-Apr-2018	31-Aug-2021	
Peak +3 months - share price (JPY)	4,470	5,380	
Peak +3 months share price vs. cycle Min share price (JPY)	3,266	2,951	
Peak +3 months share price vs. cycle Min share price (%)	271%	121%	
Peak +3 months share price vs. cycle Max share price (JPY)	-1,590	-760	
Peak +3 months share price vs. cycle Max share price (%)	-26%	-12%	
Peak +6 months	31-Jul-2018	30-Nov-2021	
Peak +6 months - share price (JPY)	3,670	5,090	
Peak +6 months share price vs. cycle Min share price (JPY)	2,466	2,661	
Peak +6 months share price vs. cycle Min share price (%)	205%	110%	
Peak +6 months share price vs. cycle Max share price (JPY)	-2,390	-1,050	
Peak +6 months share price vs. cycle Max share price (%)	-39%	-17%	
Peak +12 months	31-Jan-2019	31-May-2022	
Peak +12 months - share price (JPY)	3,060	4,485	
Peak +12 months share price vs. cycle Min share price (JPY)	1,856	2,056	
Peak +12 months share price vs. cycle Min share price (%)	154%	85%	
Peak +12 months share price vs. cycle Max share price (JPY)	-3,000	-1,655	
Peak +12 months share price vs. cycle Max share price (%)	-50%	-27%	



Upcycle 3 (Jan-2013 to Mar-2015): SMC share price peaked 4M prior to the Japan machine tool order peak; YoY growth slowdown occurred much earlier



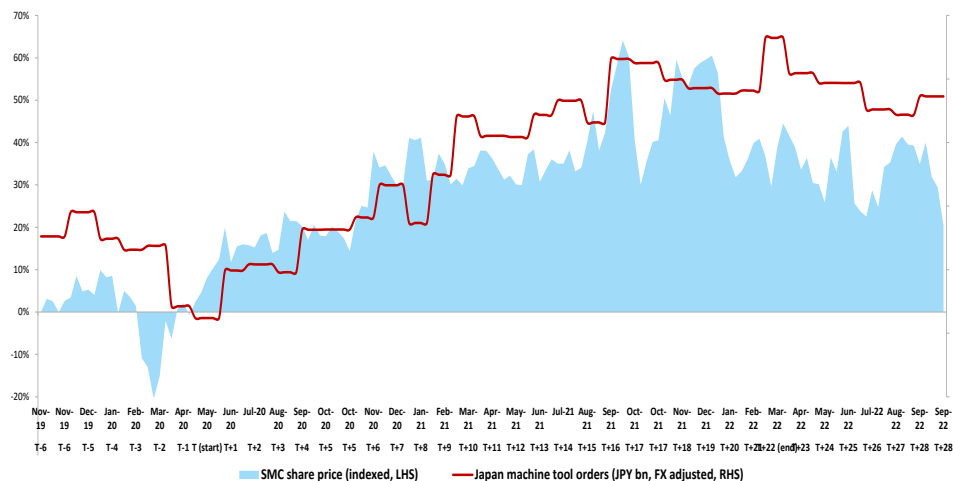
Upcycle 4 (Oct-2016 to Mar-2018): SMC share price peaked 2M before the upcycle high, and simultaneously when YoY growth reached the peak



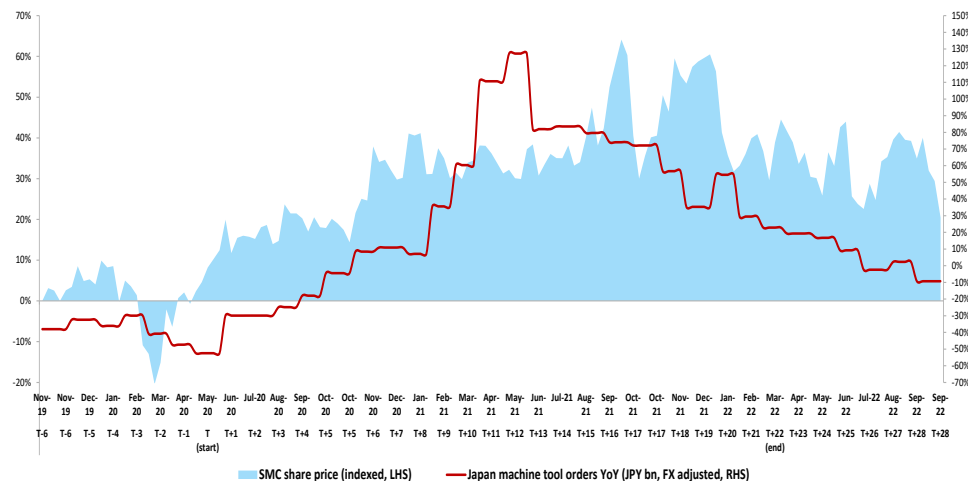


Upcycle 5 (May-2020 to Mar-2022): SMC share price peaked 3-6M prior to the order peak

Upcycle May-2020 - Mar-2022: SMC indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)

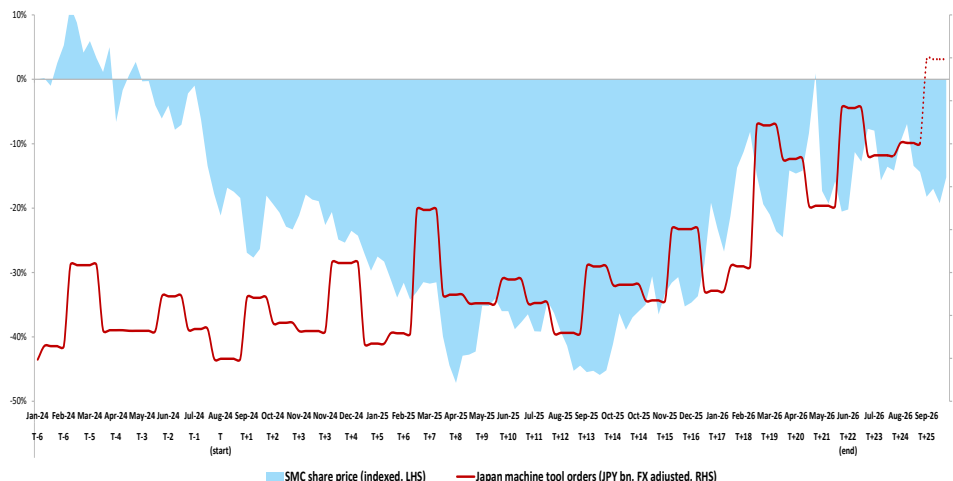


Upcycle May-2020 - Mar-2022: SMC indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)

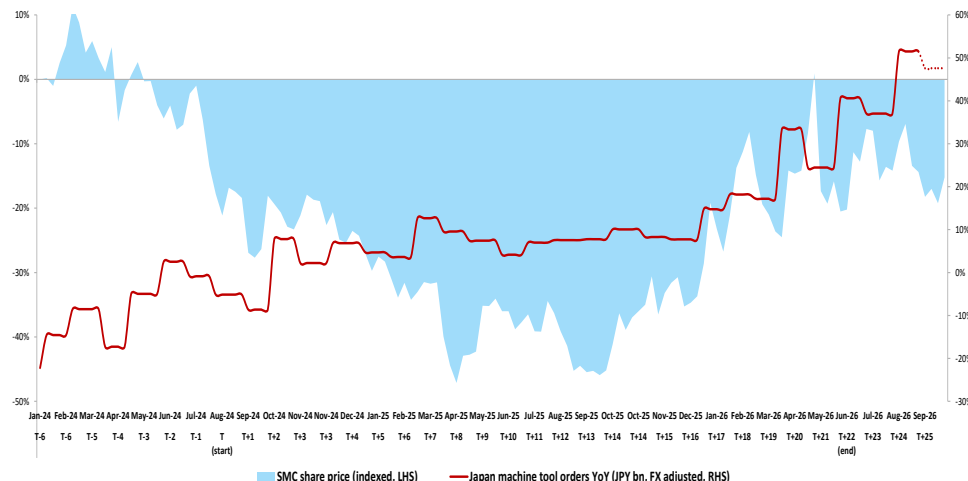


Upcycle 6 (Aug-2024 to Jun-2026): SMC share price already corrected heavily when orders still trending upwards

Upcycle Aug-2024 - Jun-2026: SMC indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)



Upcycle Aug-2024 - Jun-2026: SMC indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)





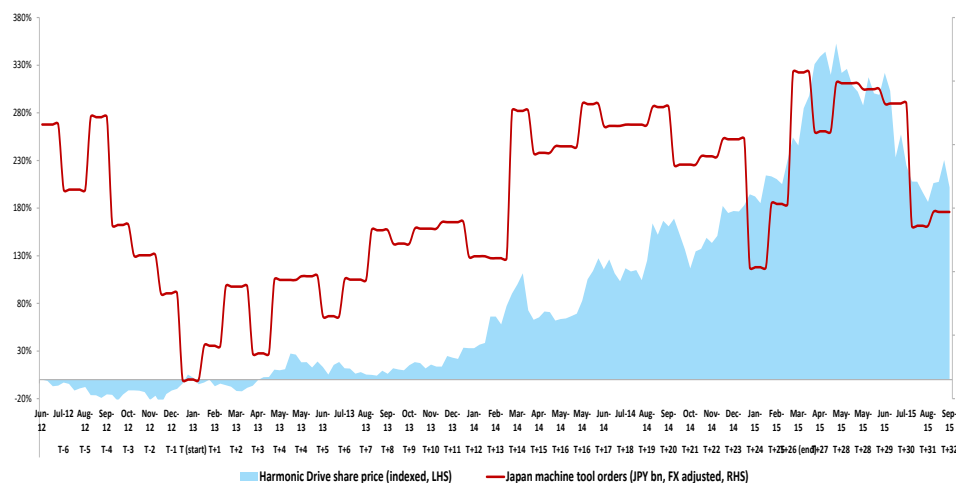
Under the assumption that Japan machine tool orders YoY growth has peaked in August 2026, SMC's share price has corrected by close to -20% from its current cycle peak in September, inline with the share price corrections observed in the past two upcycles

SMC	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBE)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBE)
Peak - orders YoY	47.6%	127.2%	51.5%
SMC share price performance			
Min - Share price (JPY) across the cycle	23,200	37,620	42,310
Date of minimum	08-Jul-2016 (T-3 month)	16-Mar-2020 (T-2 month)	22-Apr-2025 (T+8 month)
Max - Share price (JPY) across the cycle	55,460	79,340	84,560
Date of maximum	29-Jan-2018 (T+15 month)	17-Sep-2021 (T+16 month)	12-May-2026 (T+21 month)
Increase in share price from min to max (JPY)	32,260	41,720	42,250
Increase in share price from min to max (%)	139%	111%	100%
Japan machine tool orders (FX adjusted) YoY peak	31-Jan-2018 (T+15)	31-May-2021 (T+12)	31-Aug-2026 (T+24)
Share price (JPY) at the peak date	53,520	65,520	70,710
Peak-date share price vs. cycle Min share price (JPY)	30,320	27,900	28,400
Peak-date share price vs. cycle Min share price (%)	131%	74%	67%
Peak-date share price vs. cycle Max share price (JPY)	-1,940	-13,820	-13,850
Peak-date share price vs. cycle Max share price (%)	-3%	-17%	-16%
Peak +1 month	28-Feb-2018	30-Jun-2021	25-Sept-2026 (latest date)
Peak +1 month - share price (JPY)	45,030	65,650	69,620
Peak +1 month share price vs. cycle Min share price (JPY)	21,830	28,030	27,310
Peak +1 month share price vs. cycle Min share price (%)	94%	75%	65%
Peak +1 month share price vs. cycle Max share price (JPY)	-10,430	-13,690	-14,940
Peak +1 month share price vs. cycle Max share price (%)	-19%	-17%	-18%
Peak +2 months	30-Mar-2018	30-Jul-2021	
Peak +2 months - share price (JPY)	43,070	64,800	
Peak +2 months share price vs. cycle Min share price (JPY)	19,870	27,180	
Peak +2 months share price vs. cycle Min share price (%)	86%	72%	
Peak +2 months share price vs. cycle Max share price (JPY)	-12,390	-14,540	
Peak +2 months share price vs. cycle Max share price (%)	-22%	-18%	
Peak +3 months	27-Apr-2018	31-Aug-2021	
Peak +3 months - share price (JPY)	41,720	70,510	
Peak +3 months share price vs. cycle Min share price (JPY)	18,520	32,890	
Peak +3 months share price vs. cycle Min share price (%)	80%	87%	
Peak +3 months share price vs. cycle Max share price (JPY)	-13,740	-8,830	
Peak +3 months share price vs. cycle Max share price (%)	-25%	-11%	
Peak +6 months	31-Jul-2018	30-Nov-2021	
Peak +6 months - share price (JPY)	37,480	72,650	
Peak +6 months share price vs. cycle Min share price (JPY)	14,280	35,030	
Peak +6 months share price vs. cycle Min share price (%)	62%	93%	
Peak +6 months share price vs. cycle Max share price (JPY)	-17,980	-6,690	
Peak +6 months share price vs. cycle Max share price (%)	-32%	-8%	
Peak +12 months	31-Jan-2019	31-May-2022	
Peak +12 months - share price (JPY)	35,730	66,610	
Peak +12 months share price vs. cycle Min share price (JPY)	12,530	28,990	
Peak +12 months share price vs. cycle Min share price (%)	54%	77%	
Peak +12 months share price vs. cycle Max share price (JPY)	-19,730	-12,730	
Peak +12 months share price vs. cycle Max share price (%)	-36%	-16%	

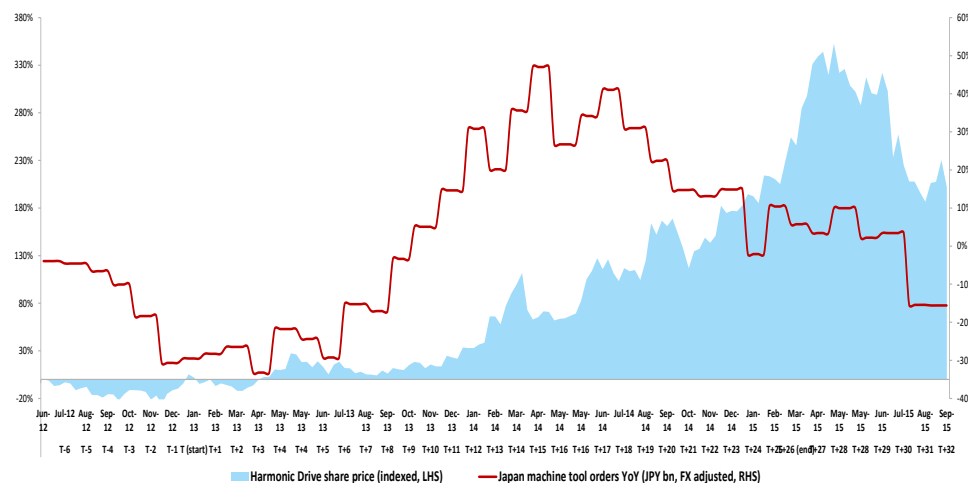


Upcycle 3 (Jan-2013 to Mar-2015): Harmonic share price peaked 2M post the order peak; YoY growth slowdown had minimal impact

Upcycle Jan-2013 - Mar-2015: Harmonic Drive indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)

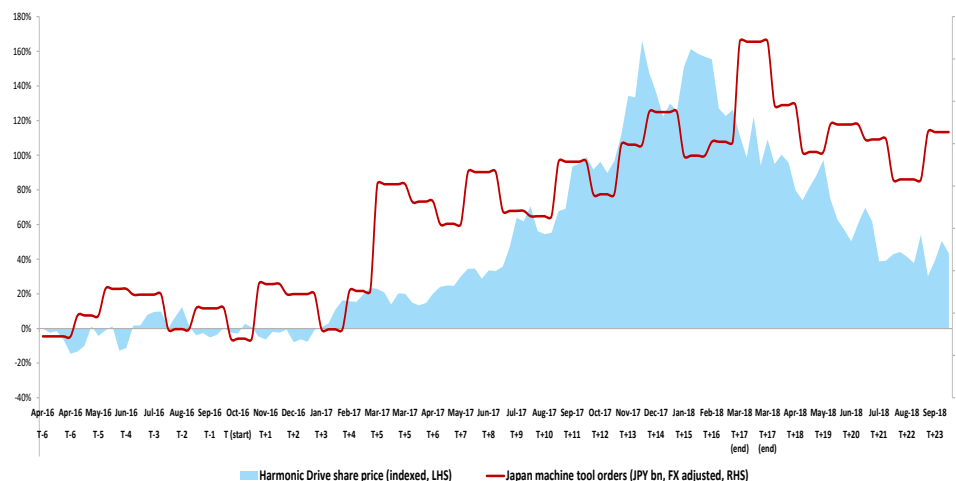


Upcycle Jan-2013 - Mar-2015: Harmonic Drive indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)

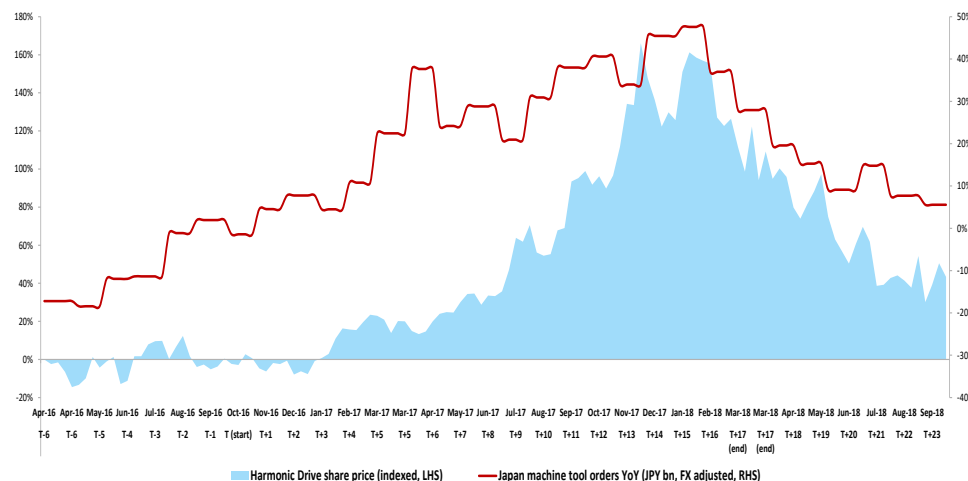


Upcycle 4 (Oct-2016 to Mar-2018): Harmonic share price peaked 2M before the upcycle high, and simultaneously when YoY growth reached the peak

Upcycle Oct-2016 - Mar-2018: Harmonic Drive indexed share price return vs Japan machine tool orders (JPY'bn, FX adjusted)



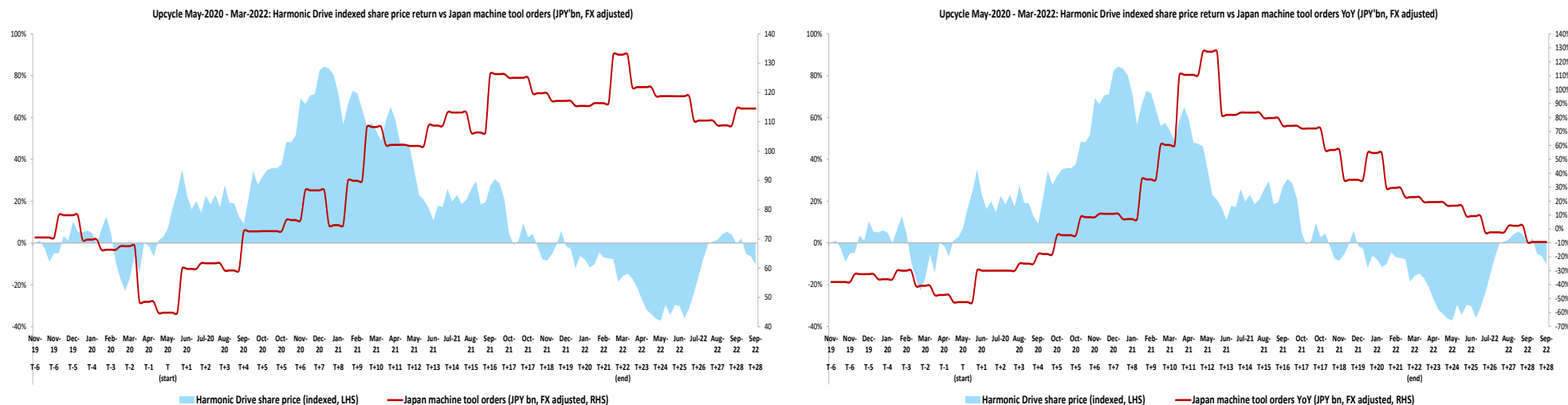
Upcycle Oct-2016 - Mar-2018: Harmonic Drive indexed share price return vs Japan machine tool orders YoY (JPY'bn, FX adjusted)



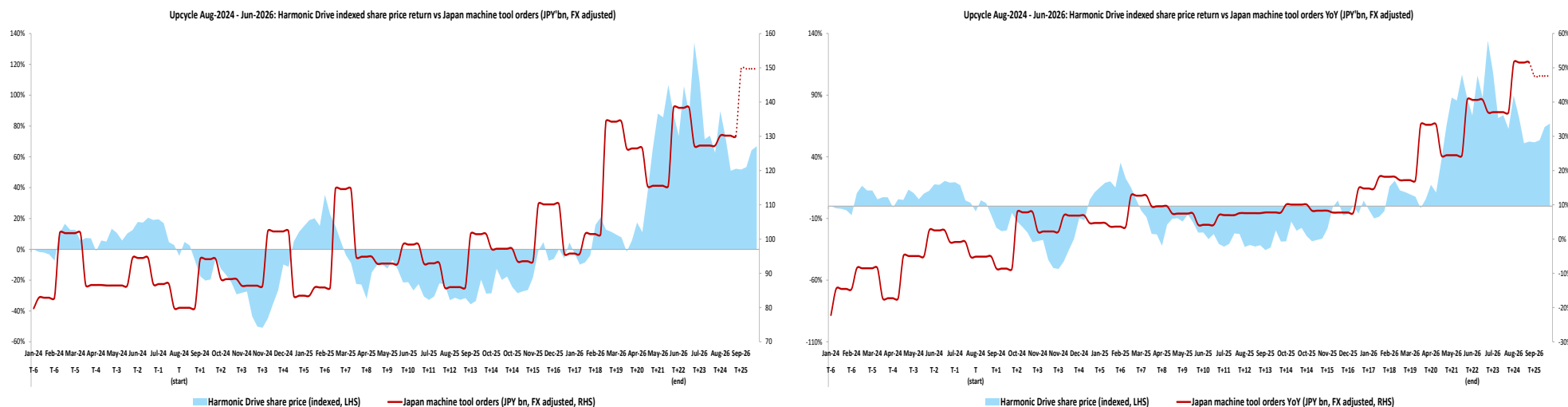
Harmonic Drive Systems (6324.T): BUY, TP ¥9,900



Upcycle 5 (May-2020 to Mar-2022): Harmonic share price peaked 15M prior to the order peak, and 5M prior to the YoY growth peak



Upcycle 6 (Aug-2024 to Jun-2026): Harmonic share price peaked 1M prior to the YoY growth peak



Harmonic Drive Systems (6324.T): BUY, TP ¥9,900



Under the assumption that Japan machine tool orders YoY growth has peaked in August 2026, HDS's share price has corrected by close to -30% from its current cycle peak in September, broadly inline with the share price corrections observed in the past two upcycles

Harmonic Drive Systems	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	Oct-2016 (T) - Mar-2018 (T+17)	May-2020 (T) - Mar-2022 (T+22)	Aug-2024 (T) - Sep-2026 (T+25, DBe)
Upcycle duration	18 months	23 months	26 months
Share-price search window	Apr-2016 (T-6) - Mar-2018 (T+17)	Nov-2019 (T-6) - Mar-2022 (T+22)	Feb-2024 (T-6) - Sep-2026 (T+25)
Japan machine tool orders (FX adjusted) YoY			
Peak month	Jan-2018 (T+15)	May-2021 (T+12)	Aug-2026 (T+24, DBe)
Peak - orders YoY	47.6%	127.2%	51.5%
Harmonic Drive Systems			
Min - Share price (JPY) across the cycle	2,491	3,815	1,853
Date of minimum	02-May-2016 (T-5 month)	12-Mar-2020 (T-2 month)	29-Nov-2024 (T+3 month)
Max - Share price (JPY) across the cycle	8,140	9,380	8,890
Date of maximum	16-Jan-2018 (T+15 month)	28-Dec-2020 (T+7 month)	06-Jul-2026 (T+23 month)
Increase in share price from min to max (JPY)	5,649	5,565	7,037
Increase in share price from min to max (%)	227%	146%	380%
Japan machine tool orders (FX adjusted) YoY peak	31-Jan-2018 (T+15)	31-May-2021 (T+12)	31-Aug-2026 (T+24)
Share price (JPY) at the peak date	7,510	5,900	5,790
Peak-date share price vs. cycle Min share price (JPY)	5,019	2,085	3,937
Peak-date share price vs. cycle Min share price (%)	201%	55%	212%
Peak-date share price vs. cycle Max share price (JPY)	-630	-3,480	-3,100
Peak-date share price vs. cycle Max share price (%)	-8%	-37%	-35%
Peak +1 month	28-Feb-2018	30-Jun-2021	25-Sept-2026 (latest date)
Peak +1 month - share price (JPY)	6,560	6,120	6,300
Peak +1 month share price vs. cycle Min share price (JPY)	4,069	2,305	4,447
Peak +1 month share price vs. cycle Min share price (%)	163%	60%	240%
Peak +1 month share price vs. cycle Max share price (JPY)	-1,580	-3,260	-2,590
Peak +1 month share price vs. cycle Max share price (%)	-19%	-35%	-29%
Peak +2 months	30-Mar-2018	30-Jul-2021	
Peak +2 months - share price (JPY)	6,110	6,040	
Peak +2 months share price vs. cycle Min share price (JPY)	3,619	2,225	
Peak +2 months share price vs. cycle Min share price (%)	145%	58%	
Peak +2 months share price vs. cycle Max share price (JPY)	-2,030	-3,340	
Peak +2 months share price vs. cycle Max share price (%)	-25%	-36%	
Peak +3 months	27-Apr-2018	31-Aug-2021	
Peak +3 months - share price (JPY)	5,250	6,090	
Peak +3 months share price vs. cycle Min share price (JPY)	2,759	2,275	
Peak +3 months share price vs. cycle Min share price (%)	111%	60%	
Peak +3 months share price vs. cycle Max share price (JPY)	-2,890	-3,290	
Peak +3 months share price vs. cycle Max share price (%)	-36%	-35%	
Peak +6 months	31-Jul-2018	30-Nov-2021	
Peak +6 months - share price (JPY)	4,380	4,685	
Peak +6 months share price vs. cycle Min share price (JPY)	1,889	870	
Peak +6 months share price vs. cycle Min share price (%)	76%	23%	
Peak +6 months share price vs. cycle Max share price (JPY)	-3,760	-4,695	
Peak +6 months share price vs. cycle Max share price (%)	-46%	-50%	
Peak +12 months	31-Jan-2019	31-May-2022	
Peak +12 months - share price (JPY)	3,790	3,315	
Peak +12 months share price vs. cycle Min share price (JPY)	1,299	-500	
Peak +12 months share price vs. cycle Min share price (%)	52%	-13%	
Peak +12 months share price vs. cycle Max share price (JPY)	-4,350	-6,065	
Peak +12 months share price vs. cycle Max share price (%)	-53%	-65%	



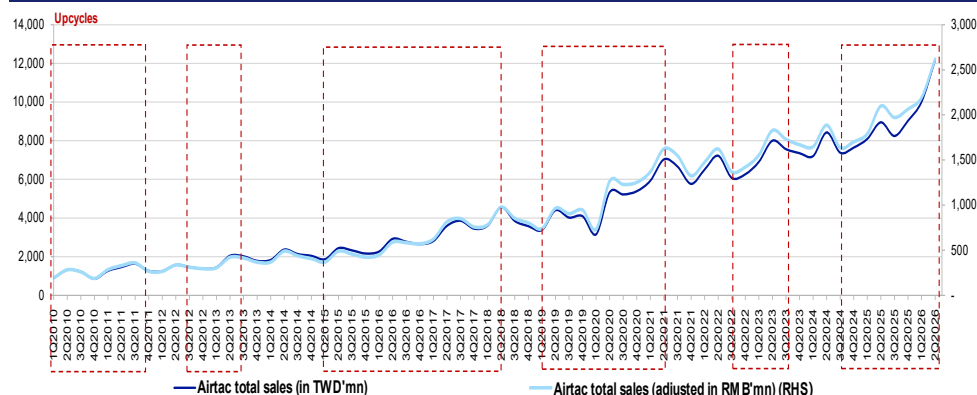
Airtac: Structurally growing
sales warrants higher share price;
YoY growth tests near-term
share price momentum

Airtac (1590.TW): BUY, TP TWD1,350

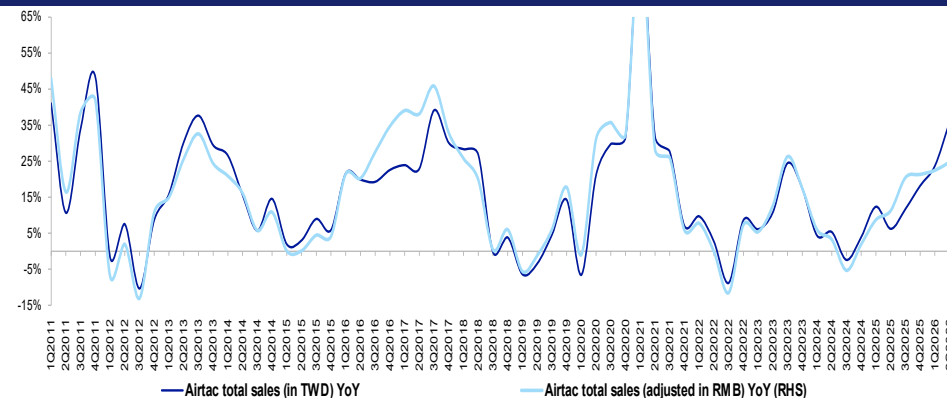
Airtac's cumulative sales increase has already exceeded historical levels



Airtac quarterly sales – structurally growing

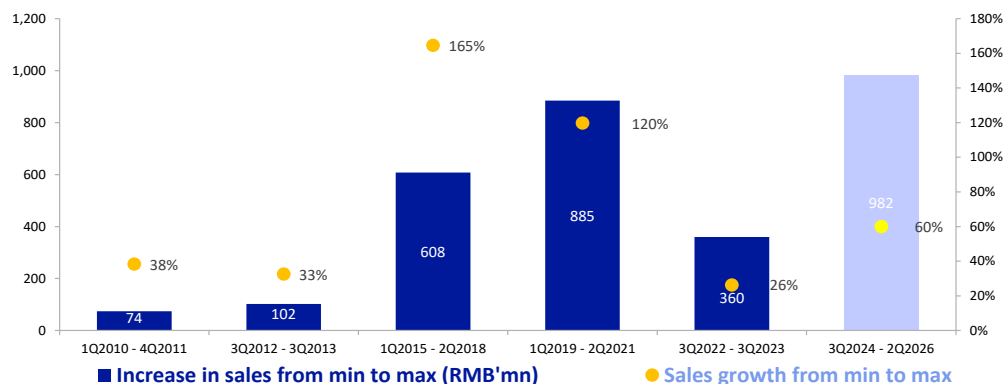


Airtac quarterly sales YoY



Airtac sales and share price performance– pattern from past upcycles

Airtac	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	1Q2010 - 4Q2011	3Q2012 - 3Q2013	1Q2015 - 2Q2018	1Q2019 - 2Q2021	3Q2022 - 3Q2023	3Q2024 - 2Q2026
Upcycle duration	21	12	39	27	12	21
Reported sales in TWD						
Start quarter	1Q2010	3Q2012	1Q2015	1Q2019	3Q2022	3Q2024
Min - Sales (TWD'mn)	900	1,466	1,867	3,379	6,077	7,381
End quarter	4Q2011	3Q2013	2Q2018	2Q2021	3Q2023	2Q2026
Max - Sales (TWD'mn)	1,269	2,019	4,561	7,035	7,564	12,164
Increase in sales from min to max (TWD'mn)	368	552	2,694	3,657	1,487	4,784
Sales growth from min to max	41%	38%	144%	108%	24%	65%
Adjusted sales in RMB						
Start quarter	1Q2010	3Q2012	1Q2015	1Q2019	3Q2022	3Q2024
Min - Sales (RMB'mn)	193	312	369	739	1,369	1,637
End quarter	4Q2011	3Q2013	2Q2018	2Q2021	3Q2023	2Q2026
Max - Sales (RMB'mn)	267	414	977	1,624	1,729	2,619
Increase in sales from min to max (RMB'mn)	74	102	608	885	360	982
Sales growth from min to max	38%	33%	165%	120%	26%	60%
Increase in sales from min to max - Upcycle 6 as % of ...	1329%	965%	162%	111%	273%	



Airtac has delivered consistent sales growth over time, reflecting structural market expansion, strong management execution, and steady market share gains.

We find the duration of Airtac's upcycles to be relatively short, typically lasting between one and two years.

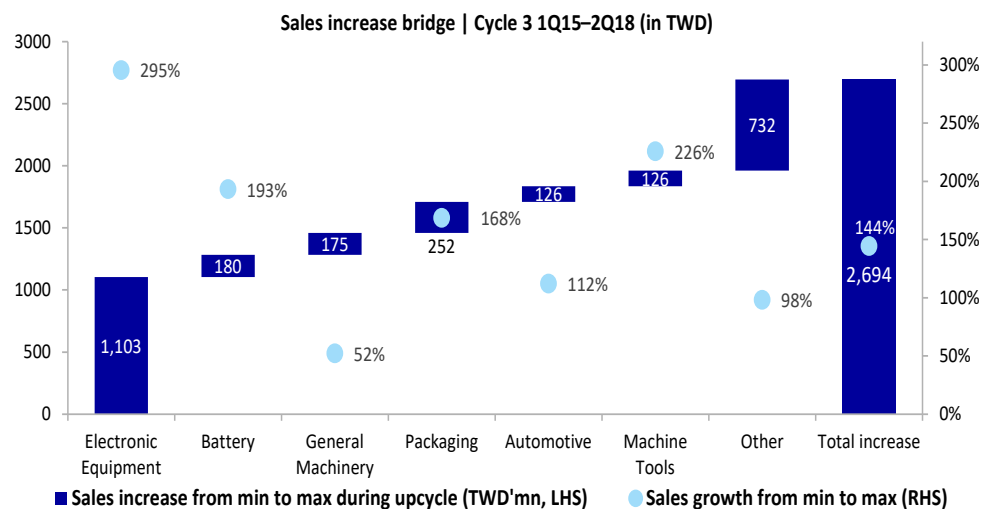
Based on historical patterns, the cumulative sales increase during this upcycle has already exceeded previous historical levels.

However, as Airtac continues to launch new products, sales could potentially have further room for upside, in our view.

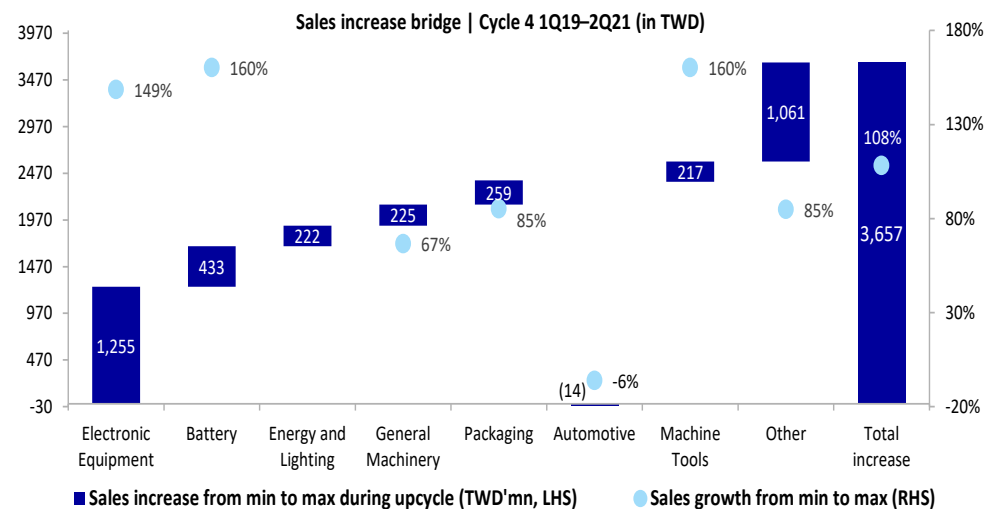
Airtac: Battery, electronic and other equipment have been the main contributors for sales during the current upcycle



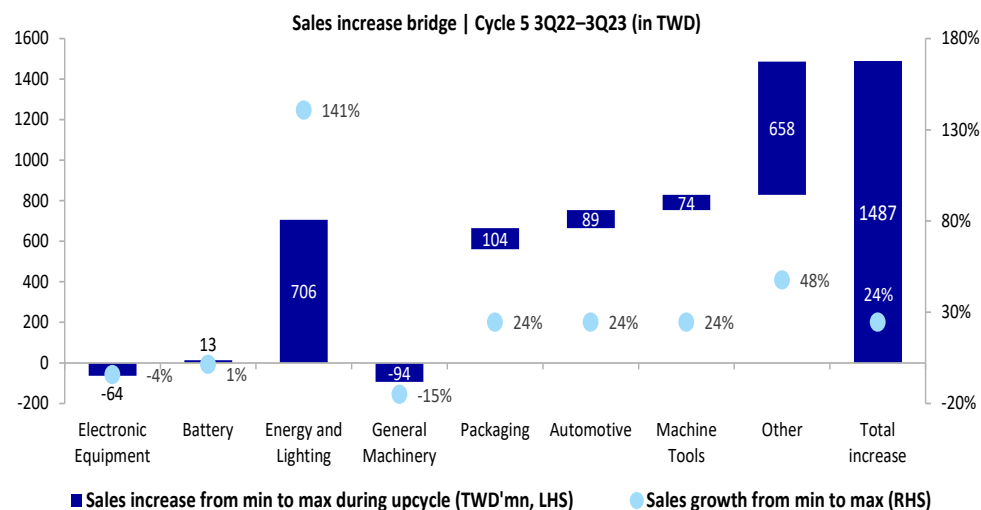
Upcycle 3 (1Q15 to 2Q18): Driven mainly by Electronic equipment due to the Apple product cycle, in our view



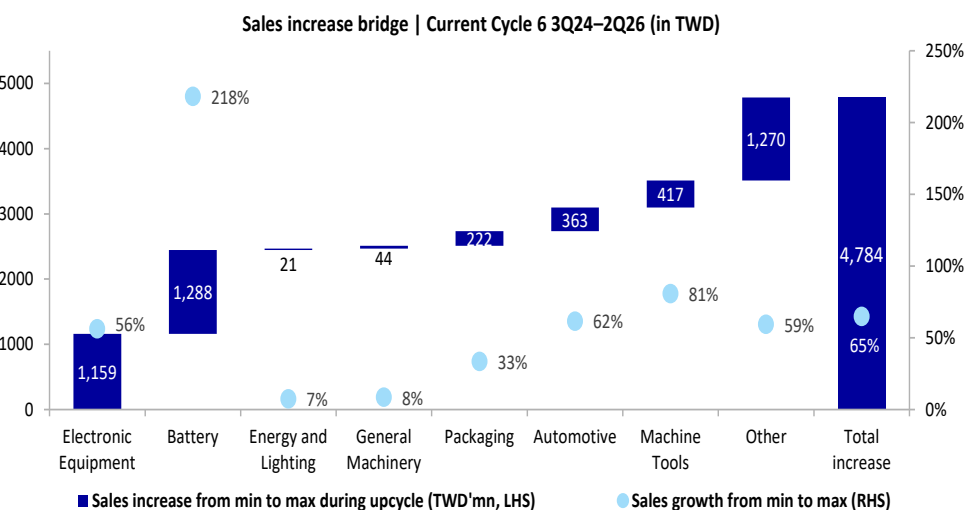
Upcycle 4 (1Q19 to 2Q21): Broad-based contributions from various end-markets, led by electronic, battery and other equipment



Upcycle 5 (3Q22 to 2Q23): Solar and other equipment was the key driver



Current upcycle 6 (3Q24 to 2Q26): Battery, other equipment and electronic have been the main contributors



Airtac: Structurally growing sales warrants higher share price; YoY growth tests near-term share price momentum



Our correlation analysis indicates that Airtac's share price is highly correlated with absolute sales value, exhibiting a correlation of over 90%. Meanwhile, share price performance shows a 30-40% correlation with YoY sales growth.

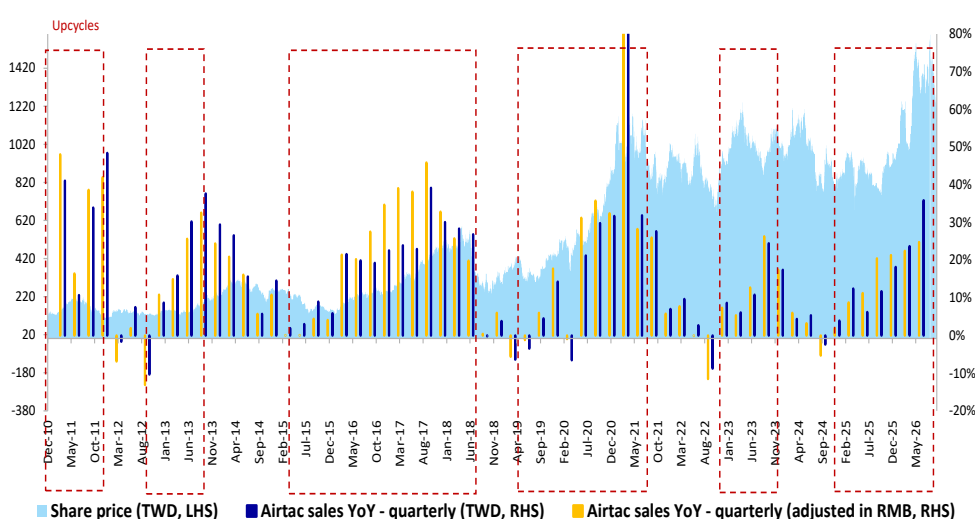
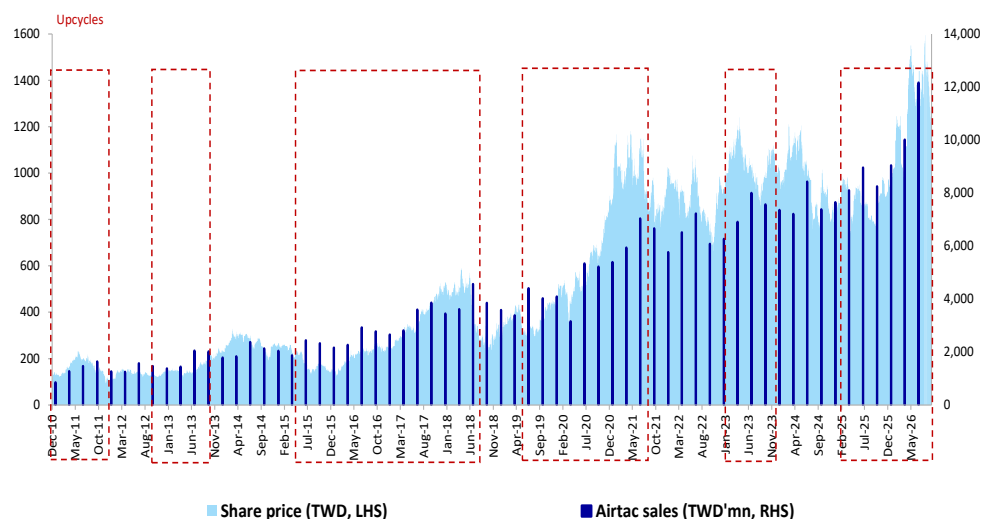
Given Airtac's structural sales growth, driven by market share gains and new product launches, we believe this justifies a higher share price over the long term.

In the near term, however, as we believe the sales growth could be peaking in 2Q26/3Q26E, the potentially decelerating YoY sales growth could test the share price return momentum.

Airtac share price correlates well with sales value, and YoY sales growth

Relationship (Airtac) (since 2011)	Stock lead	Correlation
Quarterly sales (in TWD)		
Quarterly sales (in TWD) vs share price	Same Quarter	0.93
Quarterly sales (in TWD) vs share price	1Q Lead	0.92
Quarterly Sales (in TWD) vs 3M share price return	Same Quarter	-0.06
Quarterly Sales (in TWD) vs 3M share price return	1Q Lead	-0.08
Quarterly sales (in TWD) YoY		
Quarterly sales (in TWD) YoY vs share price	Same Quarter	0.05
Quarterly sales (in TWD) YoY vs share price	1Q Lead	-0.03
Quarterly sales (in TWD) YoY vs 3M share price return	Same Quarter	0.28
Quarterly sales (in TWD) YoY vs 3M share price return	1Q Lead	0.32
Quarterly sales (adjusted in RMB) YoY		
Quarterly sales (adjusted in RMB) YoY vs share price	Same Quarter	0.01
Quarterly sales (adjusted in RMB) YoY vs share price	1Q Lead	-0.07
Quarterly sales (adjusted in RMB) YoY vs 3M share price return	Same Quarter	0.30
Quarterly sales (adjusted in RMB) YoY vs 3M share price return	1Q Lead	0.37

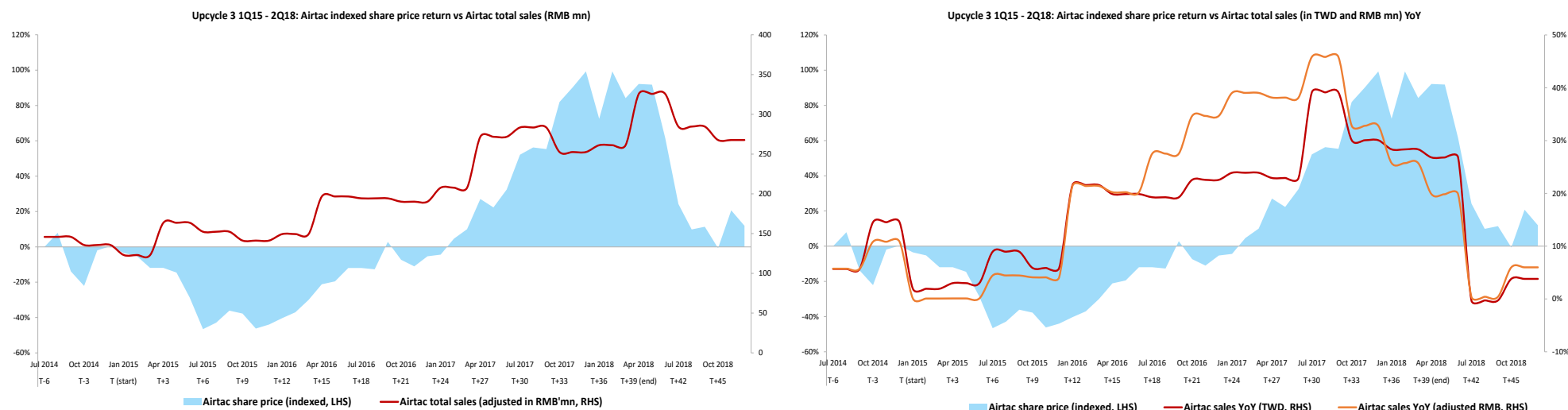
Airtac share price vs sales value and YoY sales growth over time



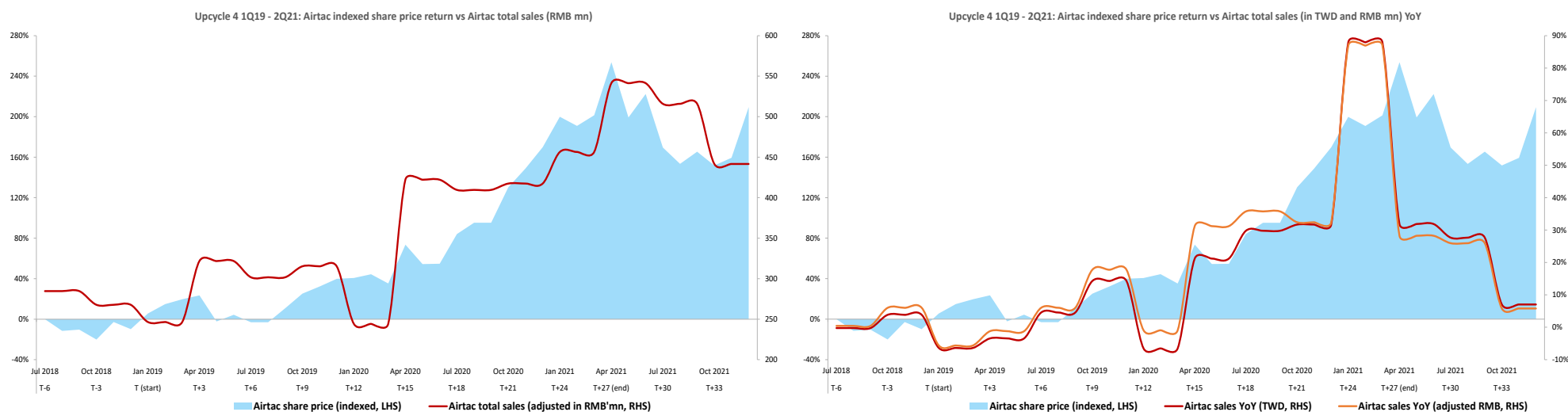
Airtac: Share price performance during past upcycles (1)



Upcycle 3 (1Q15 to 2Q18): Share price peaked when sales reached the peak, and 6M after sales YoY growth peaked



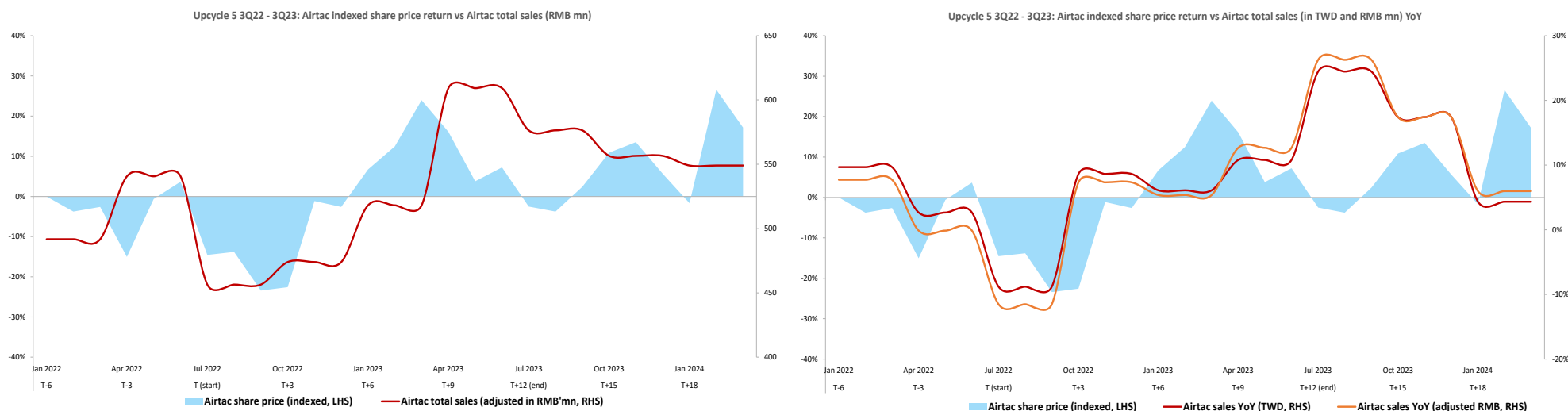
Upcycle 4 (1Q19 to 2Q21): Share price peaked when sales reached the peak, and 3M after sales YoY growth peaked



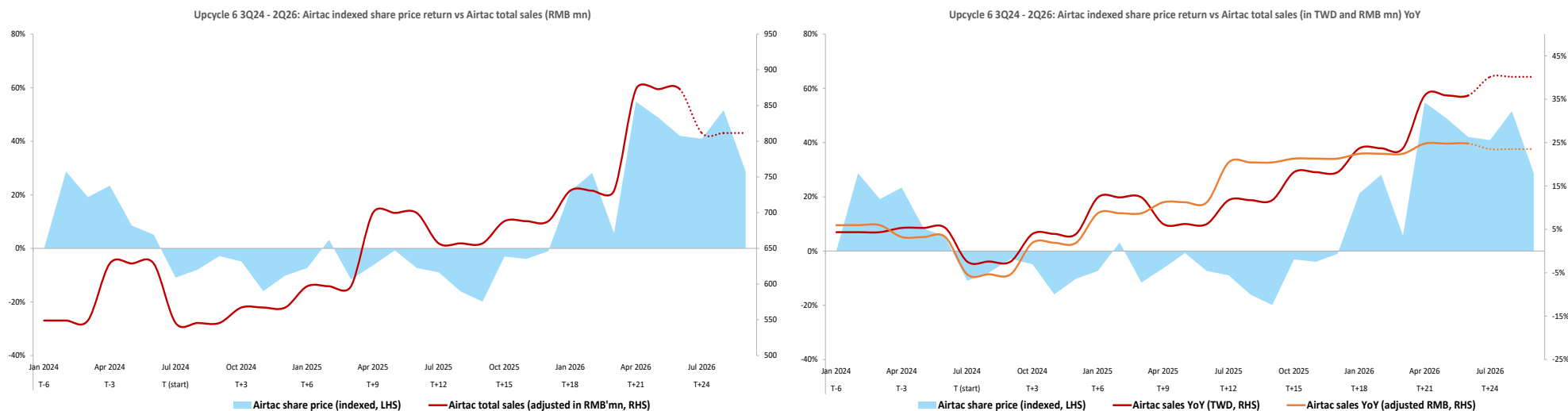
Airtac: Share price performance during past upcycles (2)



Upcycle 5 (3Q22 to 3Q23): Share price peaked 1M prior to the sales peak and 5M prior to the YoY sales growth peak



Upcycle 6 (3Q24 to 2Q26): Share price peaking when the YoY growth also peaking



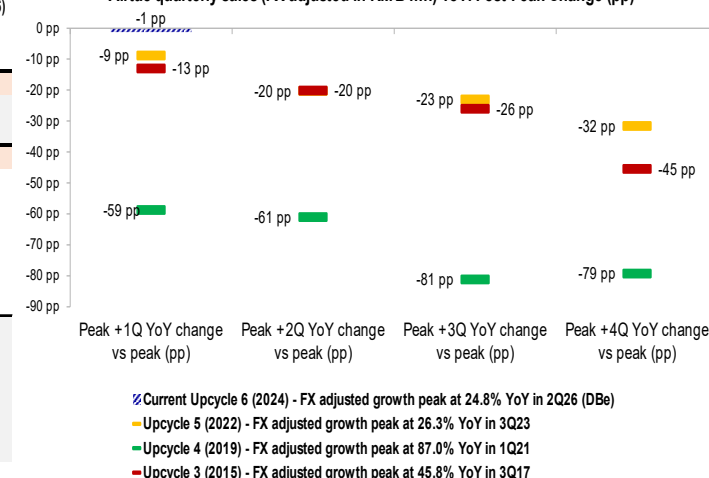
Airtac: Share price correction appears overdone



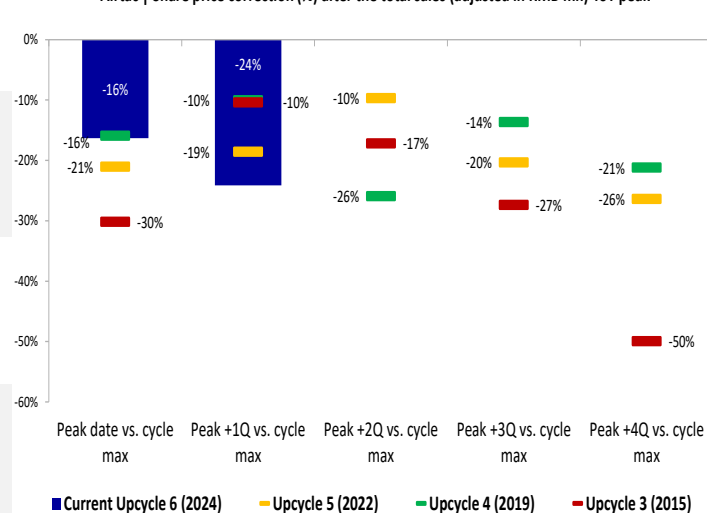
Under the assumption that FX-adjusted sales YoY growth has peaked in 2Q26, SMC's share price has corrected by over -20% from its current cycle peak in September, more severe than the share price corrections observed in the past upcycles

Airtac	Upcycle 3	Upcycle 4	Upcycle 5	Upcycle 6
Upcycle time	1Q2015 (T - T+2) - 2Q2018 (T+39 - T+41)	1Q2019 (T - T+2) - 2Q2021 (T+27 - T+29)	3Q2022 (T - T+2) - 3Q2023 (T+12 - T+14)	3Q2024 (T - T+2) - 3Q2026 (T+24 - T+26)
Upcycle duration	14 quarters	10 quarters	5 quarters	9 quarters
Share-price search window	Jul-2014 (T-6) - Jun-2018 (T+41)	Jul-2018 (T-6) - Jun-2021 (T+29)	Jan-2022 (T-6) - Sep-2023 (T+14)	Jan-2024 (T-6) - Sep-2026 (T+26)
Airtac quarterly sales (FX adjusted, RMB'mn) YoY				
Peak quarter	3Q2017	1Q2021	3Q2023	2Q2026
Peak - sales YoY	45.8%	87.0%	26.3%	24.8%
Airtac share price performance				
Min - Share price (TWD) across the cycle	127	231	653	675
Date of minimum	14-Jan-2016 (T+12 month)	11-Oct-2018 (T-3 month)	25-Oct-2022 (T+3 month)	09-Apr-2025 (T+9 month)
Max - Share price (TWD) across the cycle	590	1,181	1,240	1,595
Date of maximum	18-Apr-2018 (T+39 month)	28-Apr-2021 (T+27 month)	12-Apr-2023 (T+9 month)	12-Aug-2026 (T+25 month)
Increase in share price from min to max (TWD)	463	950	587	920
Increase in share price from min to max (%)	364%	412%	90%	136%
Airtac quarterly sales (FX adjusted, RMB'mn) YoY peak				
Share price (TWD) at the peak date	412	993	979	1,335
Peak-date share price vs. cycle Min share price (TWD)	285	763	326	660
Peak-date share price vs. cycle Min share price (%)	224%	330%	50%	98%
Peak-date share price vs. cycle Max share price (TWD)	-178	-188	-261	-260
Peak-date share price vs. cycle Max share price (%)	-30%	-16%	-21%	-16%
Peak +1 quarter				
Peak +1 quarter - share price (TWD)	529	1,063	1,010	1,210
Peak +1 quarter share price vs. cycle Min share price (TWD)	402	832	357	535
Peak +1 quarter share price vs. cycle Min share price (%)	316%	360%	55%	79.3%
Peak +1 quarter share price vs. cycle Max share price (TWD)	-61	-119	-230	-385
Peak +1 quarter share price vs. cycle Max share price (%)	-10%	-10%	-19%	-24%
Peak +2 quarters				
Peak +2 quarters - share price (TWD)	489	875	1,120	
Peak +2 quarters share price vs. cycle Min share price (TWD)	362	644	467	
Peak +2 quarters share price vs. cycle Min share price (%)	285%	279%	72%	
Peak +2 quarters share price vs. cycle Max share price (TWD)	-101	-306	-120	
Peak +2 quarters share price vs. cycle Max share price (%)	-17%	-26%	-10%	
Peak +3 quarters				
Peak +3 quarters - share price (TWD)	428	1,020	988	
Peak +3 quarters share price vs. cycle Min share price (TWD)	301	789	335	
Peak +3 quarters share price vs. cycle Min share price (%)	237%	342%	51%	
Peak +3 quarters share price vs. cycle Max share price (TWD)	-162	-161	-252	
Peak +3 quarters share price vs. cycle Max share price (%)	-27%	-14%	-20%	
Peak +4 quarters				
Peak +4 quarters - share price (TWD)	296	931	913	
Peak +4 quarters share price vs. cycle Min share price (TWD)	168	700	260	
Peak +4 quarters share price vs. cycle Min share price (%)	133%	303%	40%	
Peak +4 quarters share price vs. cycle Max share price (TWD)	-295	-250	-327	
Peak +4 quarters share price vs. cycle Max share price (%)	-50%	-21%	-26%	

Airtac quarterly sales (FX adjusted in RMB'mn) YoY: Post-Peak Change (pp)



Airtac | Share price correction (%) after the total sales (adjusted in RMB'mn) YoY peak





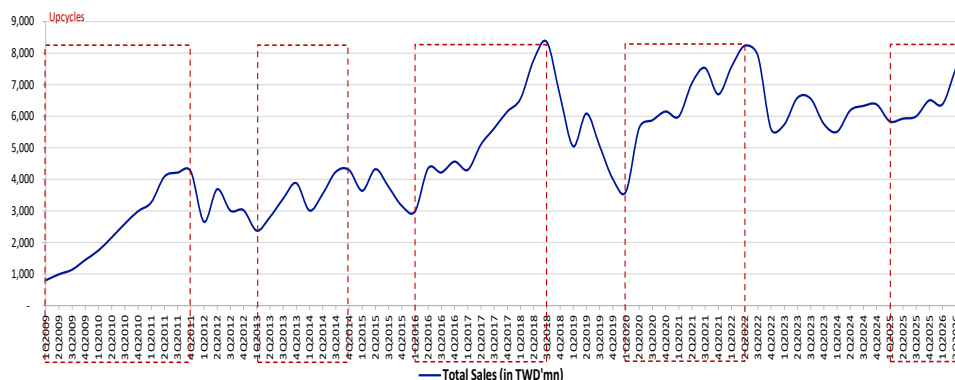
Hiwin: Early stage of the upcycle
implying further upside

Hiwin (2049.TW): BUY, TP TWD440

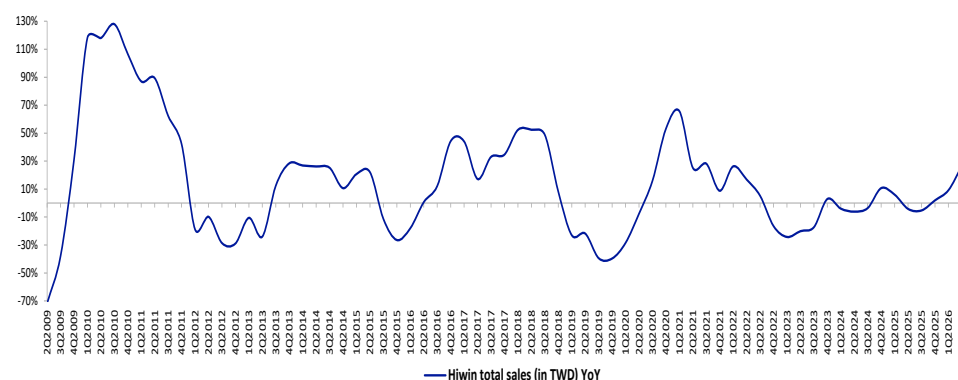
Hiwin – Early stage of the upcycle implying further upside



Hiwin quarterly sales – Still below historical levels

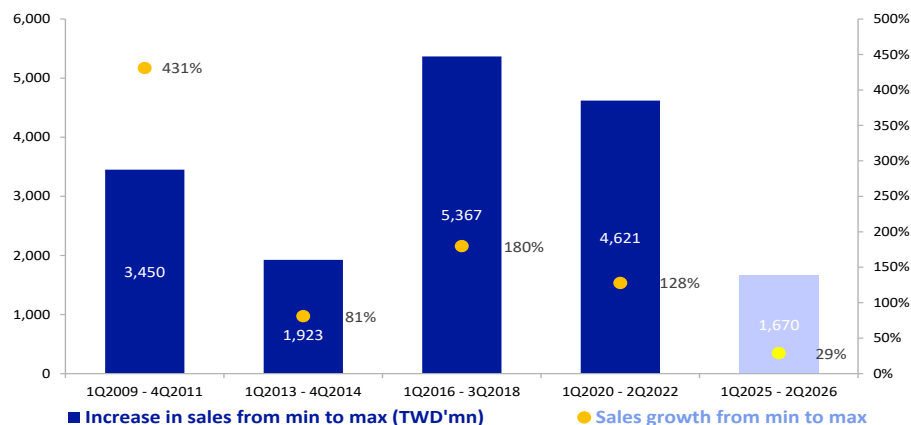


Hiwin quarterly sales YoY



Hiwin sales and share price performance – pattern from past upcycles

Hiwin	Upcycle 1	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5
Revenue performance					
Upcycle time	1Q2009 - 4Q2011	1Q2013 - 4Q2014	1Q2016 - 3Q2018	1Q2020 - 2Q2022	1Q2025 - 2Q2026
Upcycle duration	33	21	30	27	15
Start quarter	1Q2009	1Q2013	1Q2016	1Q2020	1Q2025
Min - Sales (TWD'mn)	800	2,374	2,984	3,614	5,839
End quarter	4Q2011	4Q2014	3Q2018	2Q2022	2Q2026
Max - Sales (TWD'mn)	4,250	4,297	8,352	8,235	7,508
Increase in sales from min to max (TWD'mn)	3,450	1,923	5,367	4,621	1,670
Sales growth from min to max	431%	81%	180%	128%	29%
Increase in sales from min to max - Upcycle 5 as % of ...	48%	87%	31%	36%	



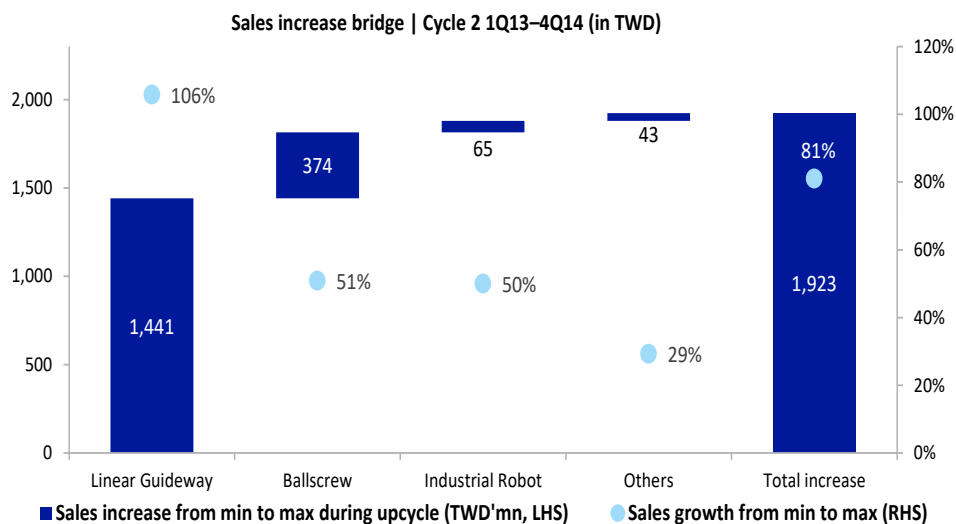
We believe Hiwin is in the early stages of its current upcycle. Historical patterns suggest the company is only 30-35% into this upcycle.

By product (detailed on the following slide), the linear guide and screw businesses have recovered by only 20-30% from their troughs during the current upcycle. This is significantly below the 100%+ recovery seen in previous cycles, implying substantial further room for upside, in our view.

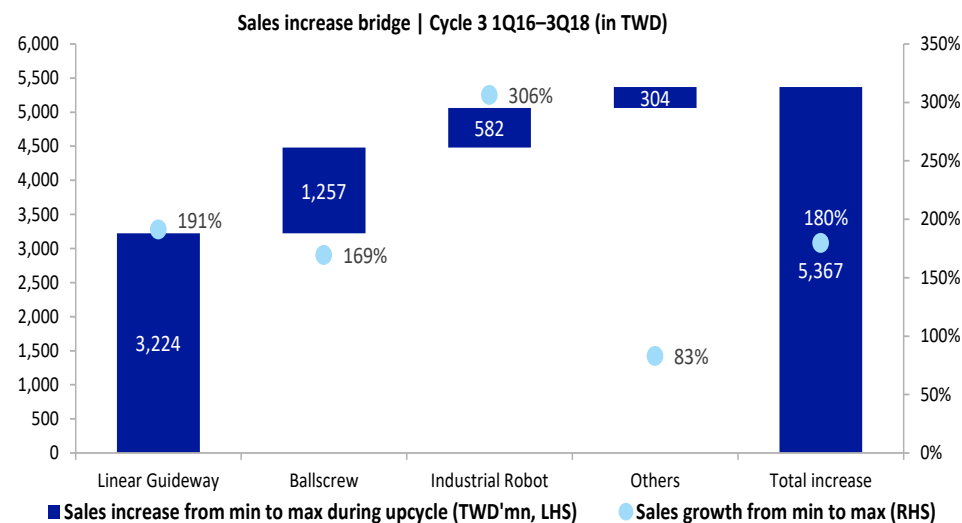
Hiwin: Further room for recovery for linear guide and screw businesses; robot to also help boost sales



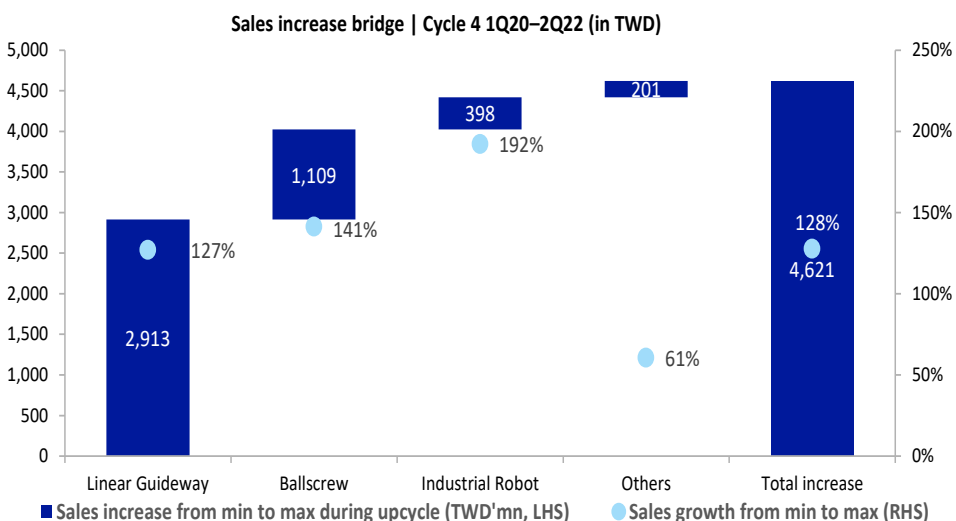
Upcycle 2 (1Q13 to 4Q14)



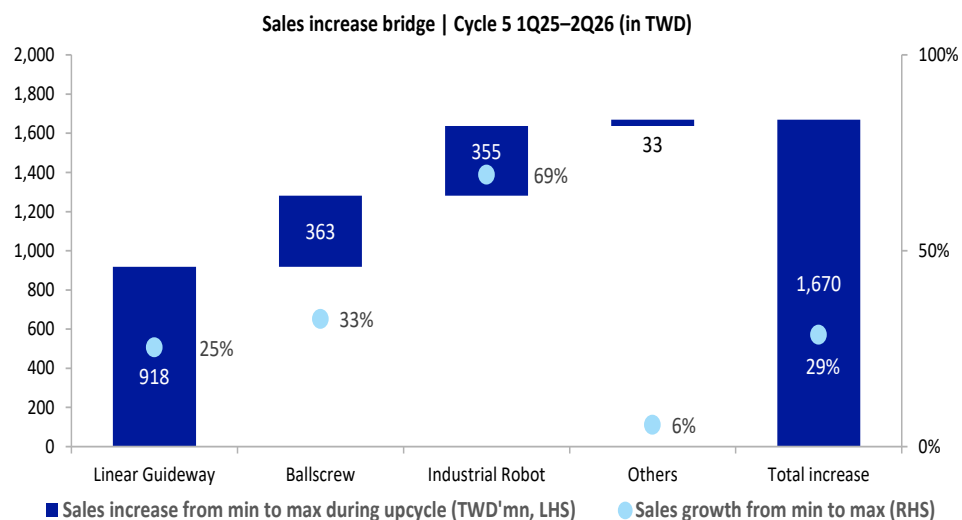
Upcycle 3 (1Q16 to 3Q18)



Upcycle 4 (1Q20 to 2Q22)



Current upcycle 5 (1Q25 to 2Q26)



Hiwin: Correlation implies share price upside



Similar to Airtac, our correlation analysis indicates that Hiwin's share price exhibits a ~75% correlation with absolute sales value. Meanwhile, the share price movement shows a ~40% correlation with YoY sales growth.

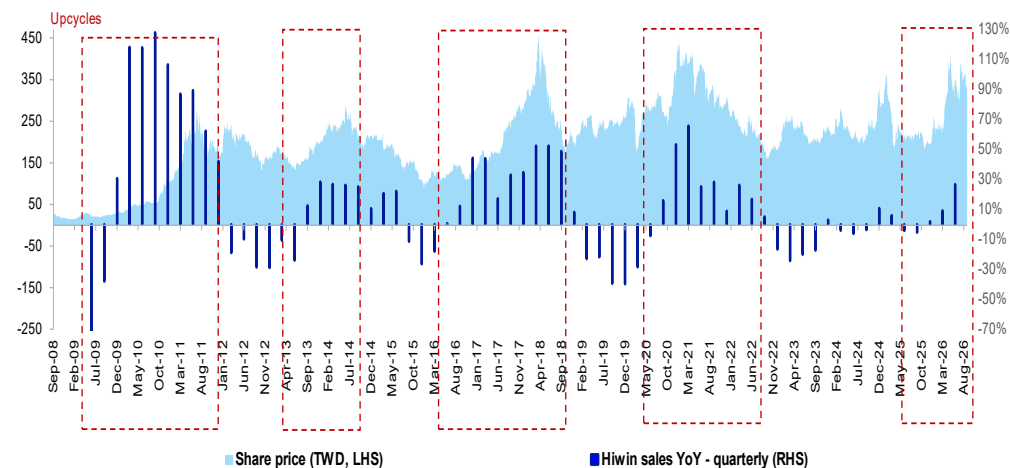
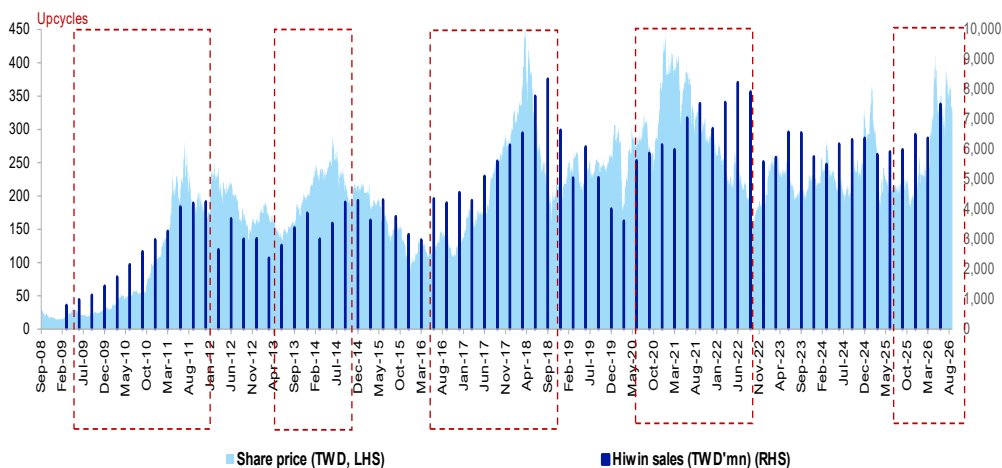
As we believe Hiwin is in the early stages of its upcycle and sales are poised to grow further, this should translate into a higher share price.

In the near term, we forecast accelerating YoY sales growth in 3Q26E versus 2Q26, which should support share price momentum.

Hiwin share price correlates with sales value, and YoY sales growth

Relationship (Hiwin) (since 2012)	Stock lead	Correlation
Quarterly sales		
Quarterly sales vs share price	Same Quarter	0.54
Quarterly sales vs share price	1Q Lead	0.60
Quarterly Sales vs 3M share price return	Same Quarter	-0.05
Quarterly Sales vs 3M share price return	1Q Lead	0.05
Quarterly sales YoY		
Quarterly sales YoY vs share price	Same Quarter	0.44
Quarterly sales YoY vs share price	1Q Lead	0.34
Quarterly sales YoY vs 3M share price return	Same Quarter	0.16
Quarterly sales YoY vs 3M share price return	1Q Lead	0.29

Hiwin share price vs sales value and YoY sales growth over time

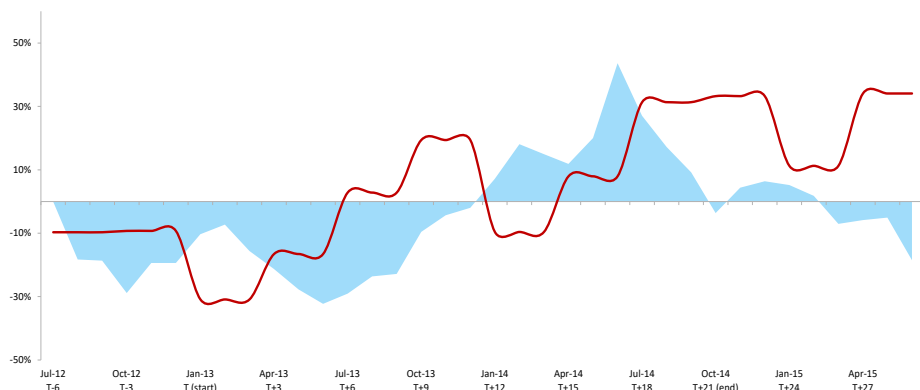


Hiwin: Share price performance during past upcycles (1)



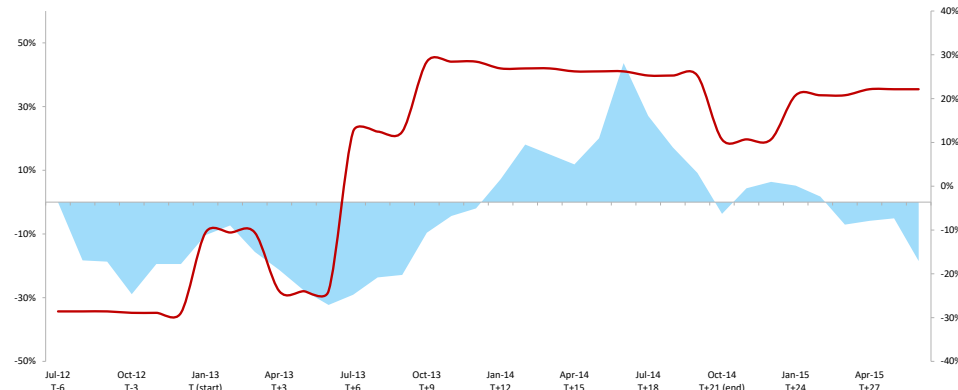
Upcycle 2 (1Q13 to 4Q13): Share price peaked 2M prior to the sales peak, and 3M prior to the sales YoY growth peak

Upcycle 2 1Q13 - 4Q14: Hiwin share price return (vs T-6) vs Hiwin monthly sales proxy (TWD mn)



Hiwin share price (indexed, LHS) Hiwin total sales (in TWD'mn, RHS)

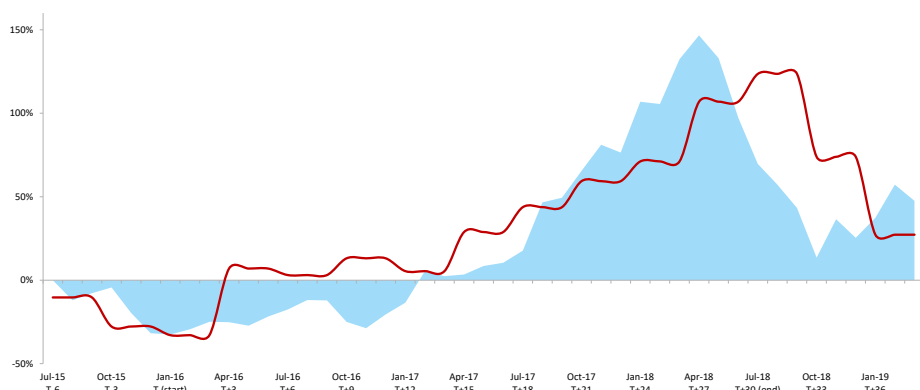
Upcycle 2 1Q13 - 4Q14: Hiwin share price return (vs T-6) vs Hiwin quarterly sales YoY



Hiwin share price (indexed, LHS) Airtac total salesYoY (RHS)

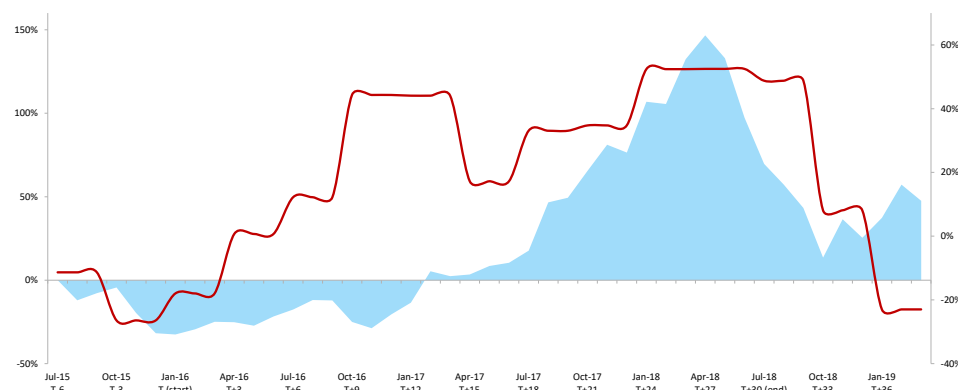
Upcycle 3 (1Q16 to 3Q18): Share price peaked 3M prior to the sales peak, and coincided with the YoY growth peak

Upcycle 3 1Q16 - 3Q18: Hiwin share price return (vs T-6) vs Hiwin monthly sales proxy (TWD mn)



Hiwin share price (indexed, LHS) Hiwin total sales (in TWD'mn, RHS)

Upcycle 3 1Q16 - 3Q18: Hiwin share price return (vs T-6) vs Hiwin quarterly sales YoY



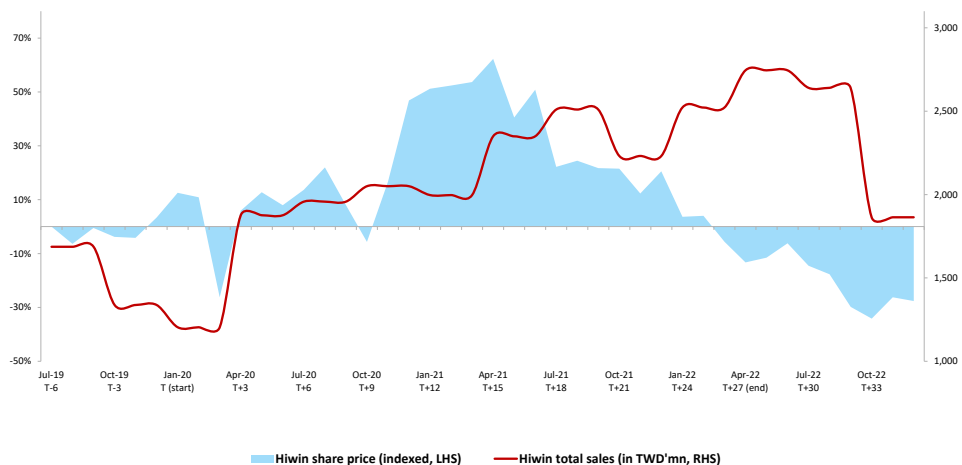
Hiwin share price (indexed, LHS) Airtac total salesYoY (RHS)

Hiwin: Share price performance during past upcycles (2)

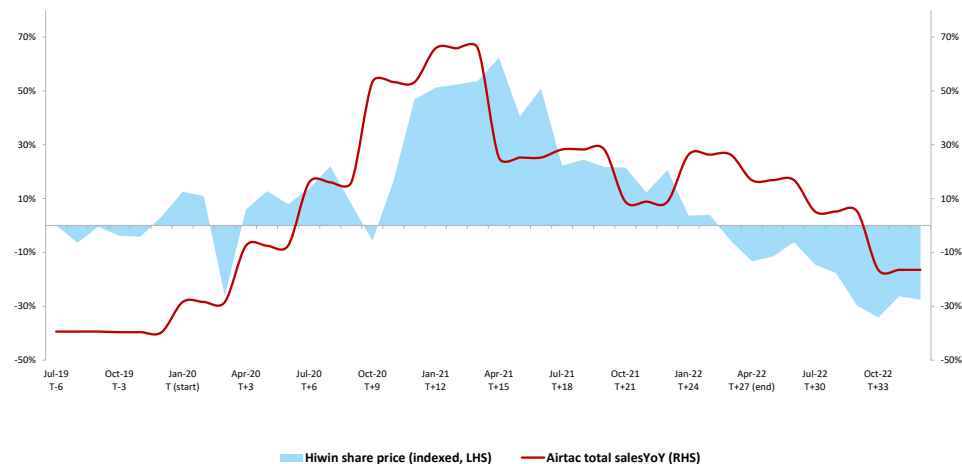


Upcycle 4 (1Q20 to 2Q22): Share price peaked 12M prior to the sales peak, and 1M post the YoY sales growth peak

Upcycle 4 1Q20 - 2Q22: Hiwin share price return (vs T-6) vs Hiwin monthly sales proxy (TWD mn)

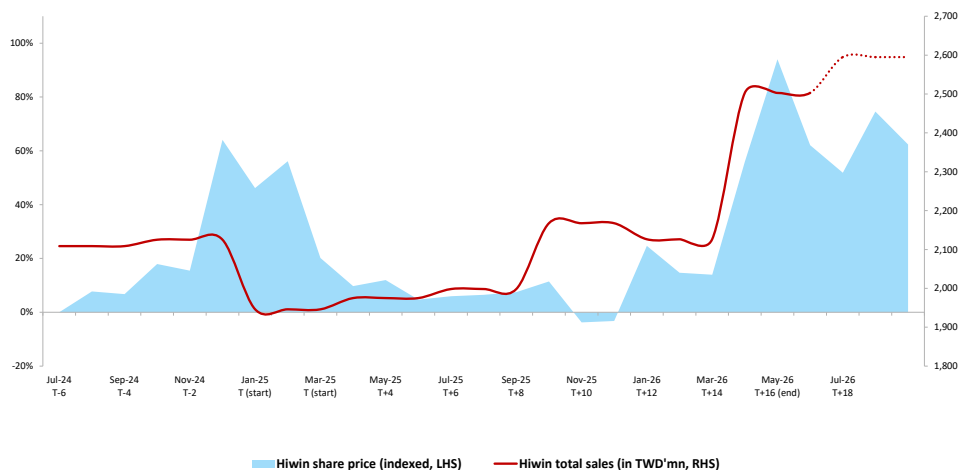


Upcycle 4 1Q20 - 2Q22: Hiwin share price return (vs T-6) vs Hiwin quarterly sales YoY

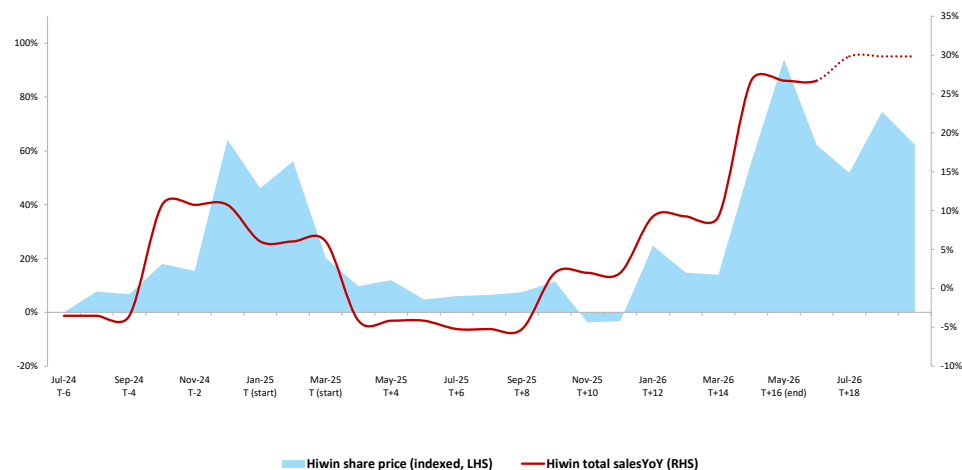


Upcycle 5 (1Q25 to 2Q26): Share price already peaked in April 2026, although we believe neither the sales or sales YoY growth has peaked yet

Upcycle 5 1Q25 - 2Q26: Hiwin share price return (vs T-6) vs Hiwin monthly sales proxy (TWD mn)



Upcycle 5 1Q25 - 2Q26: Hiwin share price return (vs T-6) vs Hiwin quarterly sales YoY

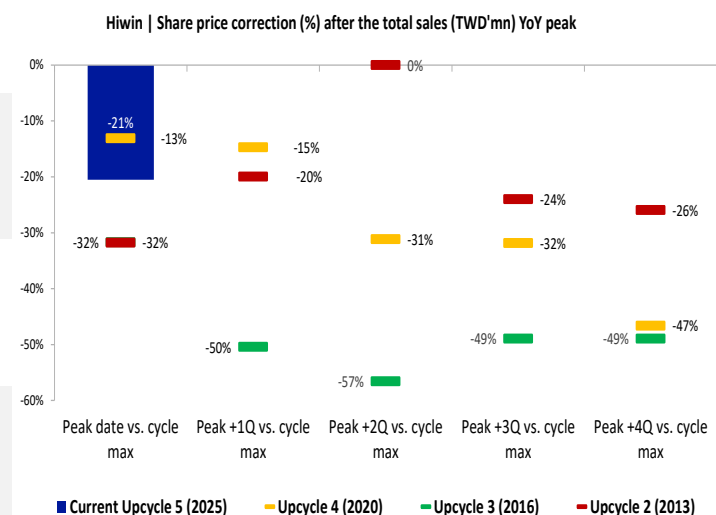
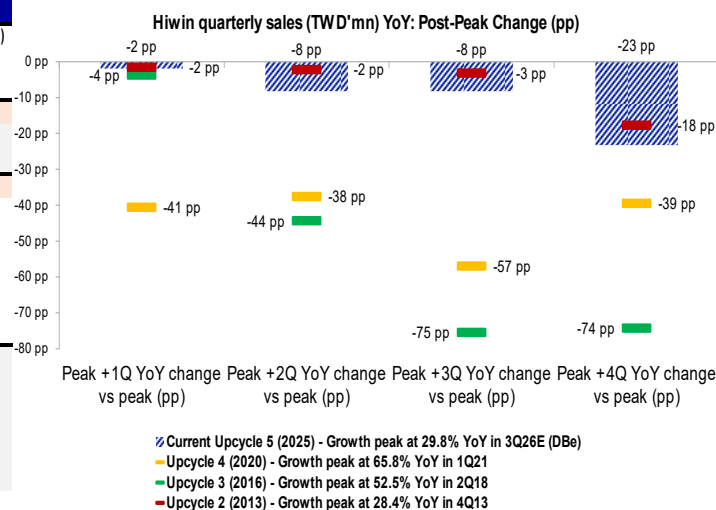


Hiwin: Share price correction factors in a more severe sales growth deceleration



We expect mild sales growth deceleration at Hiwin, whilst its share price correct has more than priced in the deceleration, in our view

Hiwin	Upcycle 2	Upcycle 3	Upcycle 4	Upcycle 5
Upcycle time	1Q2013 (T - T+2) - 4Q2014 (T+21 - T+23)	1Q2016 (T - T+2) - 3Q2018 (T+30 - T+32)	1Q2020 (T - T+2) - 2Q2022 (T+27 - T+29)	1Q2025 (T - T+2) - 3Q2026 (T+18 - T+20)
Upcycle duration	8 quarters	11 quarters	10 quarters	7 quarters
Share-price search window	Jul-2012 (T-6) - Dec-2014 (T+23)	Jul-2015 (T-6) - Sep-2018 (T+32)	Jul-2019 (T-6) - Jun-2022 (T+29)	Jul-2024 (T-6) - Sep-2026 (T+20)
Hiwin quarterly sales (TWD'mn) YoY				
Peak quarter	4Q2013	2Q2018	1Q2021	3Q2026 (DBe)
Peak - sales YoY	28.4%	52.5%	65.8%	29.8%
Hiwin share price performance				
Min - Share price (TWD) across the cycle	129	88	166	179
Date of minimum	05-Nov-2012 (T-2 month)	15-Jan-2016 (T)	19-Mar-2020 (T+2 month)	09-Apr-2025 (T+3 month)
Max - Share price (TWD) across the cycle	290	458	446	410
Date of maximum	30-Jun-2014 (T+17 month)	18-Apr-2018 (T+27 month)	21-Jan-2021 (T+12 month)	26-May-2026 (T+16 month)
Increase in share price from min to max (TWD)	161	370	279	231
Increase in share price from min to max (%)	125%	422%	168%	129%
Hiwin quarterly sales (TWD'mn) YoY peak	4Q2013 (T+9 - T+11)	2Q2018 (T+27 - T+29)	1Q2021 (T+12 - T+14)	3Q2026E (T+18 - T+20)
Share price (TWD) at the peak date	198	313	387	326 (25-Sep-2026, latest date)
Peak-date share price vs. cycle Min share price (TWD)	69	225	221	147
Peak-date share price vs. cycle Min share price (%)	54%	257%	133%	82%
Peak-date share price vs. cycle Max share price (TWD)	-92	-145	-58	-84
Peak-date share price vs. cycle Max share price (%)	-32%	-32%	-13%	-21%
Peak +1 quarter	1Q2014	3Q2018	2Q2021	
Peak +1 quarter - share price (TWD)	232	227	380	
Peak +1 quarter share price vs. cycle Min share price (TWD)	104	140	214	
Peak +1 quarter share price vs. cycle Min share price (%)	80%	159%	128%	
Peak +1 quarter share price vs. cycle Max share price (TWD)	-58	-231	-65	
Peak +1 quarter share price vs. cycle Max share price (%)	-20%	-50%	-15%	
Peak +2 quarters	2Q2014	4Q2018	3Q2021	
Peak +2 quarters - share price (TWD)	290	199	307	
Peak +2 quarters share price vs. cycle Min share price (TWD)	161	111	140	
Peak +2 quarters share price vs. cycle Min share price (%)	125%	127%	84%	
Peak +2 quarters share price vs. cycle Max share price (TWD)	0	-259	-139	
Peak +2 quarters share price vs. cycle Max share price (%)	0%	-57%	-31%	
Peak +3 quarters	3Q2014	1Q2019	4Q2021	
Peak +3 quarters - share price (TWD)	221	234	304	
Peak +3 quarters share price vs. cycle Min share price (TWD)	92	146	137	
Peak +3 quarters share price vs. cycle Min share price (%)	71%	167%	83%	
Peak +3 quarters share price vs. cycle Max share price (TWD)	-70	-224	-142	
Peak +3 quarters share price vs. cycle Max share price (%)	-24%	-49%	-32%	
Peak +4 quarters	4Q2014	2Q2019	1Q2022	
Peak +4 quarters - share price (TWD)	215	234	238	
Peak +4 quarters share price vs. cycle Min share price (TWD)	86	146	72	
Peak +4 quarters share price vs. cycle Min share price (%)	67%	167%	43%	
Peak +4 quarters share price vs. cycle Max share price (TWD)	-75	-224	-208	
Peak +4 quarters share price vs. cycle Max share price (%)	-26%	-49%	-47%	





Previous reports

Initiations and key reports:

- **Humanoid Robot (III):** [Six visions for 2026 – Scaling, iterating and diversifying – 6 Mar 2026](#)
- **China's 15th Five-Year Plan (I):** [Innovation and self-reliance; Emerging and future industries – 8 Jan 2026](#)
- **APAC Automation & Machinery:** [2026 outlook: More positives than negatives - 24 Nov 2025](#)
- **China Macro:** [The upcoming AI Boom in China - 26 Mar 2025](#)
- **Humanoid Robot (II):** [Go big and go home - 14 Feb 2025](#)
- **APAC Automation & Machinery Initiation:** [Stay selective; Humanoid robots lift growth prospects - 23 July 2024](#)
- **APAC Industrial Automation Initiation:** [Identifying winners amid rising competition from China - 19 Feb 2024](#)

Japan machine tool order reports:

- **Japan machine tool order update:** [August 2026 \(Preliminary\): Growth hit a recent high – 14 Sep 2026](#)
- **Japan machine tool order update:** [July 2026 \(Detailed\): Another strong month – 27 Aug 2026](#)
- **Japan machine tool order update:** [June 2026 \(detailed\) – 30 July 2026](#)
- **Japan machine tool order update:** [May 2026 \(detailed\) – 26 June 2026](#)
- **Japan machine tool order update:** [Apr 2026 \(detailed\) – 27 May 2026](#)

China automation market reports:

- **China Automation Market Monthly tracker:** [Decent August but growth moderates from 2Q26 peak; 2H26E on a deceleration path – 8 Sep 2026](#)
- **China Automation Market Monthly tracker:** [July slowed; Likely growth deceleration in 2H26E; K-shaped dynamics continued – 18 Aug 2026](#)
- **China Automation Market Monthly tracker:** [June stayed solid, finishing 2Q26 stronger than 1Q26; Growth could decelerate in 2H26E – 14 July 2026](#)
- **China Automation Market Monthly tracker:** [June could slightly decelerate; Further price increases likely in 2H26E – 5 June 2026](#)
- **China Automation Market Monthly tracker:** [Broad-based acceleration in April; Shortage emerging; May to remain robust – 21 May 2026](#)

Automation company reports:

- **HIWIN:** [Positive momentum to continue; BUY – 19 Aug 2026](#)
- **Harmonic Drive Systems:** [Beat and raise in FY1Q; Semis and medical strong; AI robot growing; BUY – 7 Aug 2026](#)
- **SMC:** [FY1Q ahead; Strong semis boosting orders; Upside to FY targets; BUY – 7 Aug 2026](#)
- **FANUC:** [Component shortages constrained strong demand; DB channel checks – 3 Aug 2026](#)
- **Yaskawa Electric:** [Profits missed due to system migration; Underlying momentum solid; Buy – 13 Jul 2026](#)
- **Airtac:** [Preview: Strong 1Q26; FX supports; No price increase yet; Distributor feedback – 13 Apr 2026](#)
- **THK:** [3Q25 miss; FY guidance downgrade; Shareholder return losing steam; Sell – 12 Nov 2025](#)
- **Annual benchmarking of global peers:** [Some hopes and some gaps – 17 Sep 2025](#)

Appendix 1

Important Disclosures



*Other information available upon request

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst about the subject issuers and the securities of those issuers. In addition, the undersigned lead analyst has not and will not receive any compensation for providing a specific recommendation or view in this report. **IRIS ZHENG, XU LIU**



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively "Deutsche Bank"). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls ideas on the Research Website (<https://research.db.com/Research/>), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call idea represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls ideas, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.



The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (<https://research.db.com/Research/>) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates – these are common in emerging markets. The index fixings may – by construction – lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited – up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the "Characteristics and Risks of Standardized Options", at <https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document>. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at <https://research.db.com/Research/> on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.



European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the "Additional Information" section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to "professional investors" as defined in Part 1 of Schedule 1 of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929, email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: <https://www.db.com/ir/en/annual-reports.htm>

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. "Moody's", "Standard & Poor's", and "Fitch" mentioned in this report are not registered credit rating agencies in Japan unless Japan or "Nippon" is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, +65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.



Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. . DSSA registered office is at Faisaliah Tower, 17th floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for "wholesale clients" within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australian specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD000000000521304.xhtml . Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG.

It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2025 Deutsche Bank AG